SOLUTIONS MANUAL
Bioprocess Engineering Principles

Pauline M. Doran

$$\eta_{il} = \frac{\tanh \phi_l}{\phi_l}$$

$$\mu = \frac{\mu_{max}\, s}{K_S + s}$$

$$\dot{Q} = -k A \frac{dT}{dy}$$

Peclet number

Effective diffusivity

Apparent viscosity

$$k = A e^{-E/RT}$$

$$\dot{\gamma} = k N_i$$

$$Re_i = \frac{N_i D_i^2 \rho}{K \dot{\gamma}^{n-1}}$$

$$\frac{1}{UA} = \frac{1}{h_h A} + \frac{B}{kA} + \frac{1}{h_c A}$$

Specific growth rate

Thiele modulus

$$\sum_{\text{input streams}} (Mh) - \sum_{\text{output streams}} (Mh) - Q + W_s = \Delta E$$

Fouling factor

Ideal gas law

$$C_{AS}^* = K_F C_A^{*\,1/n}$$

Steady-state mass balance

SIGMA FACTOR

$$t_d = \frac{\ln 2}{\mu}$$

Heat of reaction

$$Nu = 0.87\, Re_i^{0.62}\, Pr^{0.33} \left(\frac{\mu_b}{\mu_w}\right)^{0.14}$$

$$\mathcal{D}_{Az} \frac{\partial^2 C_A}{\partial z^2} + u \frac{-\partial C_A}{\partial z} = \frac{\partial C_A}{\partial t} + \left(\frac{1-\varepsilon}{\varepsilon}\right) \frac{\partial C_{AS}}{\partial t}$$

Dimensional homogeneity

$$k_L a \left(C_{AL}^* - C_{AL}\right) = q_O x$$

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SOLUTIONS MANUAL

Bioprocess Engineering Principles

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ISBN 0 7334 1547 4

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Table of Contents

Solutions

		Page
Chapter 2	*Introduction to Engineering Calculations*	1
Chapter 3	*Presentation and Analysis of Data*	9
Chapter 4	*Material Balances*	17
Chapter 5	*Energy Balances*	41
Chapter 6	*Unsteady-State Material and Energy Balances*	54
Chapter 7	*Fluid Flow and Mixing*	76
Chapter 8	*Heat Transfer*	86
Chapter 9	*Mass Transfer*	98
Chapter 10	*Unit Operations*	106
Chapter 11	*Homogeneous Reactions*	122
Chapter 12	*Heterogeneous Reactions*	139
Chapter 13	*Reactor Engineering*	151

NOTE

All equations, tables, figures, page numbers, etc., mentioned in this manual refer to the textbook, *Bioprocess Engineering Principles.*

2

Introduction to Engineering Calculations

2.1 Unit conversion

(a)

From Table A.9 (Appendix A): 1 cP = 10^{-3} kg m^{-1} s^{-1}

1 m = 100 cm

Therefore:

$$1.5\times10^{-6}\,\text{cP} = 1.5\times10^{-6}\,\text{cP}\cdot\left|\frac{10^{-3}\,\text{kg m}^{-1}\,\text{s}^{-1}}{1\,\text{cP}}\right|\cdot\left|\frac{1\,\text{m}}{100\,\text{cm}}\right| = 1.5\times10^{-11}\,\text{kg s}^{-1}\,\text{cm}^{-1}$$

Answer: 1.5 × 10^{-11} kg s^{-1} cm^{-1}

(b)

From Table A.8 (Appendix A): 1 hp (British) = 42.41 Btu min^{-1}

Therefore:

$$0.122\,\text{hp} = 0.122\,\text{hp}\cdot\left|\frac{42.41\,\text{Btu min}^{-1}}{1\,\text{hp}}\right| = 5.17\,\text{Btu min}^{-1}$$

Answer: 5.17 Btu min^{-1}

(c)

From Table A.5 (Appendix A): 1 mmHg = 1.316 × 10^{-3} atm

From Table A.1 (Appendix A): 1 ft = 0.3048 m

From Table A.7 (Appendix A): 1 l atm = 9.604 × 10^{-2} Btu

From Table A.8 (Appendix A): 1 Btu min^{-1} = 2.391 × 10^{-2} metric horsepower

1 m = 100 cm

1 l = 1000 cm^3

1 h = 60 min

Therefore:

$$670\,\text{mmHg ft}^3 = 670\,\text{mmHg ft}^3\cdot\left|\frac{1.316\times10^{-3}\,\text{atm}}{1\,\text{mmHg}}\right|\cdot\left|\frac{9.604\times10^{-2}\,\text{Btu}}{1\,\text{l atm}}\right|\cdot\left|\frac{0.3048\,\text{m}}{1\,\text{ft}}\right|^3\cdot\left|\frac{100\,\text{cm}}{1\,\text{m}}\right|^3\cdot\left|\frac{1\,\text{l}}{1000\,\text{cm}^3}\right|\cdot\left|\frac{2.391\times10^{-2}\,\text{metric horsepower}}{1\,\text{Btu min}^{-1}}\right|\cdot\left|\frac{1\,\text{h}}{60\,\text{min}}\right| = 9.56\times10^{-4}\,\text{metric horsepower h}$$

Answer: 9.56 × 10^{-4} metric horsepower h

(d)

From Table A.7 (Appendix A): 1 Btu = 0.2520 kcal

From Table A.3 (Appendix A): 1 lb = 453.6 g

Therefore:

$$345\,\text{Btu lb}^{-1} = 345\,\text{Btu lb}^{-1}\cdot\left|\frac{0.2520\,\text{kcal}}{1\,\text{Btu}}\right|\cdot\left|\frac{1\,\text{lb}}{453.6\,\text{g}}\right| = 0.192\,\text{kcal g}^{-1}$$

Answer: 0.192 kcal g^{-1}

2.2 Unit conversion

Case 1

Convert to units of kg, m, s.

From Table A.3 (Appendix A): 1 lb = 0.4536 kg

From Table A.2 (Appendix A): 1 $ft^3 = 2.832 \times 10^{-2}$ m^3
From Table A.9 (Appendix A): 1 cP = 10^{-3} kg m^{-1} s^{-1}
1 m = 100 cm = 1000 mm
Therefore, using Eq. (2.1):

$$Re = \frac{D u \rho}{\mu} = \frac{\left(2\text{ mm} \cdot \left|\frac{1\text{ m}}{1000\text{ mm}}\right|\right)\left(3\text{ cm s}^{-1} \cdot \left|\frac{1\text{ m}}{100\text{ cm}}\right|\right)\left(25\text{ lb ft}^{-3} \cdot \left|\frac{0.4536\text{ kg}}{1\text{ lb}}\right| \cdot \left|\frac{1\text{ ft}^3}{2.832\times10^{-2}\text{ m}^3}\right|\right)}{10^{-6}\text{ cP} \cdot \left|\frac{10^{-3}\text{ kg m}^{-1}\text{ s}^{-1}}{1\text{ cP}}\right|} = 2.4\times10^7$$

Answer: 2.4×10^7

Case 2
Convert to units of kg, m, s.
From Table A.1 (Appendix A): 1 in. = 2.54×10^{-2} m
From Table A.9 (Appendix A): 1 lb_m ft^{-1} h^{-1} = 4.134×10^{-4} kg m^{-1} s^{-1}
1 h = 3600 s
Therefore, using Eq. (2.1):

$$Re = \frac{D u \rho}{\mu} = \frac{\left(1\text{ in.} \cdot \left|\frac{2.54\times10^{-2}\text{ m}}{1\text{ in.}}\right|\right)\left(1\text{ m s}^{-1}\right)\left(12.5\text{ kg m}^{-3}\right)}{0.14\times10^{-4}\text{ lb}_m\text{ s}^{-1}\text{ ft}^{-1} \cdot \left|\frac{4.134\times10^{-4}\text{ kg m}^{-1}\text{ s}^{-1}}{1\text{ lb}_m\text{ ft}^{-1}\text{ h}^{-1}}\right| \cdot \left|\frac{3600\text{ s}}{\text{h}}\right|} = 1.5\times10^4$$

Answer: 1.5×10^4

2.3 Dimensionless groups and property data

From the *Chemical Engineers' Handbook*, the diffusivity of oxygen in water at 25°C is 2.5×10^{-5} cm^2 s^{-1}. Assuming this is the same at 28°C, $\mathcal{D} = 2.5 \times 10^{-5}$ cm^2 s^{-1}. Also, from the *Chemical Engineers' Handbook*, the density of water at 28°C is $\rho_L = 0.9962652$ g cm^{-3}, and the viscosity of water at 28°C is $\mu_L = 0.87$ cP. The density of oxygen at 28°C and 1 atm pressure can be calculated using the ideal gas law. As molar density is the same as n/V, from Eq. (2.32):

$$\rho_G = \frac{n}{V} = \frac{p}{RT}$$

Temperature in the ideal gas equation is absolute temperature; therefore, from Eq. (2.24):

$$T = (28 + 273.15)\text{ K} = 301.15\text{ K}$$

From Table 2.5, $R = 82.057$ cm^3 atm K^{-1} $gmol^{-1}$. Substituting parameter values into the density equation gives:

$$\rho_G = \frac{p}{RT} = \frac{1\text{ atm}}{\left(82.057\text{ cm}^3\text{ atm K}^{-1}\text{ gmol}^{-1}\right)(301.15\text{ K})} = 4.05\times10^{-5}\text{ gmol cm}^{-3}$$

From the atomic weights in Table B.1 (Appendix B), the molecular weight of oxygen is 32.0. Converting the result for ρ_G to mass terms:

$$\rho_G = 4.05\times10^{-5}\text{ gmol cm}^{-3} \cdot \left|\frac{32.0\text{ g}}{1\text{ gmol}}\right| = 1.30\times10^{-3}\text{ g cm}^{-3}$$

From Table A.9 (Appendix A): 1 cP = 10^{-2} g cm^{-1} s^{-1}; from Table A.1 (Appendix A): 1 ft = 0.3048 m = 30.48 cm. The parameter values and conversion factors, together with $D_b = 2$ mm = 0.2 cm, can now be used to calculate the dimensionless groups in the equation for the Sherwood number.

$$Gr = \frac{D_b^3 \rho_G (\rho_L - \rho_G) g}{\mu_L^2} = \frac{(0.2\text{ cm})^3\left(1.30\times10^{-3}\text{ g cm}^{-3}\right)\left(0.9962652 - 1.30\times10^{-3}\right)\text{g cm}^{-3}\left(32.17\text{ ft s}^{-2} \cdot \left|\frac{30.48\text{ cm}}{1\text{ ft}}\right|\right)}{\left(0.87\text{ cP} \cdot \left|\frac{10^{-2}\text{ g cm}^{-1}\text{ s}^{-1}}{1\text{ cP}}\right|\right)^2}$$

$$Gr = 134$$

$$Sc = \frac{\mu_L}{\rho_L \mathcal{D}} = \frac{0.87\,\text{cP}\,.\left|\dfrac{10^{-2}\,\text{g cm}^{-1}\,\text{s}^{-1}}{1\,\text{cP}}\right|}{\left(0.9962652\,\text{g cm}^{-3}\right)\left(2.5\times10^{-5}\,\text{cm}^2\,\text{s}^{-1}\right)} = 349$$

Therefore:

$$Sh = 0.31\,Gr^{1/3}\,Sc^{1/3} = 0.31\,(134)^{1/3}\,(349)^{1/3} = 11.2$$

From the equation for *Sh*:

$$k_L = \frac{Sh\,\mathcal{D}}{D_b} = \frac{(11.2)\left(2.5\times10^{-5}\,\text{cm}^2\,\text{s}^{-1}\right)}{0.2\,\text{cm}} = 1.40\times10^{-3}\,\text{cm s}^{-1}$$

Answer: 1.40×10^{-3} cm s^{-1}

2.4 Mass and weight

From the definition of density on p 16, mass is equal to volume multiplied by density. Therefore:

$$\text{Mass of water} = (10\,\text{ft}^3)\,(62.4\,\text{lb}_m\,\text{ft}^{-3}) = 624\,\text{lb}_m$$

From p 16, weight is the force with which a body is attracted to the centre of the earth by gravity. According to Newton's law (p 15), this force is equal to the mass of the body multiplied by the gravitational acceleration.

(a)

From p 15, at sea level and 45° latitude, gravitational acceleration $g = 32.174$ ft s^{-2}. Therefore:

$$\text{Weight} = 624\,\text{lb}_m\,(32.174\,\text{ft s}^{-2}) = 2.008\times10^4\,\text{lb}_m\,\text{ft s}^{-2}$$

Converting these units to lb$_f$, from Eq. (2.16), 1 lb$_f$ = 32.174 lb$_m$ ft s^{-2}; therefore:

$$\text{Weight} = 2.008\times10^4\,\text{lb}_m\,\text{ft s}^{-2}\,.\left|\frac{1\,\text{lb}_f}{32.174\,\text{lb}_m\,\text{ft s}^{-2}}\right| = 624\,\text{lb}_f$$

Answer: 624 lb$_f$. *When $g = 32.174$ ft s^{-2}, the number of lb mass is equal to the number of lb force.*

(b)

From Table A.1 (Appendix A): 1 m = 3.281 ft. Using the same procedure as in **(a)**:

$$\text{Weight} = 624\,\text{lb}_m\left(9.76\,\text{m s}^{-2}\,.\left|\frac{3.281\,\text{ft}}{1\,\text{m}}\right|\right) = 1.998\times10^4\,\text{lb}_m\,\text{ft s}^{-2}$$

Converting to lb$_f$:

$$\text{Weight} = 1.998\times10^4\,\text{lb}_m\,\text{ft s}^{-2}\,.\left|\frac{1\,\text{lb}_f}{32.174\,\text{lb}_m\,\text{ft s}^{-2}}\right| = 621\,\text{lb}_f$$

Answer: 621 lb$_f$

2.5 Dimensionless numbers

First, evaluate the units of the groups $(C_p\,\mu/k)$ and $(D\,G/\mu)$:

$$\text{Units of}\left(\frac{C_p\,\mu}{k}\right) = \frac{\left(\text{Btu lb}^{-1}\,^\circ\text{F}^{-1}\right)\text{lb h}^{-1}\,\text{ft}^{-1}}{\text{Btu h}^{-1}\,\text{ft}^{-2}\left(^\circ\text{F ft}^{-1}\right)^{-1}} = 1$$

$$\text{Units of}\left(\frac{D\,G}{\mu}\right) = \frac{(\text{ft})\,\text{lb h}^{-1}\,\text{ft}^{-2}}{\text{lb h}^{-1}\,\text{ft}^{-1}} = 1$$

Therefore, these groups are dimensionless. For the equation to be dimensionally homogeneous, $(h/C_p G)$ must also be dimensionless; the units of h must therefore cancel the units of $C_p G$.

$$\text{Units of } h = \text{units of } C_p G = (\text{Btu lb}^{-1}\,^\circ\text{F}^{-1})(\text{lb h}^{-1}\,\text{ft}^{-2}) = \text{Btu}\,^\circ\text{F}^{-1}\,\text{h}^{-1}\,\text{ft}^{-2}$$

The dimensions of h can be deduced from its units. From Table A.7 (Appendix A), Btu is a unit of energy with dimensions = L^2MT^{-2}. °F is a unit of temperature which, from Table 2.1, has the dimensional symbol Θ. h is a unit of time with dimension = T; ft is a unit of length with dimension = L. Therefore:

$$\text{Dimensions of } h = L^2MT^{-2}\,\Theta^{-1}\,T^{-1}\,L^{-2} = MT^{-3}\Theta^{-1}$$

Answer: Units = Btu $^\circ F^{-1}\,h^{-1}\,ft^{-2}$; dimensions = $MT^{-3}\Theta^{-1}$

2.6 Dimensional homogeneity and g_c

From Table A.8 (Appendix A), dimensions of $P = L^2MT^{-3}$
Dimensions of $g = LT^{-2}$
Dimensions of $\rho = ML^{-3}$
Dimensions of $D_i = L$
From p 11, the dimensions of rotational speed, $N_i = T^{-1}$; from p 15, the dimensions of $g_c = 1$. Therefore:

$$\text{Dimensions of } \frac{P\,g}{\rho\,N_i^3\,D_i^5} = \frac{(L^2MT^{-3})(LT^{-2})}{(ML^{-3})(T^{-1})^3\,L^5} = LT^{-2}$$

As N_P is a dimensionless number, equation (i) is not dimensionally homogeneous and therefore cannot be correct.

$$\text{Dimensions of } \frac{P\,g_c}{\rho\,N_i^3\,D_i^5} = \frac{(L^2MT^{-3})(1)}{(ML^{-3})(T^{-1})^3\,L^5} = 1$$

Equation (ii) is dimensionally homogeneous and therefore likely to be correct.

Answer: (ii)

2.7 Molar units

From the atomic weights in Table B.1 (Appendix B), the molecular weight of NaOH is 40.0.

(a)
From Eq. (2.19):

$$\text{lb moles NaOH} = \frac{20.0\text{ lb}}{40.0\text{ lb lbmol}^{-1}} = 0.50\text{ lbmol}$$

Answer: 0.50 lbmol

(b)
From Table A.3 (Appendix A): 1 lb = 453.6 g. Therefore:

$$20.0\text{ lb} = 20.0\text{ lb} \cdot \left|\frac{453.6\text{ g}}{1\text{ lb}}\right| = 9072\text{ g}$$

From Eq. (2.18):

$$\text{gram moles NaOH} = \frac{9072\text{ g}}{40.0\text{ g gmol}^{-1}} = 227\text{ gmol}$$

Answer: 227 gmol

(c)
From p 16, 1 kgmol = 1000 gmol. Therefore, from **(b)**:

$$\text{kg moles NaOH} = 227\ \text{gmol} \cdot \left| \frac{1\ \text{kgmol}}{1000\ \text{gmol}} \right| = 0.227\ \text{kgmol}$$

Answer: 0.227 kgmol

2.8 Density and specific gravity

(a)

From p 16, the density of water at 4°C can be taken as exactly 1 g cm^{-3}. Therefore, for a substance with specific gravity $1.5129^{20°C}_{4°C}$, the density at 20°C is 1.5129 g cm^{-3}.

(i)

1 kg = 1000 g

1 m = 100 cm

Therefore:

$$\text{Density} = 1.5129\ \text{g cm}^{-3} \cdot \left| \frac{1\ \text{kg}}{1000\ \text{g}} \right| \cdot \left| \frac{100\ \text{cm}}{1\ \text{m}} \right|^3 = 1512.9\ \text{kg m}^{-3}$$

Answer: 1512.9 kg m^{-3}

(ii)

From the atomic weights in Table B.1 (Appendix B), the molecular weight of nitric acid (HNO_3) is 63.0. In 1 cm^3 HNO_3, from Eq. (2.18):

$$\text{gram moles} = \frac{1.5129\ \text{g}}{63.0\ \text{g gmol}^{-1}} = 0.0240\ \text{gmol}$$

Therefore, the molar density is 0.0240 gmol cm^{-3}. From the definition of specific volume on p 16:

$$\text{Molar specific volume} = \frac{1}{\text{molar density}} = \frac{1}{0.0240\ \text{gmol cm}^{-3}} = 41.67\ \text{cm}^3\ \text{gmol}^{-1}$$

Answer: 41.67 cm^3 $gmol^{-1}$

(b)

(i)

From p 16, as density is defined as the mass per unit volume, the mass flow rate is equal to the volumetric flow rate multiplied by the density:

$$\text{Mass flow rate} = \left(50\ \text{cm}^3\ \text{min}^{-1}\right)\left(1.6\ \text{g cm}^{-3}\right) = 80\ \text{g min}^{-1}$$

Answer: 80 g min^{-1}

(ii)

From the atomic weights in Table B.1 (Appendix B), the molecular weight of carbon tetrachloride, CCl_4, is 153.8. Using the mass flow rate from (a):

$$\text{Molar flow rate} = 80\ \text{g min}^{-1} \cdot \left| \frac{1\ \text{gmol}}{153.8\ \text{g}} \right| = 0.52\ \text{gmol min}^{-1}$$

Answer: 0.52 gmol min^{-1}

2.9 Molecular weight

From p 17, the composition of air is close to 21% oxygen and 79% nitrogen. For gases at low pressures, this means 21 mol% O_2 and 79 mol% N_2. Therefore, in 1 gmol air, there are 0.21 gmol O_2 and 0.79 gmol N_2 From the atomic weights in Table B.1 (Appendix B), the molecular weights of O_2 and N_2 are 32.0 and 28.0, respectively. The molecular weight of air is equal to the number of grams in 1 gmol:

$$1\ \text{gmol air} = 0.21\ \text{gmol O}_2 \cdot \left| \frac{32.0\ \text{g}}{1\ \text{gmol}} \right| + 0.79\ \text{gmol N}_2 \cdot \left| \frac{28.0\ \text{g}}{1\ \text{gmol}} \right| = 28.8\ \text{g}$$

Answer: 28.8

2.10 Mole fraction

The molecular weights can be obtained from Table B.7 (Appendix B): water 18.0; ethanol 46.1; methanol 32.0; glycerol 92.1; acetic acid 60.1; benzaldehyde 106.1. In 100 g solution, there are 30 g water, 25 g ethanol, 15 g methanol, 12 g glycerol, 10 g acetic acid, 8 g benzaldehyde, and no other components. Therefore:

$$\text{Moles water} = 30\ \text{g} \cdot \left|\frac{1\ \text{gmol}}{18.0\ \text{g}}\right| = 1.67\ \text{gmol}$$

$$\text{Moles ethanol} = 25\ \text{g} \cdot \left|\frac{1\ \text{gmol}}{46.1\ \text{g}}\right| = 0.54\ \text{gmol}$$

$$\text{Moles methanol} = 15\ \text{g} \cdot \left|\frac{1\ \text{gmol}}{32.0\ \text{g}}\right| = 0.47\ \text{gmol}$$

$$\text{Moles glycerol} = 12\ \text{g} \cdot \left|\frac{1\ \text{gmol}}{92.1\ \text{g}}\right| = 0.13\ \text{gmol}$$

$$\text{Moles acetic acid} = 10\ \text{g} \cdot \left|\frac{1\ \text{gmol}}{60.1\ \text{g}}\right| = 0.17\ \text{gmol}$$

$$\text{Moles benzaldehyde} = 8\ \text{g} \cdot \left|\frac{1\ \text{gmol}}{106.1\ \text{g}}\right| = 0.08\ \text{gmol}$$

The total number of moles is 1.67 + 0.54 + 0.47 + 0.13 + 0.17 + 0.08 = 3.06 gmol. From Eq. (2.20):

$$\text{Mole fraction water} = \frac{1.67}{3.06} = 0.55$$

$$\text{Mole fraction ethanol} = \frac{0.54}{3.06} = 0.18$$

$$\text{Mole fraction methanol} = \frac{0.47}{3.06} = 0.15$$

$$\text{Mole fraction glycerol} = \frac{0.13}{3.06} = 0.04$$

$$\text{Mole fraction acetic acid} = \frac{0.17}{3.06} = 0.06$$

$$\text{Mole fraction benzaldehyde} = \frac{0.08}{3.06} = 0.03$$

Answer: 0.55 water; 0.18 ethanol; 0.15 methanol; 0.04 glycerol; 0.06 acetic acid; 0.03 benzaldehyde

2.11 Temperature scales

From Eq. (2.27):

$$-40 = 1.8\,T(°\text{C}) + 32$$
$$T(°\text{C}) = -40$$

From Eq. (2.25):

$$T(°\text{R}) = -40 + 459.67$$
$$T(°\text{R}) = 420$$

From Eq. (2.24) and the result for T (°C):

$$T(\text{K}) = -40 + 273.15$$
$$T(\text{K}) = 233$$

Answer: 40°C; 420°R; 233 K

2.12 Pressure scales

(a)

Assume that the atmospheric pressure is 14.7 psi. From Eq. (2.28):

$$\text{Absolute pressure} = 15\ \text{psi} + 14.7\ \text{psi} = 29.7\ \text{psi}$$

From Table A.5 (Appendix A): 1 psi = 6.805×10^{-2} atm. Therefore:

$$\text{Absolute pressure} = 29.7 \text{ psi} \cdot \left| \frac{6.805 \times 10^{-2} \text{ atm}}{1 \text{ psi}} \right| = 2.02 \text{ atm}$$

Answer: 29.7 psi; 2.02 atm

(b)
From p 19, vacuum pressure is the pressure below atmospheric. If the atmospheric pressure is 14.7 psi:

$$\text{Absolute pressure} = 14.7 \text{ psi} - 3 \text{ psi} = 11.7 \text{ psi}$$

Answer: 11.7 psi

2.13 Stoichiometry and incomplete reaction

(a)
The molecular weights are calculated from Table B.1 (Appendix B): penicillin = 334.4; glucose = 180.2. The maximum theoretical yield from the stoichiometric equation is 1 gmol of penicillin for every 1.67 gmol of glucose. This is equivalent to 334.4 g penicillin per $1.67 \times 180.2 = 300.9$ g glucose, or 1.1 g g^{-1}.

Answer: 1.1 g g^{-1}

(b)
The maximum theoretical yield in **(a)** is obtained when all the glucose consumed is directed into penicillin production according to the stoichiometric equation. If only 6% of the glucose is used in this way, the actual yield of penicillin from glucose is much lower, at 334.4 g penicillin per ($300.9 \times {}^{100}/_{6}$) g glucose, or 0.067 g g^{-1}.

Answer: 0.067 g g^{-1}

(c)
From the atomic weights in Table B.1 (Appendix B), the molecular weight of phenylacetic acid is 136.2.
(i)
The only possible limiting substrates are glucose and phenylacetic acid. Using a basis of 1 l medium, if (50 – 5.5) = 44.5 g glucose are consumed but only 6% is available for penicillin synthesis, the mass of glucose used in the penicillin reaction is $44.5 \times {}^{6}/_{100} = 2.67$ g. This is equivalent to ${}^{2.67 \text{ g}}/_{180.2 \text{ g gmol}^{-1}} = 1.48 \times 10^{-2}$ gmol glucose available for penicillin synthesis. At the same time, 4 g or ${}^{4 \text{ g}}/_{136.2 \text{ g gmol}^{-1}} = 2.94 \times 10^{-2}$ gmol phenylacetic acid is available which, according to the stoichiometric equation, requires $1.67 \times 2.94 \times 10^{-2} = 4.91 \times 10^{-2}$ gmol glucose for complete reaction. As the gmol glucose required is greater than the gmol glucose available after growth and maintenance activities, glucose is the limiting substrate.

Answer: Glucose

(ii)
Of the 44.5 g l^{-1} glucose consumed, 24% or 10.7 g l^{-1} is used for growth. In a 100-litre tank, the total mass of glucose consumed for growth is therefore 1070 g or 1.07 kg.

Answer: 1.07 kg

(iii)
From **(i)**, 1.48×10^{-2} gmol glucose is used in the penicillin reaction per litre. According to the stoichiometry, this produces ${}^{1.48 \times 10^{-2}}/_{1.67} = 8.86 \times 10^{-3}$ gmol penicillin per litre. Therefore, in a 100-litre tank, 0.886 gmol or 0.886 gmol × 334.4 g $gmol^{-1}$ = 296 g penicillin are formed.

Answer: 296 g

(iv)
If, from **(i)**, 1.48×10^{-2} gmol l^{-1} glucose is used in the penicillin reaction, ${}^{1.48 \times 10^{-2}}/_{1.67} = 8.86 \times 10^{-3}$ gmol l^{-1} phenylacetic acid must also be used. This is equivalent to 8.86×10^{-3} gmol $l^{-1} \times 136.2$ g $gmol^{-1} = 1.21$ g l^{-1} phenylacetic acid. As 4 g l^{-1} are provided, (4 – 1.21) = 2.79 g l^{-1} phenylacetic acid must remain.

Answer: 2.79 g l^{-1}

2.14 Stoichiometry, yield and the ideal gas law

(a)
Adding up the numbers of C, H, O and N atoms on both sides of the equation shows that the equation is balanced.

Answer: Yes

(b)
The molecular weights are calculated from Table B.1 (Appendix B).
Cells: 91.5
Hexadecane: 226.4
From the stoichiometry, as 1 gmol of hexadecane is required to produce 1.65 gmol of cells, the maximum yield is 1.65 gmol × 91.5 g gmol^{-1} = 151 g cells per 226.4 g hexadecane, or 0.67 g g^{-1}.

Answer: 0.67 g g^{-1}

(c)
From the atomic weights in Table B.1 (Appendix B), the molecular weight of oxygen is 32.0. From the stoichiometry, 16.28 gmol of oxygen is required to produce 1.65 gmol of cells which, from **(b)**, is equal to 151 g cells. The maximum yield is therefore 151 g cells per (16.28 gmol × 32.0 g gmol^{-1}) = 521 g oxygen, or 0.29 g g^{-1}.

Answer: 0.29 g g^{-1}

(d)
Production of 2.5 kg cells is equivalent to 2500 g = $^{2500\ g}/_{91.5\ g\ gmol\text{-}1}$ = 27.3 gmol cells. The minimum amounts of substrates are required when 100% of the hexadecane is converted according to the stoichiometric equation.
(i)
From the stoichiometry, production of 27.3 gmol cells requires $^{27.3}/_{1.65}$ = 16.5 gmol = 16.5 gmol × 226.4 g gmol^{-1} = 3736 g = 3.74 kg hexadecane.

Answer: 3.74 kg

(ii)
From the answer in **(d)(i)**, the concentration of hexadecane required is 3.74 kg in 3 m^3, or 1.25 kg m^{-3}.

Answer: 1.25 kg m^{-3}

(iii)
According to the stoichiometric equation, production of 27.3 gmol cells requires 27.3 × 16.28/1.65 = 269.4 gmol oxygen. As air at low pressure contains close to 21 mol% oxygen (p 17), the total moles of air required is $^{269.4}/_{0.21}$ = 1282.9 gmol. The volume of air required can be calculated using the ideal gas law. From Eq. (2.32):

$$V = \frac{n R T}{p}$$

Temperature in the ideal gas equation is absolute temperature; from Eq. (2.24):

$$T = (20 + 273.15)\ \text{K} = 293.15\ \text{K}$$

From Table 2.5, R = 82.057 cm^3 atm K^{-1} gmol^{-1}. Substituting these values into the equation for V gives:

$$V = \frac{(1282.9\ \text{gmol})\left(82.057\ \text{cm}^3\ \text{atm K}^{-1}\ \text{gmol}^{-1}\right)(293.15\ \text{K})}{1\ \text{atm}} \cdot \left|\frac{1\ \text{m}}{100\ \text{cm}}\right|^3 = 31\ \text{m}^3$$

Answer: 31 m^3

3

Presentation and Analysis of Data

3.1 Combination of errors

$C^*_{AL} = 0.25 \text{ mol m}^{-3} \pm 4\% = 0.25 \pm 0.010 \text{ mol m}^{-3}$
$C_{AL} = 0.183 \text{ mol m}^{-3} \pm 4\% = 0.183 \pm 0.0073 \text{ mol m}^{-3}$
$OTR = 0.011 \text{ mol m}^{-3} \text{ s}^{-1} \pm 5\%$
For subtraction, absolute errors are added. Therefore:

$$C^*_{AL} - C_{AL} = (0.25 - 0.183) \pm (0.010 + 0.0073) \text{ mol m}^{-3} = 0.067 \pm 0.0173 \text{ mol m}^{-3} = 0.067 \text{ mol m}^{-3} \pm 25.8\%$$

For division, relative errors are added. Therefore:

$$k_L a = \frac{0.011 \text{ mol m}^{-3} \text{ s}^{-1}}{0.067 \text{ mol m}^{-3}} \pm (25.8 + 5)\% = 0.16 \text{ s}^{-1} \pm 31\% = 0.16 \pm 0.05 \text{ s}^{-1}$$

Answer: 31%. *This example illustrates how a combination of small measurement errors can result in a relatively large uncertainty in the final result.*

3.2 Mean and standard deviation

(a)
The best estimate is the mean, $\bar{x}$. From Eq. (3.1):

$$\bar{x} = \frac{5.15 + 5.45 + 5.50 + 5.35}{4} = 5.36$$

Answer: 5.36

(b)
Calculate the standard deviation from Eq. (3.2):

$$\sigma = \sqrt{\frac{(5.15 - 5.36)^2 + (5.45 - 5.36)^2 + (5.50 - 5.36)^2 + (5.35 - 5.36)^2}{4 - 1}} = 0.15$$

Answer: The standard deviation is 0.15. *Note that standard error, which can be calculated from the standard deviation, is a more direct indication of the precision of a mean.*

(c)

$$\bar{x} = \frac{5.15 + 5.45}{2} = 5.30$$

Standard deviation is not appropriate for expressing the accuracy of a mean evaluated using only two samples. In this case the maximum error, i.e. the difference between the mean and either of the two measured values, might be used instead. The maximum error in this example is $(5.30 - 5.15) = 0.15$.

Answer: 5.30; an indication of the accuracy is ± 0.15

(d)

$$\bar{x} = \frac{5.15 + 5.45 + 5.50 + 5.35 + 5.15 + 5.45 + 5.50 + 5.35}{8} = 5.36$$

$$\sigma = \sqrt{\frac{2(5.15-5.36)^2+2(5.45-5.36)^2+2(5.50-5.36)^2+2(5.35-5.36)^2}{8-1}} = 0.14$$

Answer: The best estimate of optimal pH is unchanged at 5.36, but the standard deviation is slightly lower at 0.14. *This example illustrates that although the standard deviation decreases as the number of measurements is increased, σ is not strongly dependent on n. The best way to improve the reliability of the mean is to ensure that the individual measurements are as intrinsically accurate as possible, rather than repeat the measurement many times.*

3.3 Linear and non-linear models

(a)

$x_1 = 1; y_1 = 10$

$x_2 = 8; y_2 = 0.5$

A straight line plot of y versus x on linear coordinates means that the data can be represented using Eq. (3.6). From Eqs (3.7) and (3.8):

$$A = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{0.5-10}{8-1} = -1.36$$

$$B = y_1 - A\,x_1 = 10 - (-1.36)\,1 = 11.4$$

Answer: $y = -1.36\,x + 11.4$

(b)

$x_1 = 3.2; y_1 = 14.5$

$x_2 = 8.9; y_2 = 38.5$

A straight line plot of y versus $x^{1/2}$ on linear coordinates means that the data can be represented using the equation:

$$y = A\,x^{1/2} + B$$

with A and B given by the equations:

$$A = \frac{y_2 - y_1}{x_2^{1/2} - x_1^{1/2}} = \frac{38.5-14.5}{8.9^{1/2} - 3.2^{1/2}} = 20.1$$

$$B = y_1 - A\,x_1^{1/2} = 14.5 - 20.1\,(3.2^{1/2}) = -21.5$$

Answer: $y = 20.1\,x^{1/2} - 21.5$

(c)

$x_1 = 5; y_1 = 6$

$x_2 = 1; y_2 = 3$

A straight line plot of $1/y$ versus x^2 on linear coordinates means that the data can be represented using the equation:

$$1/y = A\,x^2 + B$$

with A and B given by the equations:

$$A = \frac{1/y_2 - 1/y_1}{x_2^2 - x_1^2} = \frac{1/3 - 1/6}{1^2 - 5^2} = -6.9 \times 10^{-3}$$

$$B = 1/y_1 - A\,x_1^2 = 1/6 - \left(-6.9 \times 10^{-3}\right)(5^2) = 0.34$$

Answer: $1/y = -6.9 \times 10^{-3}\,x^2 + 0.34$

(d)

$x_1 = 0.5; y_1 = 25$

$x_2 = 550; y_2 = 2600$

A straight line plot of y versus x on log–log coordinates means that the data can be represented using Eq. (3.10). From Eqs (3.13) and (3.14):

$$A = \frac{(\ln y_2 - \ln y_1)}{(\ln x_2 - \ln x_1)} = \frac{\ln 2600 - \ln 25}{\ln 550 - \ln 0.5} = 0.663$$

$$\ln B = \ln y_1 - A \ln x_1 = \ln 25 - (0.663) \ln 0.5 = 3.678$$

$$B = e^{3.678} = 39.6$$

Answer: $y = 39.6\, x^{0.663}$

(e)
$x_1 = 1.5$; $y_1 = 2.5$
$x_2 = 10$; $y_2 = 0.036$
A straight line plot of y versus x on semi-log coordinates means that the data can be represented using Eq. (3.15). From Eqs (3.17) and (3.18):

$$A = \frac{(\ln y_2 - \ln y_1)}{(x_2 - x_1)} = \frac{\ln 0.036 - \ln 2.5}{10 - 1.5} = -0.50$$

$$\ln B = \ln y_1 - A\, x_1 = \ln 2.5 - (-0.50)\, 1.5 = 1.666$$

$$B = e^{1.666} = 5.29$$

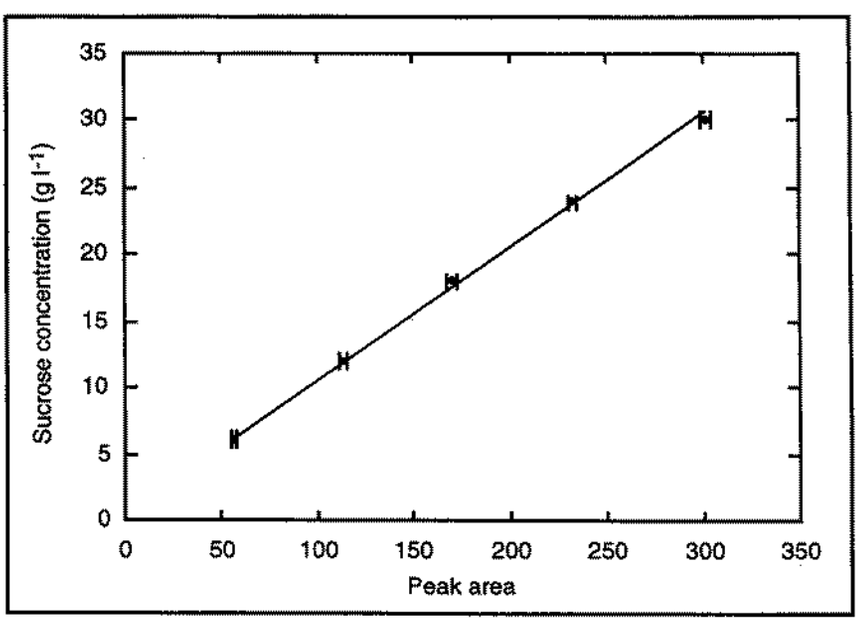
Answer: $y = 5.29\, e^{-0.50\, x}$

3.4 Linear curve fitting

(a)
The results determined using Eqs (3.1) and (3.2) are listed below.

Sucrose concentration (g l^{-1})	Mean peak area	Standard deviation
6.0	56.84	1.21
12.0	112.82	2.06
18.0	170.63	2.54
24.0	232.74	1.80
30.0	302.04	2.21

(b)



(c)
The linear least squares fit of the data is:

$$y = 0.098\,x + 0.83$$

Answer: $y = 0.098\,x + 0.83$, where y is sucrose concentration in g l^{-1} and x is peak area.

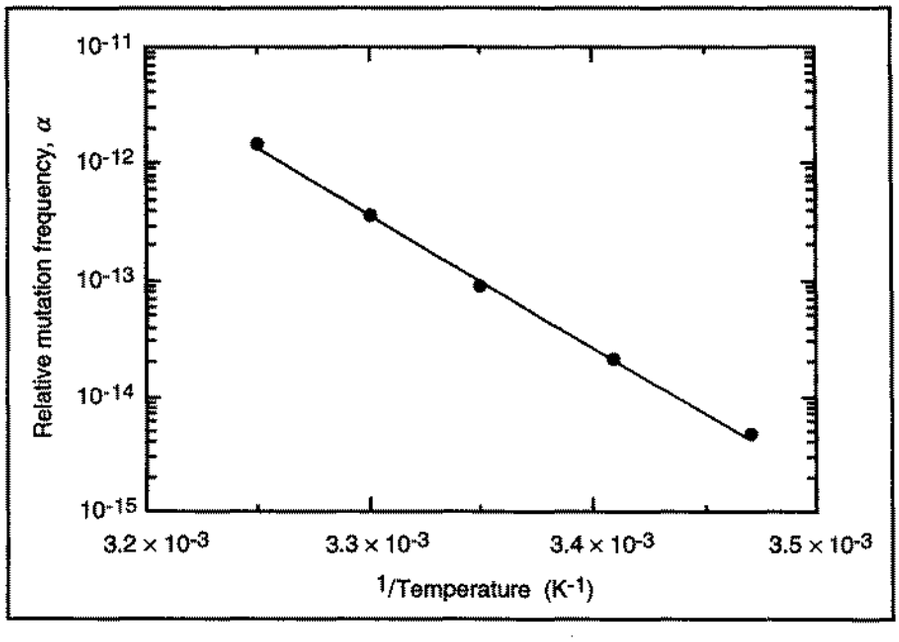
(d)
For $x = 209.86$, the equation in (c) gives a sucrose concentration of 21.4 g l^{-1}.

Answer: 21.4 g l^{-1}

3.5 Non-linear model: calculation of parameters

(a)
The proposed model equation has the general form of Eq. (3.15); therefore, if the model is suitable, a plot of α versus $1/T$ on semi-logarithmic coordinates will give a straight line. As T in the equation is absolute temperature, °C must first be converted to degrees Kelvin using Eq. (2.24). The data are listed and plotted below.

Temperature (°C)	Temperature (K)	$1/T$ (K^{-1})	Relative mutation frequency, α
15	288.15	3.47×10^{-3}	4.4×10^{-15}
20	293.15	3.41×10^{-3}	2.0×10^{-14}
25	298.15	3.35×10^{-3}	8.6×10^{-14}
30	303.15	3.30×10^{-3}	3.5×10^{-13}
35	308.15	3.25×10^{-3}	1.4×10^{-12}



As the data give a straight line on semi-logarithmic coordinates, the model can be considered to fit the data well.

(b)
The equation for the straight line in (a) is:

$$y = 9.66 \times 10^{24}\, e^{-26{,}121x}$$

where y is relative mutation frequency and x is reciprocal temperature in units of K^{-1}. For dimensional homogeneity the exponent must be dimensionless (p 12), so that –26,121 has units of K, and E/R in the model equation is equal to 26,121 K. From Table 2.5, $R = 8.3144$ J $gmol^{-1}$ K^{-1}; therefore:

$$E = (26{,}121\ \text{K})\,(8.3144\ \text{J gmol}^{-1}\ \text{K}^{-1}) = 217{,}180.4\ \text{J gmol}^{-1} = 217.2\ \text{kJ gmol}^{-1}$$

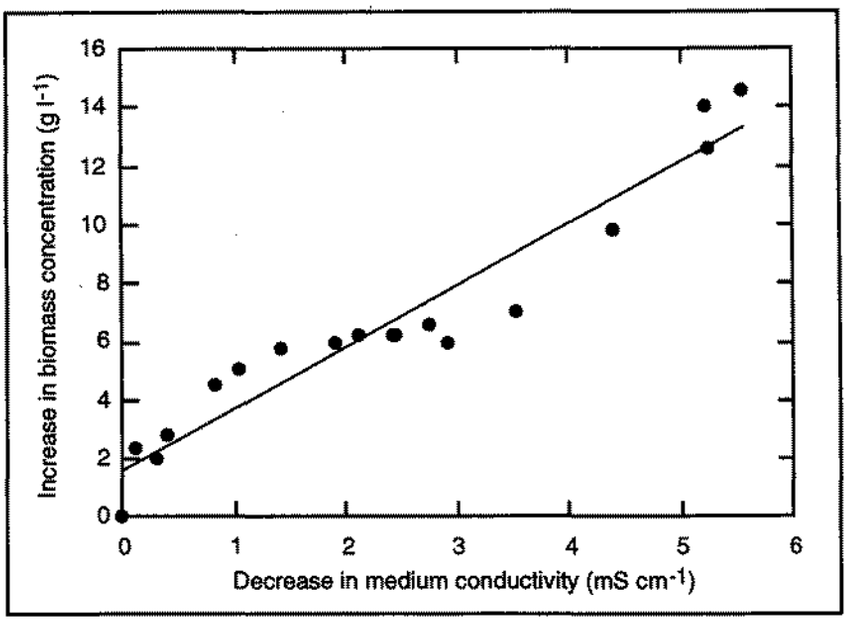
Answer: 217.2 kJ $gmol^{-1}$

(c)
From the equation in **(b)** for the straight line, α_0 is equal to 9.66×10^{24}.

Answer: 9.66×10^{24}

3.6 Linear regression: distribution of residuals

(a)



The linear least squares fit of the data is:

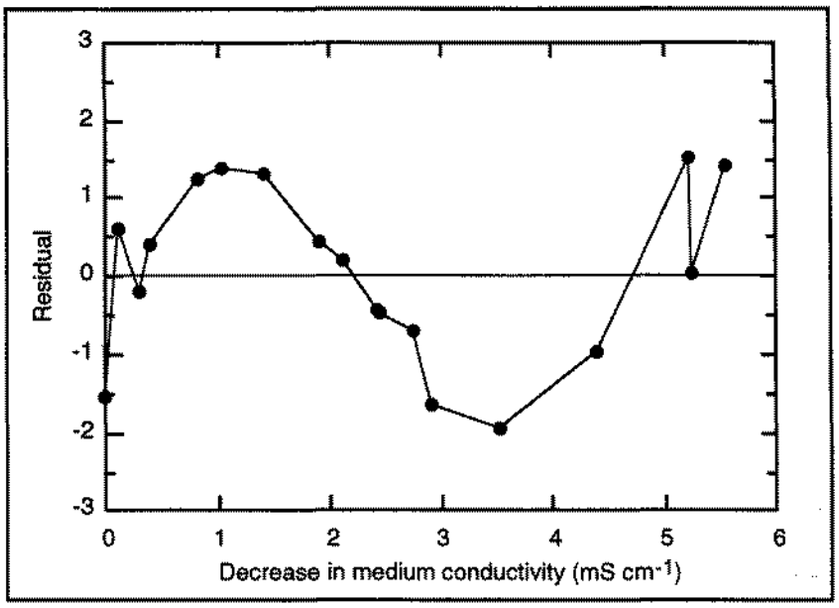
$$y = 1.58 + 2.10\,x$$

where y is increase in biomass concentration in g l^{-1} and x is decrease in medium conductivity in mS cm^{-1}.

(b)
The residuals are calculated as the difference between the measured values for increase in biomass concentration and the values for y obtained from the equation in **(a)**.

Decrease in medium conductivity (mS cm^{-1})	Residual
0	–1.58
0.12	0.57
0.31	–0.23
0.41	0.36
0.82	1.20
1.03	1.36
1.40	1.28
1.91	0.41
2.11	0.19
2.42	–0.46
2.44	–0.50
2.74	–0.73
2.91	–1.69
3.53	–1.99
4.39	–1.00
5.21	1.48
5.24	0.02
5.55	1.37

These results are plotted below.

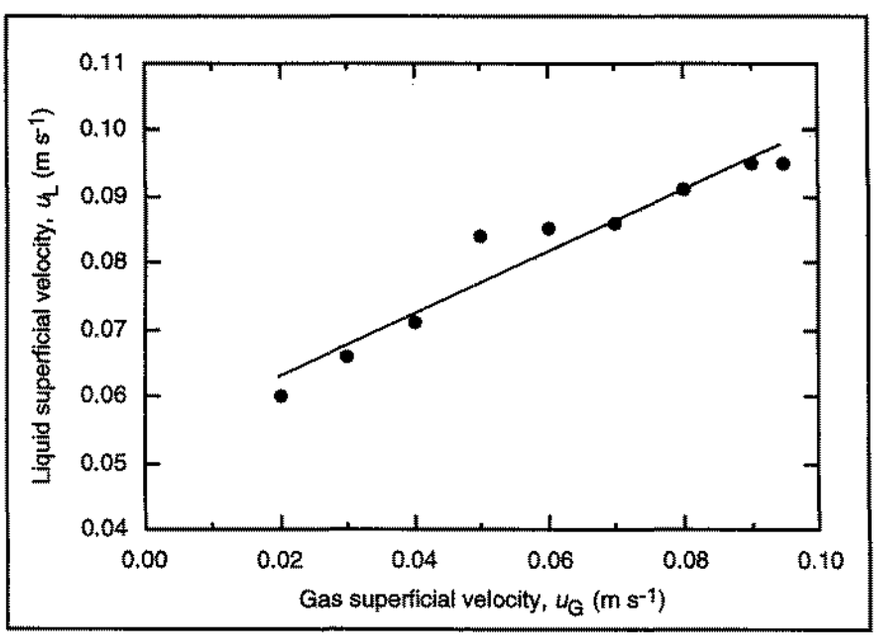


The residuals are not randomly distributed: they are mainly positive at low values of decrease in medium conductivity, then negative, then positive again. Therefore, the straight line fit of the data cannot be considered a very good one.

3.7 Discriminating between rival models

(a)

The results are plotted using linear coordinates below.

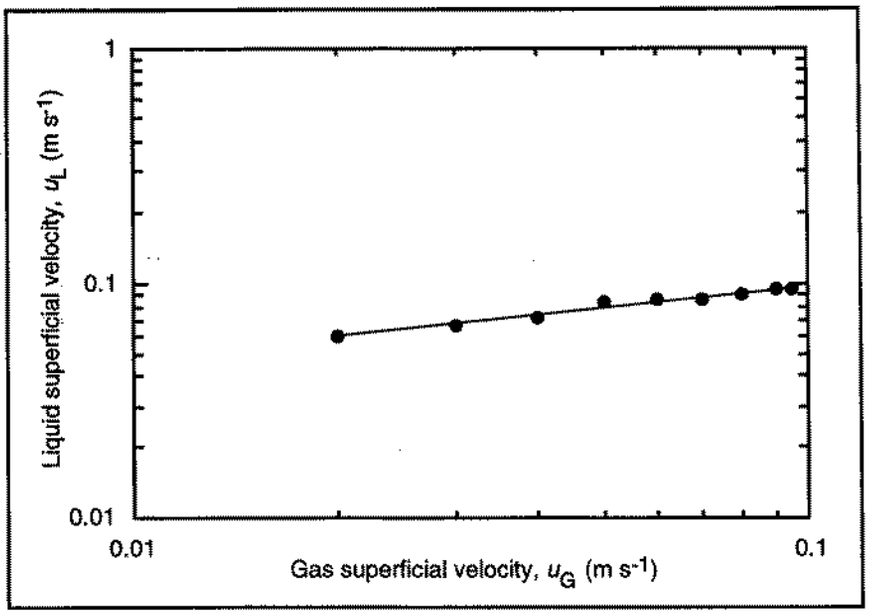


The data are reasonably well fitted using a linear model. The linear least squares equation for the straight line fit is:

$$y = 0.054 + 0.466\,x$$

where y is liquid superficial velocity in m s^{-1} and x is gas superficial velocity in m s^{-1}. The sum of the squares of the residuals between the measured values for liquid superficial velocity and the values for y obtained from the above equation is 8.4×10^{-5}.

(b)
The proposed power law equation has the same form as Eq. (3.10). Therefore, if the power law model is suitable, the data should give a straight line when plotted on log–log coordinates.



The data are reasonably well fitted using a power law model. The equation for the straight line in the plot is:

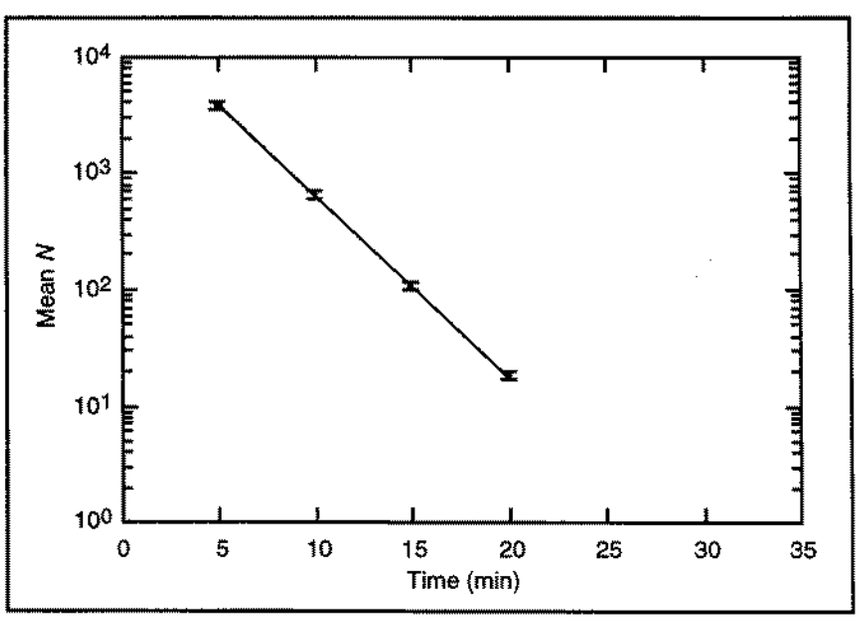
$$y = 0.199\,x^{0.309}$$

where y is liquid superficial velocity in m s^{-1} and x is gas superficial velocity in m s^{-1}. The sum of the squares of the residuals between the measured values for liquid superficial velocity and the values obtained from the above equation is 4.2×10^{-5}.

(c)
The non-linear model is better because the sum of squares of the residuals is smaller.

3.8 Non-linear model: calculation of parameters

(a), (b)
The proposed model equation has the same form as Eq. (3.15). Therefore, if the model is suitable, the data should give a straight line when plotted on semi-logarithmic coordinates.



(c), (d)
The equation for the straight line in the figure is:

$$y = 2.13 \times 10^4 \, e^{-0.353x}$$

where y is the number of viable cells and x is time in min. As the exponent must be dimensionless to preserve dimensional homogeneity (p 12), the dimensions of k_d are T^{-1}; therefore $k_d = 0.353$ min^{-1}. The dimensions of N_0 are the same as N, i.e. N_0 is dimensionless and equal to 2.13×10^4.

Answer: $k_d = 0.353$ min^{-1}, $N_0 = 2.13 \times 10^4$; the dimensions of k_d are T^{-1}, N_0 is dimensionless

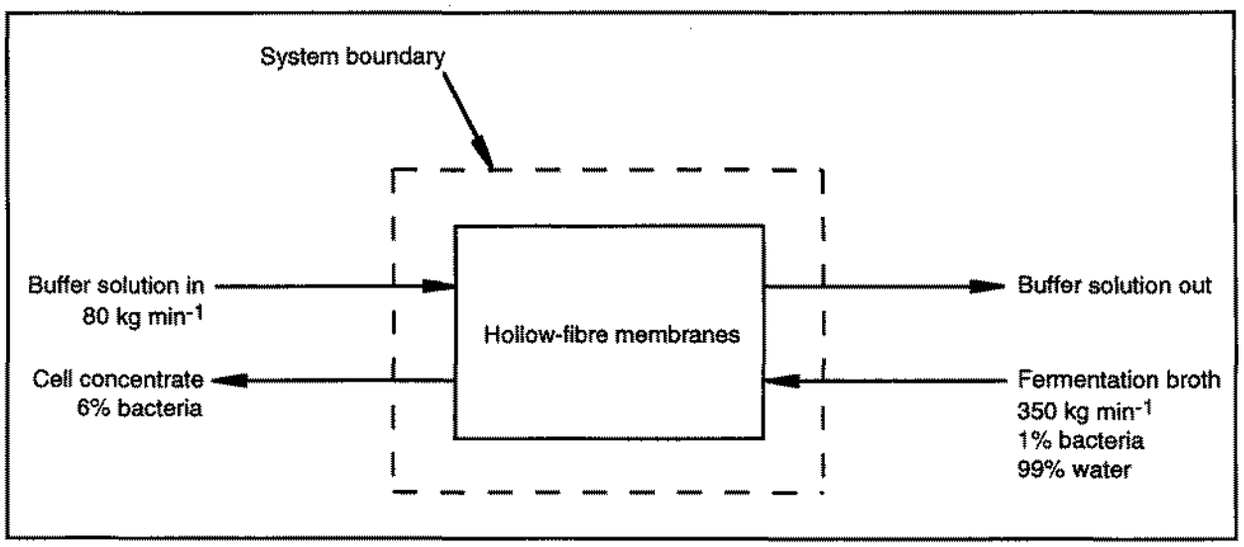
4

Material Balances

4.1 Cell concentration using membranes

1. *Assemble*

(i) *Flow sheet*



(ii) *System boundary*

The system boundary is shown on the flow sheet.

(iii) *Reaction equation*

No reaction occurs.

2. *Analyse*

(i) *Assumptions*

– steady state
– no leaks
– only water passes across the membrane

(ii) *Extra data*

No extra data are required.

(iii) *Basis*

1 min, or 350 kg fermentation broth

(iv) *Compounds involved in reaction*

No compounds are involved in reaction.

(v) *Mass-balance equation*

As there is no reaction, the appropriate mass-balance equation is Eq. (4.3):

$$\text{mass in} = \text{mass out}$$

3. *Calculate*

(i) *Calculation table*

The calculation table below shows all given quantities in kg. The total mass of cell concentrate is denoted C; the total mass of buffer out is denoted B. The columns for water refer to water originating in the fermentation broth.

Stream	*In*				*Out*			
	Water	*Bacteria*	*Buffer*	*Total*	*Water*	*Bacteria*	*Buffer*	*Total*
Fermentation broth	346.5	3.5	0	350	–	–	–	–
Buffer solution in	0	0	80	80	–	–	–	–
Cell concentrate	–	–	–	–	?	0.06 C	0	C
Buffer solution out	–	–	–	–	?	0	80	B
Total	346.5	3.5	80	430	?	0.06 C	80	$C + B$

(ii) *Mass-balance calculations*

Bacteria balance

$$3.5 \text{ kg bacteria in} = 0.06\ C \text{ kg bacteria out}$$

$$C = 58.3 \text{ kg}$$

Total mass balance

$$430 \text{ kg total mass in} = (C + B) \text{ kg total mass out}$$

Using the result for C:

$$B = 371.7 \text{ kg}$$

Water balance

$$346.5 \text{ kg water in} = \text{water out}$$

$$\text{Water out} = 346.5 \text{ kg}$$

These calculations allow completion of the mass-balance table with all quantities in kg.

Stream	*In*				*Out*			
	Water	*Bacteria*	*Buffer*	*Total*	*Water*	*Bacteria*	*Buffer*	*Total*
Fermentation broth	346.5	3.5	0	350	–	–	–	–
Buffer solution in	0	0	80	80	–	–	–	–
Cell concentrate	–	–	–	–	54.8	3.50	0	58.3
Buffer solution out	–	–	–	–	291.7	0	80	371.7
Total	346.5	3.5	80	430	346.5	3.50	80	430

(iii) *Check the results*

All columns and rows of the completed table add up correctly.

4. *Finalise*

(a)

After rounding to three significant figures, the total flow rate of buffer solution out of the annulus is 372 kg min^{-1}.

Answer: 372 kg min^{-1}

(b)

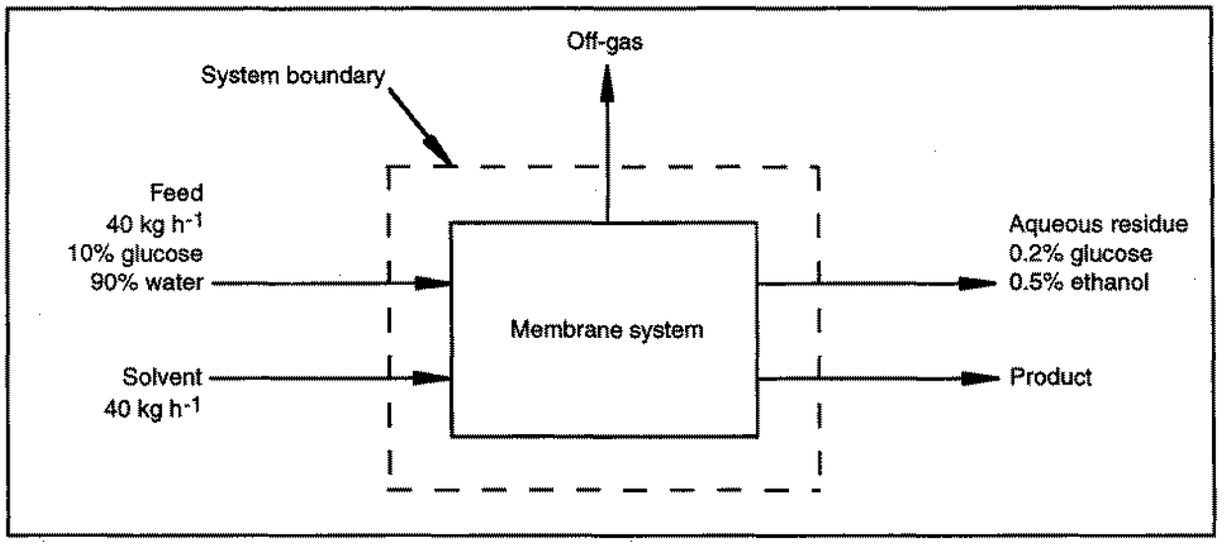
The total flow rate of cell concentrate from the membrane tubes is 58.3 kg min^{-1}.

Answer: 58.3 kg min^{-1}

4.2 Membrane reactor

1. *Assemble*

(i) *Flow sheet*



(ii) *System boundary*

The system boundary is shown on the flow sheet.

(iii) *Reaction equation*

$$C_6H_{12}O_6 \rightarrow 2\,C_2H_6O + 2\,CO_2$$

2. *Analyse*

(i) *Assumptions*

– steady state
– no leaks
– yeast cells do not grow or dislodge from the membrane
– no evaporation
– all CO_2 produced leaves in the off-gas
– no side reactions

(ii) *Extra data*

Molecular weights (Table B.1, Appendix B): glucose = 180.2
ethanol = 46.1
CO_2 = 44.0

(iii) *Basis*

1 h, or 40 kg feed solution

(iv) *Compounds involved in reaction*

Glucose, ethanol and carbon dioxide are involved in the reaction.

(v) *Mass-balance equations*

For glucose, ethanol and carbon dioxide, the appropriate mass-balance equation is Eq. (4.2):

$$\text{mass in} + \text{mass generated} = \text{mass out} + \text{mass consumed}$$

For water, solvent and total mass, the appropriate mass-balance equation is Eq. (4.3):

$$\text{mass in} = \text{mass out}$$

3. *Calculate*

(i) *Calculation table*

The calculation table below shows all given quantities in kg. The total mass of aqueous residue is denoted R; the total mass of product out is denoted P; the total mass of carbon dioxide out is denoted G.

Stream	*In*						*Out*					
	Glucose	*Ethanol*	CO_2	*Solvent*	H_2O	*Total*	*Glucose*	*Ethanol*	CO_2	*Solvent*	H_2O	*Total*
Feed	4	0	0	0	36	40	–	–	–	–	–	–
Solvent	0	0	0	40	0	40	–	–	–	–	–	–
Aqueous residue	–	–	–	–	–	–	0.002 R	0.005 R	0	0	?	R
Product	–	–	–	–	–	–	0	?	0	?	0	P
Off-gas	–	–	–	–	–	–	0	0	?	0	0	G
Total	4	0	0	40	36	80	0.002 R	?	?	?	?	$R + P + G$

(ii) *Mass-balance calculations*

Solvent balance

Solvent is a tie component.

$$40 \text{ kg solvent in} = \text{solvent out}$$

$$\text{Solvent out} = 40 \text{ kg}$$

Water balance

Water is a tie component.

$$36 \text{ kg water in} = \text{water out}$$

$$\text{Water out} = 36 \text{ kg}$$

As water appears on the Out side of the table only in the aqueous residue stream:

$$0.002\,R + 0.005\,R + 36 \text{ kg} = R$$

$$R = 36.254 \text{ kg}$$

Therefore, the residual glucose in the aqueous residue stream = 0.002 R = 0.073 kg; the ethanol in the aqueous residue stream = 0.005 R = 0.181 kg.

Glucose balance

$$4 \text{ kg glucose in} + 0 \text{ kg glucose generated} = 0.073 \text{ kg glucose out} + \text{glucose consumed}$$

$$\text{Glucose consumed} = 3.927 \text{ kg}$$

Converting the glucose consumed to molar terms:

$$3.927 \text{ kg glucose} = 3.927 \text{ kg} \cdot \left| \frac{1 \text{ kgmol}}{180.2 \text{ kg}} \right| = 0.0218 \text{ kgmol}$$

From the reaction stoichiometry, conversion of this amount of glucose generates 2 × 0.0218 = 0.0436 kgmol ethanol and 2 × 0.0218 = 0.0436 kgmol CO_2. Converting these molar quantities to mass:

$$0.0436 \text{ kgmol ethanol} = 0.0436 \text{ kgmol} \cdot \left| \frac{46.1 \text{ kg}}{1 \text{ kgmol}} \right| = 2.010 \text{ kg}$$

$$0.0436 \text{ kgmol } CO_2 = 0.0436 \text{ kgmol} \cdot \left| \frac{44.0 \text{ kg}}{1 \text{ kgmol}} \right| = 1.92 \text{ kg}$$

CO_2 balance

$$0 \text{ kg } CO_2 \text{ in} + 1.92 \text{ kg } CO_2 \text{ generated} = CO_2 \text{ out} + 0 \text{ kg } CO_2 \text{ consumed}$$

$$CO_2 \text{ out} = 1.92 \text{ kg} = G$$

Ethanol balance

$$0 \text{ kg ethanol in} + 2.010 \text{ kg ethanol generated} = \text{ethanol out} + 0 \text{ kg ethanol consumed}$$

$$\text{Ethanol out} = 2.010 \text{ kg}$$

Ethanol leaves the system only in the product and aqueous residue streams. Therefore:

Ethanol out in the product stream = (2.010 – 0.181) kg = 1.829 kg

As the product stream consists of ethanol and solvent only:

$$P = (1.829 + 40)\ \text{kg} = 41.829\ \text{kg}$$

These calculations allow completion of the mass-balance table with all quantities in kg.

Stream	*In*						*Out*					
	Glucose	*Ethanol*	*CO_2*	*Solvent*	*H_2O*	*Total*	*Glucose*	*Ethanol*	*CO_2*	*Solvent*	*H_2O*	*Total*
Feed	4	0	0	0	36	40	–	–	–	–	–	–
Solvent	0	0	0	40	0	40	–	–	–	–	–	–
Aqueous residue	–	–	–	–	–	–	0.073	0.181	0	0	36	36.254
Product	–	–	–	–	–	–	0	1.829	0	40	0	41.829
Off-gas	–	–	–	–	–	–	0	0	1.92	0	0	1.92
Total	4	0	0	40	36	80	0.073	2.010	1.92	40	36	80.00

(iii) *Check the results*
All columns and rows of the completed table add up correctly.

4. *Finalise*
(a)
1.829 kg ethanol are contained in 41.829 kg of product stream. The ethanol concentration is therefore $1.829/41.829 \times 100\% = 4.4\%$.

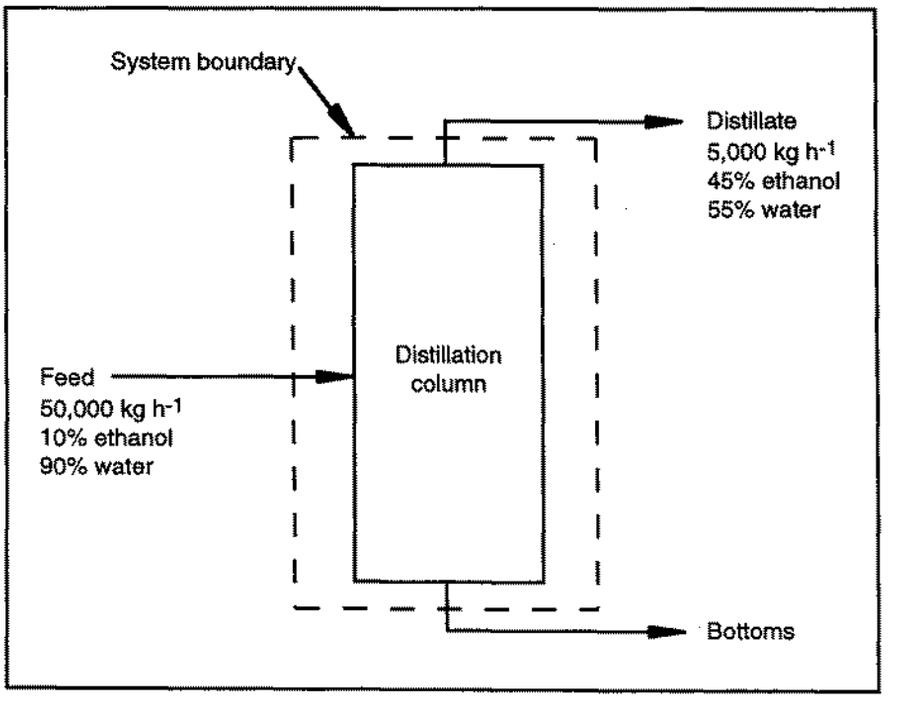
Answer: 4.4%

(b)
The mass flow rate of CO_2 is 1.92 kg h^{-1}.

Answer: 1.92 kg h^{-1}

4.3 Ethanol distillation

1. *Assemble*
(i) *Flow sheet*



(ii) *System boundary*
The system boundary is shown on the flow sheet.
(iii) *Reaction equation*
No reaction occurs.

2. *Analyse*
(i) *Assumptions*
– steady state
– no leaks
(ii) *Extra data*
No extra data are required.
(iii) *Basis*
1 h, or 50,000 kg feed
(iv) *Compounds involved in reaction*
No compounds are involved in reaction.
(v) *Mass-balance equation*
As there is no reaction, the appropriate mass-balance equation is Eq. (4.3):

$$\text{mass in} = \text{mass out}$$

3. *Calculate*
(i) *Calculation table*
The calculation table shows all given quantities in kg.

Stream	*In*			*Out*		
	Ethanol	*Water*	*Total*	*Ethanol*	*Water*	*Total*
Feed	5,000	45,000	50,000	–	–	–
Distillate	–	–	–	2,250	2,750	5,000
Bottoms	–	–	–	?	?	?
Total	5,000	45,000	50,000	?	?	?

(ii) *Mass-balance calculations*
Total mass balance

$$50{,}000 \text{ kg total mass in} = \text{total mass out}$$

$$\text{Total mass out} = 50{,}000 \text{ kg}$$

Therefore, from the total column on the Out side of the table:

$$\text{Bottoms out} = (50{,}000 - 5{,}000)\text{ kg} = 45{,}000 \text{ kg}$$

Ethanol balance

$$5{,}000 \text{ kg ethanol in} = \text{ethanol out}$$

$$\text{Ethanol out} = 5{,}000 \text{ kg}$$

From the ethanol column of the Out side of the table:

$$\text{Ethanol out in the bottoms} = (5{,}000 - 2{,}250)\text{ kg} = 2{,}750 \text{ kg}$$

Water balance

$$45{,}000 \text{ kg water in} = \text{water out}$$

$$\text{Water out} = 45{,}000 \text{ kg}$$

From the water column of the Out side of the table:

$$\text{Water out in the bottoms} = (45{,}000 - 2{,}750)\text{ kg} = 42{,}250 \text{ kg}$$

These calculations allow completion of the mass-balance table with all quantities in kg.

Stream	*In*			*Out*		
	Ethanol	*Water*	*Total*	*Ethanol*	*Water*	*Total*
Feed	5,000	45,000	50,000	–	–	–
Distillate	–	–	–	2,250	2,750	5,000
Bottoms	–	–	–	2,750	42,250	45,000
Total	5,000	45,000	50,000	5,000	45,000	50,000

(iii) *Check the results*
All columns and rows of the completed table add up correctly.

4. *Finalise*
(a)
The bottoms contains 2,750 kg ethanol and 42,250 kg water in a total of 45,000 kg. Therefore, the composition is $2{,}750/45{,}000 \times 100\% = 6.1\%$ ethanol, and $42{,}250/45{,}000 \times 100\% = 93.9\%$ water.

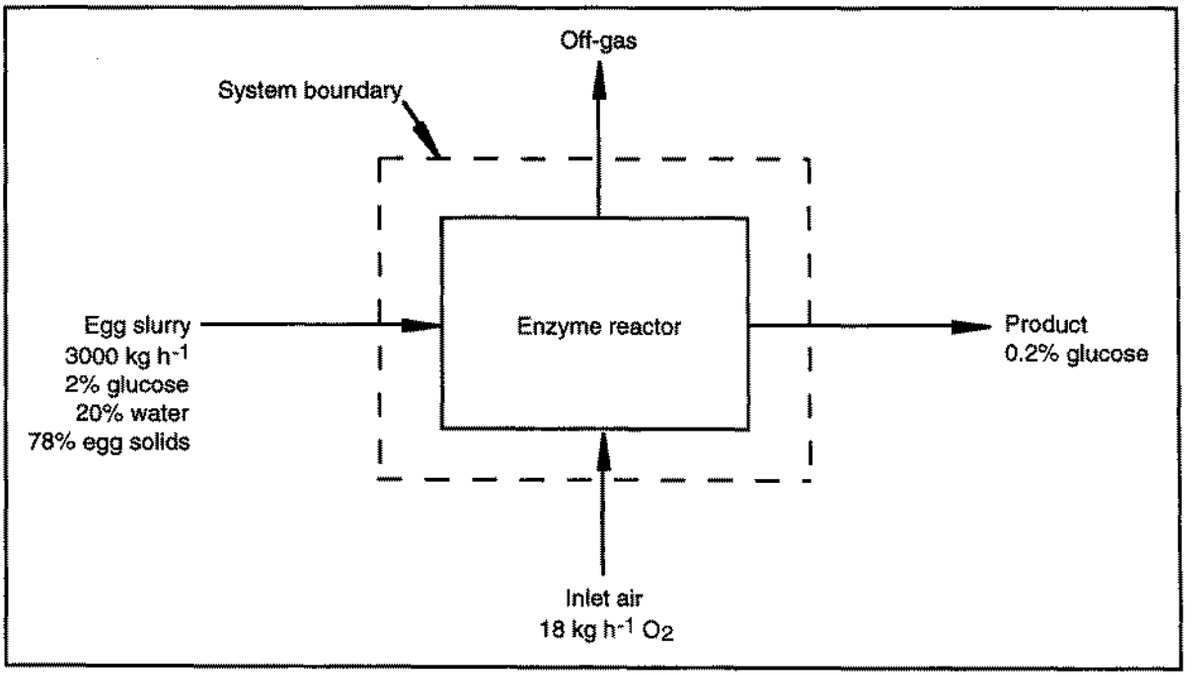
Answer: 6.1% ethanol, 93.9% water

(b)
Directly from the table, the rate of alcohol loss in the bottoms is 2,750 kg h^{-1}.

Answer: 2,750 kg h^{-1}

4.4 Removal of glucose from dried egg

1. *Assemble*
(i) *Flow sheet*



(ii) *System boundary*
The system boundary is shown on the flow sheet.
(iii) *Reaction equation*

$$C_6H_{12}O_6 + O_2 + H_2O \rightarrow C_6H_{12}O_7 + H_2O_2$$

2. *Analyse*
(i) *Assumptions*
– steady state
– no leaks
– air and off-gas are dry
– gases are at low pressure so vol% = mol%
– H_2O_2 remains in the liquid phase

(ii) *Extra data*

Molecular weights (Table B.1, Appendix B): glucose = 180.2
O_2 = 32.0
N_2 = 28.0
H_2O = 18.0
gluconic acid = 196.2
H_2O_2 = 34.0

Composition of air (p 17): 21% O_2, 79% N_2 by volume

(iii) *Basis*

1 h, or 3000 kg egg slurry

(iv) *Compounds involved in reaction*

Glucose, O_2, water, gluconic acid and H_2O_2 are involved in the reaction.

(v) *Mass-balance equations*

For glucose, O_2, water, gluconic acid and H_2O_2, the appropriate mass-balance equation is Eq. (4.2):

$$\text{mass in} + \text{mass generated} = \text{mass out} + \text{mass consumed}$$

For egg solids, N_2 and total mass, the appropriate mass-balance equation is Eq. (4.3):

$$\text{mass in} = \text{mass out}$$

3. *Calculate*

(i) *Calculation table*

The mass of N_2 accompanying 18 kg O_2 in air can be calculated from the known composition of air. Converting 18 kg O_2 to molar units:

$$18\,\text{kg}\,O_2 = 18\,\text{kg}\,O_2 \cdot \left|\frac{1\,\text{kgmol}}{32.0\,\text{kg}}\right| = 0.563\,\text{kgmol}\,O_2$$

Therefore, $^{79}/_{21} \times 0.563$ kgmol = 2.118 kgmol N_2 enter in the air stream. Converting this to mass units:

$$2.118\,\text{kgmol}\,N_2 = 2.118\,\text{kgmol}\,N_2 \cdot \left|\frac{28.0\,\text{kg}}{1\,\text{kgmol}}\right| = 59.30\,\text{kg}\,N_2$$

The calculation tables below show all known quantities in kg. The total mass of off-gas is denoted *G*; the total mass of product is denoted *P*. The In side of the mass-balance table is complete.

Stream	*In*							
	Glucose	*Water*	*Egg solids*	*O_2*	*N_2*	*Gluconic acid*	*H_2O_2*	*Total*
Egg slurry	60	600	2340	0	0	0	0	3000
Air	0	0	0	18	59.30	0	0	77.3
Off-gas	–	–	–	–	–	–	–	–
Product	–	–	–	–	–	–	–	–
Total	60	600	2340	18	59.30	0	0	3077.3

Stream	*Out*							
	Glucose	*Water*	*Egg solids*	*O_2*	*N_2*	*Gluconic acid*	*H_2O_2*	*Total*
Egg slurry	–	–	–	–	–	–	–	–
Air	–	–	–	–	–	–	–	–
Off-gas	0	0	0	?	?	0	0	*G*
Product	0.002 *P*	?	?	0	0	?	?	*P*
Total	0.002 *P*	?	?	?	?	?	?	*G* + *P*

(ii) *Mass-balance calculations*

Egg solids balance

Egg solids is a tie component.

$$2340\,\text{kg egg solids in} = \text{egg solids out}$$

$$\text{Egg solids out} = 2340\,\text{kg}$$

N_2 balance

N_2 is a tie component.

$$59.30 \text{ kg } N_2 \text{ in } = N_2 \text{ out}$$

$$N_2 \text{ out } = 59.30 \text{ kg}$$

Glucose balance

60 kg glucose in + 0 kg glucose generated = 0.002 P kg glucose out + glucose consumed

$$\text{Glucose consumed } = (60 - 0.002\,P) \text{ kg}$$

Converting the glucose consumed to molar terms:

$$\text{Glucose consumed } = (60 - 0.002\,P) \text{ kg} \cdot \left| \frac{1 \text{ kgmol}}{180.2 \text{ kg}} \right| = \frac{(60 - 0.002\,P)}{180.2} \text{ kgmol}$$

$$\text{Glucose consumed } = \left(0.333 - 1.11 \times 10^{-5}\,P\right) \text{kgmol}$$

From the reaction stoichiometry, conversion of this amount of glucose requires the same number of kgmol O_2. Converting this molar quantity to mass of O_2:

$$\left(0.333 - 1.11 \times 10^{-5}\,P\right) \text{kgmol } O_2 = \left(0.333 - 1.11 \times 10^{-5}\,P\right) \text{kgmol} \cdot \left| \frac{32.0 \text{ kg}}{1 \text{ kgmol}} \right| = \left(10.656 - 3.552 \times 10^{-4}\,P\right) \text{kg } O_2$$

O_2 balance

18 kg O_2 in + 0 kg O_2 generated = O_2 out + (10.656 – 3.552 × 10^{-4} P) kg O_2 consumed

$$O_2 \text{ out } = (18 - (10.656 - 3.552 \times 10^{-4}\,P)) \text{ kg}$$

$$O_2 \text{ out } = (7.344 + 3.552 \times 10^{-4}\,P) \text{ kg}$$

Adding this mass of O_2 to the mass of N_2 in the off-gas:

$$G = 59.30 + (7.344 + 3.552 \times 10^{-4}\,P) \text{ kg}$$

$$G = (66.64 + 3.552 \times 10^{-4}\,P) \text{ kg}$$

Total mass balance

3077.3 kg total mass in = (G + P) kg total mass out

Substituting the expression for G into the total mass balance:

$$3077.3 \text{ kg } = (66.64 + 3.552 \times 10^{-4}\,P + P) \text{ kg}$$

$$3010.7 \text{ kg } = 1.0004\,P \text{ kg}$$

$$P = 3009.6 \text{ kg}$$

Therefore, from the oxygen balance:

$$G = (66.64 + 3.552 \times 10^{-4} \times 3009.6) \text{ kg}$$

$$G = 67.71 \text{ kg}$$

and:

$$O_2 \text{ out } = (7.344 + 3.552 \times 10^{-4} \times 3009.6) \text{ kg}$$

$$O_2 \text{ out } = 8.41 \text{ kg}$$

The mass of glucose out is 0.002 × 3009.6 = 6.02 kg. The moles of glucose consumed is:

$$\text{Glucose consumed } = (0.333 - 1.11 \times 10^{-5} \times 3009.6) \text{ kgmol } = 0.300 \text{ kgmol}$$

Therefore, from stoichiometry and the molecular weights:

$$\text{Water consumed } = 0.300 \text{ kgmol} \cdot \left| \frac{18.0 \text{ kg}}{1 \text{ kgmol}} \right| = 5.40 \text{ kg}$$

$$\text{Gluconic acid generated} = 0.300\,\text{kgmol} \cdot \left|\frac{196.2\,\text{kg}}{1\,\text{kgmol}}\right| = 58.86\,\text{kg}$$

$$H_2O_2\ \text{generated} = 0.300\,\text{kgmol} \cdot \left|\frac{34\,\text{kg}}{1\,\text{kgmol}}\right| = 10.20\,\text{kg}$$

Water balance

600 kg water in + 0 kg water generated = water out + 5.40 kg water consumed

$$\text{Water out} = 594.6\,\text{kg}$$

Gluconic acid balance

0 kg gluconic acid in + 58.86 kg gluconic acid generated = gluconic acid out + 0 kg gluconic acid consumed

$$\text{Gluconic acid out} = 58.86\,\text{kg}$$

H_2O_2 balance

0 kg H_2O_2 in + 10.20 kg H_2O_2 generated = H_2O_2 out + 0 kg H_2O_2 consumed

$$H_2O_2\ \text{out} = 10.20\,\text{kg}$$

These calculations allow completion of the Out side of the mass-balance table with all quantities in kg.

Stream	*Out*							
	Glucose	*Water*	*Egg solids*	*O_2*	*N_2*	*Gluconic acid*	*H_2O_2*	*Total*
Egg slurry	–	–	–	–	–	–	–	–
Air	–	–	–	–	–	–	–	–
Off-gas	0	0	0	8.41	59.30	0	0	67.71
Product	6.02	594.6	2340	0	0	58.86	10.20	3009.6
Total	6.02	594.6	2340	8.41	59.30	58.86	10.20	3077.3

(iii) *Check the results*

All columns and rows of the completed table add up correctly to within round-off error.

4. *Finalise*

(a)

To determine which is the limiting substrate, the number of moles available of each substrate involved in the reaction must be determined. From the mass-balance table for streams in:

$$\text{Moles glucose} = 60\,\text{kg} \cdot \left|\frac{1\,\text{kg mol}}{180.2\,\text{kg}}\right| = 0.333\,\text{kgmol}$$

$$\text{Moles water} = 600\,\text{kg} \cdot \left|\frac{1\,\text{kg mol}}{18.0\,\text{kg}}\right| = 33.3\,\text{kgmol}$$

$$\text{Moles } O_2 = 18\,\text{kg} \cdot \left|\frac{1\,\text{kg mol}}{32.0\,\text{kg}}\right| = 0.563\,\text{kgmol}$$

As the substrates are required in the molar stoichiometric ratio of 1:1:1 and glucose is available in the smallest molar quantity, the extent of the reaction must be limited by glucose.

Answer: Glucose

(b)

Water and O_2 are available in excess. As only 0.333 kgmol of each will be used if the reaction proceeds to completion, from Eq. (2.34):

$$\%\ \text{excess water} = \frac{(33.3 - 0.333)\,\text{kgmol}}{0.333\,\text{kgmol}} \times 100\% = 9900\%$$

$$\%\ \text{excess } O_2 = \frac{(0.563 - 0.333)\,\text{kgmol}}{0.333\,\text{kgmol}} \times 100\% = 69\%$$

Answer: 9900% excess water; 69% excess O_2

(c)
From the completed mass-balance table, the reactor off-gas contains 8.41 kg O_2 and 59.30 kg N_2. As gas compositions are normally expressed in molar or volumetric terms (p 17), these mass values must be converted to moles:

$$8.41\,\text{kg}\,O_2 = 8.41\,\text{kg}\,O_2 \,.\left|\frac{1\,\text{kgmol}}{32.0\,\text{kg}}\right| = 0.263\,\text{kgmol}\,O_2$$

As the number of kgmol N_2 was determined in the preliminary calculations to be 2.118, the total number of moles of off-gas is (0.263 + 2.118) = 2.381 kgmol. Therefore, the composition of the off-gas is $0.263/2.381 = 0.11\ O_2$, and $2.118/2.381 = 0.89\ N_2$.

Answer: 0.11 O_2, 0.89 N_2

(d)
From the completed mass-balance table, the product stream has a total mass of 3009.6 kg and contains 6.02 kg glucose, 594.6 kg water, 2340 kg egg solids, 58.86 kg gluconic acid and 10.20 kg H_2O_2. Therefore, the composition is:

$$\frac{6.02}{3009.6} = 0.002\ \text{glucose}$$

$$\frac{594.6}{3009.6} = 0.198\ \text{water}$$

$$\frac{2340}{3009.6} = 0.778\ \text{egg solids}$$

$$\frac{58.86}{3009.6} = 0.020\ \text{gluconic acid}$$

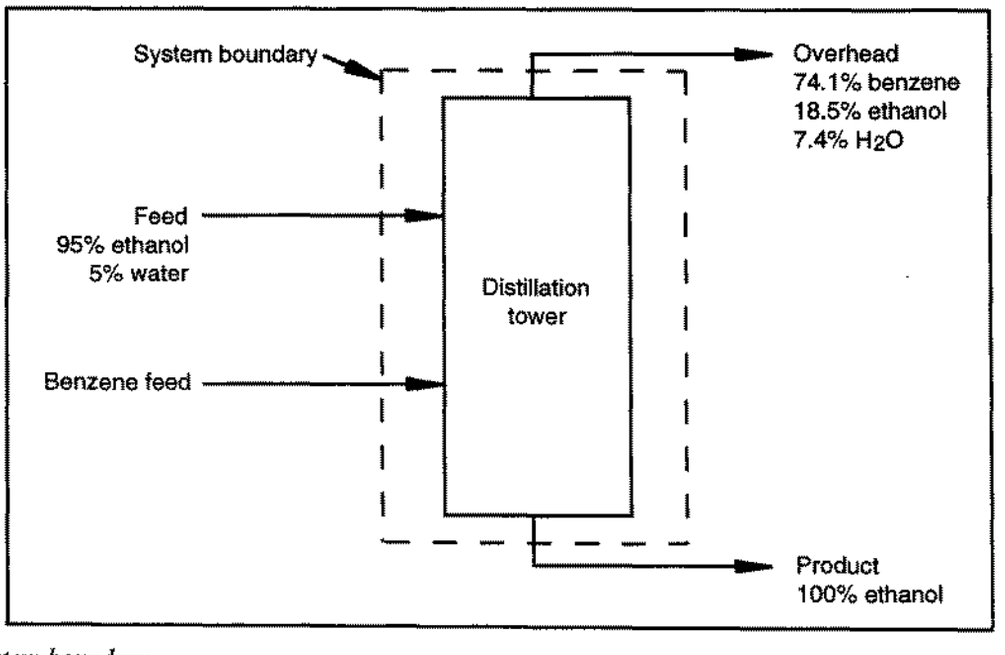
$$\frac{10.20}{3009.6} = 0.003\ H_2O_2$$

Answer: 0.002 glucose, 0.198 water, 0.778 egg solids, 0.020 gluconic acid, 0.003 H_2O_2

4.5 Azeotropic distillation

1. *Assemble*

(i) *Flow sheet*



(ii) *System boundary*
The system boundary is shown on the flow sheet.

(iii) *Reaction equation*
No reaction occurs.

2. *Analyse*
(i) *Assumptions*
– steady state
– no leaks
(ii) *Extra data*
$1000\ cm^3 = 1\ l$
$1000\ g = 1\ kg$
(iii) *Basis*
250 l ethanol product
(iv) *Compounds involved in reaction*
No compounds are involved in reaction.
(v) *Mass-balance equation*
As there is no reaction, the appropriate mass-balance equation is Eq. (4.3):

$$\text{mass in} = \text{mass out}$$

3. *Calculate*
(i) *Calculation table*
As all quantities in mass-balance calculations must be masses (rather than volumes), 250 l absolute ethanol must first be converted to mass. From the definition of density on p 16, mass is equal to volume multiplied by density:

$$250\ \text{l absolute ethanol} = 250\ \text{l} \times 0.785\ \text{g cm}^{-3} \cdot \left|\frac{1000\ \text{cm}^3}{1\ \text{l}}\right| \cdot \left|\frac{1\ \text{kg}}{1000\ \text{g}}\right| = 196.25\ \text{kg}$$

The calculation table shows all given quantities in kg. The total mass of feed in is denoted F; the total mass of benzene feed in is denoted B; the total mass of overhead out is denoted V.

Stream	*In*				*Out*			
	Ethanol	*Water*	*Benzene*	*Total*	*Ethanol*	*Water*	*Benzene*	*Total*
Feed	$0.95\ F$	$0.05\ F$	0	F	–	–	–	–
Benzene feed	0	0	B	B	–	–	–	–
Product	–	–	–	–	196.25	0	0	196.25
Overhead	–	–	–	–	$0.185\ V$	$0.074\ V$	$0.741\ V$	V
Total	$0.95\ F$	$0.05\ F$	B	$F + B$	$196.25 + 0.185\ V$	$0.074\ V$	$0.741\ V$	$196.25 + V$

(ii) *Mass-balance calculations*
Ethanol balance

$$0.95\ F\ \text{kg ethanol in} = (196.25 + 0.185\ V)\ \text{kg ethanol out}$$

$$F = (206.58 + 0.195\ V)\ \text{kg}$$

Benzene balance

$$B\ \text{kg benzene in} = 0.741\ V\ \text{kg benzene out}$$

$$B = 0.741\ V$$

Total mass balance

$$(F + B)\ \text{kg total mass in} = (196.25 + V)\ \text{kg total mass out}$$

Substituting for F and B from the ethanol and benzene balances:

$$(206.58 + 0.195\ V + 0.741\ V)\ \text{kg} = (196.25 + V)\ \text{kg}$$

$$10.33 = 0.064\ V$$

$$V = 161.4\ \text{kg}$$

Using this result in the ethanol and benzene balances gives:

$$F = 238.1\ \text{kg}$$

$$B = 119.6\,\text{kg}$$

These calculations allow completion of the mass-balance table with all quantities in kg.

Stream	*In*				*Out*			
	Ethanol	*Water*	*Benzene*	*Total*	*Ethanol*	*Water*	*Benzene*	*Total*
Feed	226.2	11.9	0	238.1	–	–	–	–
Benzene feed	0	0	119.6	119.6	–	–	–	–
Product	–	–	–	–	196.25	0	0	196.25
Overhead	–	–	–	–	29.9	11.9	119.6	161.4
Total	226.2	11.9	119.6	357.7	226.2	11.9	119.6	357.7

(iii) *Check the results*
All columns and rows of the completed table add up correctly to within round-off error.

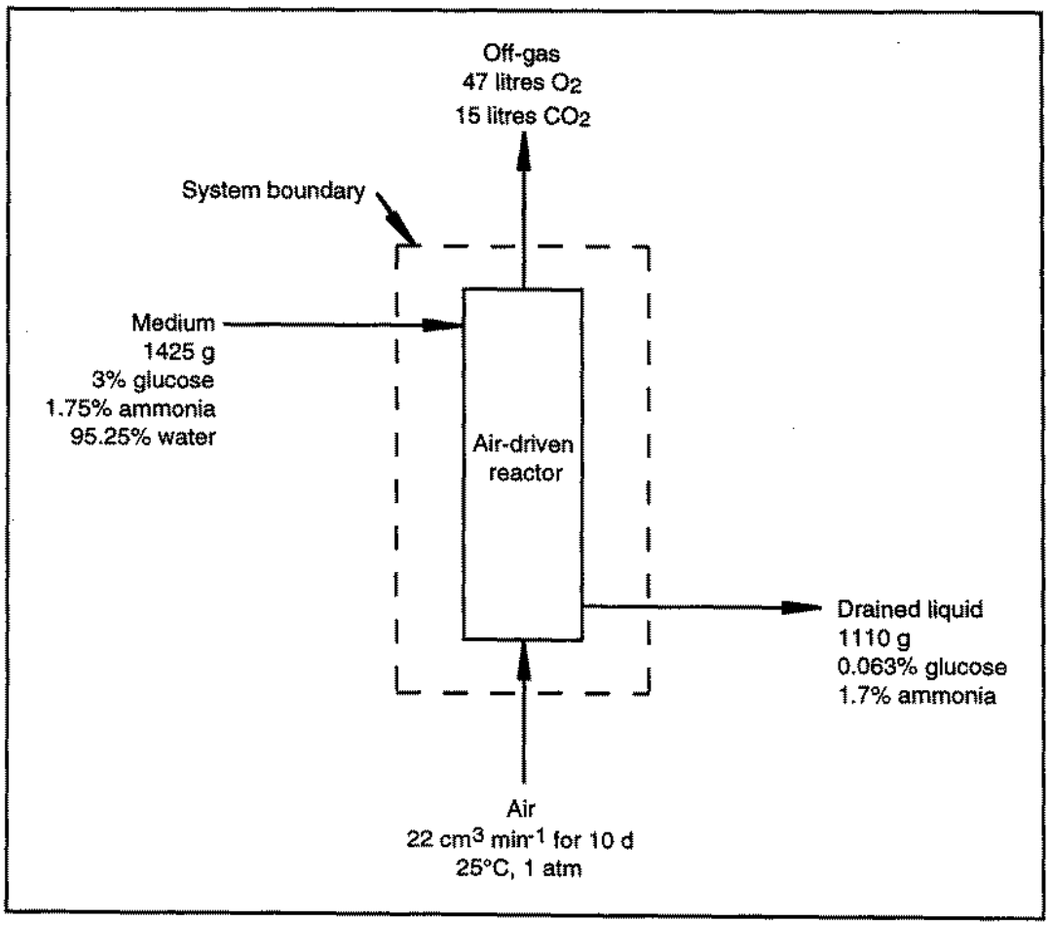
4. *Finalise*
From the completed mass-balance table, the mass of benzene required is 119.6 kg. Using the definition of density on p 16, volume is equal to mass divided by density:

$$\text{Volume of benzene} = \frac{119.6\,\text{kg}}{0.872\,\text{g cm}^{-3}} \cdot \left|\frac{1000\,\text{g}}{1\,\text{kg}}\right| \cdot \left|\frac{1\,\text{l}}{1000\,\text{cm}^3}\right| = 1371$$

Answer: 137 litres

4.6 Culture of plant roots

1. *Assemble*
(i) *Flow sheet*



(ii) *System boundary*
The system boundary is shown on the flow sheet.

(iii) *Reaction equation*
From Table B.2 (Appendix B), the molecular formula for glucose is $C_6H_{12}O_6$. The reaction equation is based on the general stoichiometric equation for aerobic growth, Eq. (4.4):

$$C_6H_{12}O_6 + a\,O_2 + b\,NH_3 \rightarrow c\,CH_\alpha O_\beta N_\delta + d\,CO_2 + e\,H_2O$$

2. *Analyse*
(i) *Assumptions*
– steady state
– no leaks
– air and off-gas are dry
– all the CO_2 produced leaves in the off-gas
– gases are at low pressure so vol% = mol%
(ii) *Extra data*
1 l = 1000 cm^3
Molecular weights (Table B.1, Appendix B): glucose = 180.2
O_2 = 32.0
N_2 = 28.0
NH_3 = 17.0
CO_2 = 44.0
H_2O = 18.0
Composition of air (p 17): 21% O_2, 79% N_2 by volume
Ideal gas constant (Table 2.5): R = 82.057 cm^3 atm K^{-1} $gmol^{-1}$
(iii) *Basis*
10 d, or 1425 g nutrient medium
(iv) *Compounds involved in reaction*
Glucose, O_2, NH_3, biomass, CO_2 and H_2O are involved in the reaction.
(v) *Mass-balance equations*
For glucose, O_2, NH_3, biomass, CO_2 and H_2O, the appropriate mass-balance equation is Eq. (4.2):

$$\text{mass in} + \text{mass generated} = \text{mass out} + \text{mass consumed}$$

For N_2 and total mass, the appropriate mass-balance equation is Eq. (4.3):

$$\text{mass in} = \text{mass out}$$

3. *Calculate*
(i) *Calculation table*
Over 10 d, the volume of air sparged into the fermenter is:

$$\text{Volume of air in} = 22\ \text{cm}^3\ \text{min}^{-1} \times 10\ \text{d} \cdot \left|\frac{60\ \text{min}}{1\ \text{h}}\right| \cdot \left|\frac{24\ \text{h}}{1\ \text{d}}\right| = 3.168 \times 10^5\ \text{cm}^3$$

Converting this gas volume to moles using the ideal gas law, Eq. (2.32), with the temperature converted from °C to degrees Kelvin using Eq. (2.24):

$$\text{Moles of air in} = n = \frac{pV}{RT} = \frac{1\ \text{atm}\left(3.168 \times 10^5\ \text{cm}^3\right)}{82.057\ \text{cm}^3\ \text{atm K}^{-1}\ \text{gmol}^{-1}\ (25 + 273.15)\ \text{K}} = 12.95\ \text{gmol}$$

From the known composition of air, the moles of O_2 in the incoming air is 0.21 × 12.95 = 2.72 gmol, and the moles of N_2 is 0.79 × 12.95 = 10.23 gmol. Converting these values to masses:

$$\text{Mass of } O_2 \text{ in} = 2.72\ \text{gmol} \cdot \left|\frac{32.0\ \text{g}}{1\ \text{gmol}}\right| = 87.04\ \text{g}$$

$$\text{Mass of } N_2 \text{ in} = 10.23\ \text{gmol} \cdot \left|\frac{28.0\ \text{g}}{1\ \text{gmol}}\right| = 286.4\ \text{g}$$

The total mass of air in is therefore (87.04 + 286.4) g = 373.44 g.

The gas volumes in the off-gas must also be converted to masses. First, convert the volumes of O_2 and CO_2 to moles using the ideal gas law, Eq. (2.32), with the temperature converted from °C to degrees Kelvin using Eq. (2.24):

$$\text{Moles of } O_2 \text{ out} = n = \frac{pV}{RT} = \frac{1\text{ atm}\,(47\text{ litres}) \cdot \left|\frac{1000\text{ cm}^3}{1\text{ l}}\right|}{82.057\text{ cm}^3\text{ atm K}^{-1}\text{ gmol}^{-1}\,(25+273.15)\text{ K}} = 1.92\text{ gmol}$$

$$\text{Moles of } CO_2 \text{ out} = n = \frac{pV}{RT} = \frac{1\text{ atm}\,(15\text{ litres}) \cdot \left|\frac{1000\text{ cm}^3}{1\text{ l}}\right|}{82.057\text{ cm}^3\text{ atm K}^{-1}\text{ gmol}^{-1}\,(25+273.15)\text{ K}} = 0.613\text{ gmol}$$

Calculate the corresponding masses:

$$\text{Mass of } O_2 \text{ out} = 1.92\text{ gmol} \cdot \left|\frac{32.0\text{ g}}{1\text{ gmol}}\right| = 61.44\text{ g}$$

$$\text{Mass of } CO_2 \text{ out} = 0.613\text{ gmol} \cdot \left|\frac{44.0\text{ g}}{1\text{ gmol}}\right| = 26.97\text{ g}$$

The calculation tables below show all known quantities in g. The In side of the mass-balance table is complete. The total mass of off-gas out is denoted G; the total biomass harvested is denoted B. As the ratio of biomass fresh weight to dry weight is 14:1, dry biomass comprises $^1/_{15} = 0.0667$ of the total biomass. Because this problem requires an integral mass balance, the biomass remaining in the fermenter after 10 d culture must also be included in the table even though it is not contained in any of the streams flowing into or out of the vessel.

Stream	*In*							
	Glucose	*O_2*	*N_2*	*NH_3*	*Dry biomass*	*CO_2*	*H_2O*	*Total*
Medium	42.75	0	0	24.94	0	0	1357.31	1425
Air	0	87.04	286.4	0	0	0	0	373.44
Drained liquid	–	–	–	–	–	–	–	–
Off-gas	–	–	–	–	–	–	–	–
Harvested biomass	–	–	–	–	–	–	–	–
Total	42.75	87.04	286.4	24.94	0	0	1357.31	1798.44

Stream	*Out*							
	Glucose	*O_2*	*N_2*	*NH_3*	*Dry biomass*	*CO_2*	*H_2O*	*Total*
Medium	–	–	–	–	–	–	–	–
Air	–	–	–	–	–	–	–	–
Drained liquid	0.699	0	0	18.87	0	0	1090.43	1110
Off-gas	0	61.44	?	0	0	26.97	0	*G*
Harvested biomass	0	0	0	0	0.0667 *B*	0	0.9333 *B*	*B*
Total	0.699	61.44	?	18.87	0.0667 *B*	26.97	1090.43 + 0.9333 *B*	1110 + *G* + *B*

(ii) *Mass-balance calculations*

N_2 balance

N_2 is a tie component.

$$286.4\text{ g } N_2 \text{ in} = N_2 \text{ out}$$

$$N_2 \text{ out} = 286.4\text{ g}$$

Using this result and adding up the row for the off-gas in the out table:

$$G = 61.44 + 286.4 + 26.97 = 374.81\text{ g}$$

Total mass balance

$$1798.44\text{ g total mass in} = (1110 + G + B)\text{ g total mass out}$$

Using the result for G:

$$B = 313.63 \text{ g}$$

Therefore, the dry biomass produced is 0.0667 × 313.63 = 20.92 g; the mass of water in the biomass is 0.9333 × 313.63 = 292.71 g.

These calculations allow completion of the Out side of the mass-balance table with all quantities in g.

Stream	*Out*							
	Glucose	*O_2*	*N_2*	*NH_3*	*Dry biomass*	*CO_2*	*H_2O*	*Total*
Medium	–	–	–	–	–	–	–	–
Air	–	–	–	–	–	–	–	–
Drained liquid	0.699	0	0	18.87	0	0	1090.43	1110
Off-gas	0	61.44	286.4	0	0	26.97	0	374.81
Harvested biomass	0	0	0	0	20.92	0	292.71	313.63
Total	0.699	61.44	286.4	18.87	20.92	26.97	1383.14	1798.44

Further mass-balance calculations allow evaluation of the masses of components consumed or generated in the reaction.

Glucose balance

42.75 g glucose in + 0 g glucose generated = 0.699 g glucose out + glucose consumed

Glucose consumed = 42.05 g

O_2 balance

87.04 g O_2 in + 0 g O_2 generated = 61.44 g O_2 out + O_2 consumed

O_2 consumed = 25.60 g

NH_3 balance

24.94 g NH_3 in + 0 g NH_3 generated = 18.87 g NH_3 out + NH_3 consumed

NH_3 consumed = 6.07 g

CO_2 balance

0 g CO_2 in + CO_2 generated = 26.97 g CO_2 out + 0 g CO_2 consumed

CO_2 generated = 26.97 g

H_2O balance

1357.31 g H_2O in + H_2O generated = (1090.43 + 0.9333 *B*) g H_2O out + 0 g H_2O consumed

Substituting the value for *B* from the total mass balance:

H_2O generated = 25.83 g

(iii) *Check the results*
All columns and rows of the completed mass-balance table add up correctly.

4. *Finalise*
(a)
Rounding to three significant figures from the completed mass-balance table, the mass of dry roots produced is 20.9 g.

Answer: 20.9 g

(b)
To determine the stoichiometry, the calculated masses of components consumed or generated in the reaction must be converted to molar quantities:

$$\text{Moles of glucose consumed} = 42.05\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{180.2\,\text{g}}\right| = 0.233\,\text{gmol}$$

$$\text{Moles of } O_2 \text{ consumed} = 25.60\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{32.0\,\text{g}}\right| = 0.800\,\text{gmol}$$

$$\text{Moles of } NH_3 \text{ consumed} = 6.07\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{17.0\,\text{g}}\right| = 0.357\,\text{gmol}$$

$$\text{Moles of } CO_2 \text{ generated} = 26.97\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{44.0\,\text{g}}\right| = 0.613\,\text{gmol}$$

$$\text{Moles of } H_2O \text{ generated} = 25.83\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{18.0\,\text{g}}\right| = 1.435\,\text{gmol}$$

$$\text{Moles of biomass generated} = \frac{20.92\,\text{g}}{\text{MW biomass}}$$

The moles of biomass generated is not yet known explicitly because the molecular formula for the dry biomass is unknown. The above molar quantities can be used as coefficients in the reaction equation:

$$0.233\,C_6H_{12}O_6 + 0.800\,O_2 + 0.357\,NH_3 \rightarrow \frac{20.92}{\text{MW biomass}}\,CH_\alpha O_\beta N_\delta + 0.613\,CO_2 + 1.435\,H_2O$$

Dividing each coefficient by 0.233 to obtain the stoichiometry per gmol glucose:

$$C_6H_{12}O_6 + 3.43\,O_2 + 1.53\,NH_3 \rightarrow \frac{89.79}{\text{MW biomass}}\,CH_\alpha O_\beta N_\delta + 2.63\,CO_2 + 6.16\,H_2O$$

The values of α, β and δ and the molecular formula for the biomass can obtained using elemental balances.

C balance: $6 = \dfrac{89.79}{\text{MW biomass}} + 2.63$

Therefore:

$$\frac{89.79}{\text{MW biomass}} = 3.37$$

This result can be used in the remaining elemental balances for completion of the stoichiometric equation.
H balance: $12 + 3 \times 1.53 = 3.37\,\alpha + 2 \times 6.16 \rightarrow \alpha = 1.27$
O balance: $6 + 2 \times 3.43 = 3.37\,\beta + 2 \times 2.63 + 6.16 \rightarrow \beta = 0.43$
N balance: $1.53 = 3.37\,\delta \rightarrow \delta = 0.45$

Answer: The complete stoichiometric equation is:

$$C_6H_{12}O_6 + 3.43\,O_2 + 1.53\,NH_3 \rightarrow 3.37\,CH_{1.27}O_{0.43}N_{0.45} + 2.63\,CO_2 + 6.16\,H_2O$$

The chemical formula for the dry roots is $CH_{1.27}O_{0.43}N_{0.45}$.

(c)
Converting to moles the mass quantities of glucose, O_2 and NH_3 available for reaction on the In side of the mass-balance table:

$$\text{Moles of glucose in} = 42.75\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{180.2\,\text{g}}\right| = 0.24\,\text{gmol}$$

$$\text{Moles of } O_2 \text{ in} = 87.04\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{32.0\,\text{g}}\right| = 2.72\,\text{gmol}$$

$$\text{Moles of } NH_3 \text{ in} = 24.93\,\text{g} \cdot \left|\frac{1\,\text{gmol}}{17.0\,\text{g}}\right| = 1.47\,\text{gmol}$$

From the stoichiometric equation, reaction of 0.24 gmol glucose requires $0.24 \times 3.43 = 0.82$ gmol O_2 and $0.24 \times 1.53 = 0.37$ gmol NH_3. As the molar quantities of O_2 and NH_3 available for reaction are in excess of these values, glucose must be the limiting substrate.

Answer: Glucose

(d)
The mass of glucose consumed is 42.05 g; the mass of dry biomass produced is 20.92 g. Therefore, the biomass yield from glucose is $20.92/42.05 = 0.50$ g g^{-1} dry weight.

Answer: 0.50 g g^{-1} dry weight

4.7 Oxygen requirement for growth on glycerol

From Table B.2 (Appendix B), the molecular formula for glycerol is $C_3H_8O_3$. From Table 4.3, the chemical formula for *Klebsiella aerogenes* can be taken as $CH_{1.75}O_{0.43}N_{0.22}$. Substituting these formulae into the general stoichiometric equation for growth, Eq. (4.4), gives:

$$C_3H_8O_3 + a\,O_2 + b\,NH_3 \rightarrow c\,CH_{1.75}O_{0.43}N_{0.22} + d\,CO_2 + e\,H_2O$$

From Table B.8 (Appendix B), the molecular weight of glycerol is 92.1. The biomass formula weight calculated from the atomic weights in Table B.1 (Appendix B) is 23.74. Taking into account the 8% ash:

$$\text{Biomass molecular weight} = \frac{23.74}{0.92} = 25.8$$

The value of the stoichiometric coefficient c can be determined from the yield $Y_{XS} = 0.4$ g g^{-1} and Eq. (4.12):

$$c = \frac{Y_{XS}\,(\text{MW substrate})}{\text{MW cells}} = \frac{0.40\text{ g g}^{-1}\left(92.1\text{ g gmol}^{-1}\right)}{25.8\text{ g gmol}^{-1}} = 1.43\text{ gmol gmol}^{-1}$$

From Table B.2 (Appendix B), the degree of reduction of glycerol relative to NH_3 is $\gamma_S = 4.67$. The degree of reduction of the biomass relative to NH_3 is:

$$\gamma_B = \frac{1\times4 + 1.75\times1 - 0.43\times2 - 0.22\times3}{1} = 4.23$$

which is also listed in Table 4.3. The theoretical oxygen demand can be determined from Eq. (4.16) with $f = 0$; from Eq. (4.13), $w = 3$ for glycerol:

$$a = {}^1/_4\,(w\,\gamma_S - c\,\gamma_B) = {}^1/_4\,(3\times4.67 - 1.43\times4.23) = 1.99$$

Therefore, 1.99 gmol oxygen are required per gmol glycerol. From the atomic weights in Table B.1 (Appendix B), the molecular weight of oxygen is 32.0. Converting a to mass terms:

$$a = 1.99\text{ gmol gmol}^{-1} = 1.99\text{ gmol gmol}^{-1}\cdot\left|\frac{32.0\text{ g}}{1\text{ gmol}}\right|\cdot\left|\frac{1\text{ gmol}}{92.1\text{ g}}\right| = 0.69\text{ g g}^{-1}$$

Answer: 0.69 g oxygen is required per g glycerol consumed

4.8 Product yield in anaerobic digestion

From Eq. (4.13), the stoichiometric equation for anaerobic growth and product formation by methane bacteria can be written as:

$$CH_3COOH + b\,NH_3 \rightarrow c\,CH_{1.4}O_{0.40}N_{0.20} + d\,CO_2 + e\,H_2O + f\,CH_4$$

From Table B.8 (Appendix B), the molecular weight of acetic acid is 60.1. From Table B.1 (Appendix B), the molecular weight of CO_2 is 44.0. The value of the stoichiometric coefficient d can be determined based on Eq. (4.14) with carbon dioxide as the product and the yield $Y_{PS} = 0.67$ kg kg$^{-1} = 0.67$ g g^{-1}:

$$d = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW }CO_2} = \frac{0.67\text{ g g}^{-1}\left(60.1\text{ g gmol}^{-1}\right)}{44.0\text{ g gmol}^{-1}} = 0.915\text{ gmol gmol}^{-1}$$

The other coefficients can be determined using this result and elemental balances.
C balance: $2 = c + d + f = c + 0.915 + f \rightarrow f = 1.085 - c$
H balance: $4 + 3\,b = 1.4\,c + 2\,e + 4\,f$
O balance: $2 = 0.40\,c + 2\,d + e = 0.40\,c + 2\times0.915 + e = 0.40\,c + 1.83 + e \rightarrow e = 0.17 - 0.40\,c$
N balance: $b = 0.20\,c$

Substituting the expressions for f, e and b from the C, O and N balances, respectively, into the H balance:

$$4 + 3 \times 0.20\,c = 1.4\,c + 2 \times (0.17 - 0.40\,c) + 4 \times (1.085 - c)$$

$$4\,c = 0.680$$

$$c = 0.170$$

Substituting this value for c into the expressions for the other coefficients gives $b = 0.034$, $e = 0.102$ and $f = 0.915$. The yield of methane is therefore 0.915 gmol per gmol acetic acid.

The maximum possible methane yield can be calculated using Eq. (4.20). From Eq. (4.13), $w = 2$ for acetic acid and $j = 1$ for methane. From Table B.2 (Appendix B), the degree of reduction of acetic acid relative to NH_3 is $\gamma_S = 4.00$, and the degree of reduction of methane relative to NH_3 is $\gamma_P = 8.00$. Substituting these values into Eq. (4.20) gives:

$$f_{max} = \frac{w\,\gamma_S}{j\,\gamma_P} = \frac{2\,(4.00)}{1\,(8.00)} = 1.0\ \text{gmol gmol}^{-1}$$

The actual methane yield of 0.915 gmol $gmol^{-1}$ therefore represents 91.5% of the theoretical maximum.

Answer: 91.5% of the theoretical maximum

4.9 Stoichiometry of single-cell protein synthesis

(a)

From Table B.2 (Appendix B), the molecular formula for glucose is $C_6H_{12}O_6$. If all carbon in the substrate is converted into biomass, production of carbon dioxide is zero. Therefore, from Eq. (4.4), the stoichiometric equation for anaerobic growth of *Cellulomonas* is:

$$C_6H_{12}O_6 + b\,NH_3 \rightarrow c\,CH_{1.56}O_{0.54}N_{0.16} + e\,H_2O$$

The stoichiometric coefficients can be determined using elemental balances.
C balance: $6 = c$
H balance: $12 + 3\,b = 1.56\,c + 2\,e$
O balance: $6 = 0.54\,c + e$
N balance: $b = 0.16\,c$
Substituting the value for c from the C balance into the O and N balances gives $e = 2.76$ and $b = 0.96$, respectively. The yield of biomass from substrate in molar terms is therefore 6 gmol $gmol^{-1}$.

Using the atomic weights in Table B.1 (Appendix B), the molecular weight of glucose is 180.2. The biomass formula weight calculated from Table B.1 (Appendix B) is 24.46. Taking into account the 5% ash:

$$\text{Biomass molecular weight} = \frac{24.46}{0.95} = 25.75$$

Therefore, in mass terms, the molar biomass yield of 6 gmol $gmol^{-1}$ = (6 × 25.75) g biomass per 180.2 g substrate = 0.86 g g^{-1}.

The maximum possible biomass yield is calculated using Eq. (4.19). From Eq. (4.13), $w = 6$ for glucose. From Table B.2 (Appendix B), the degree of reduction of glucose relative to NH_3 is $\gamma_S = 4.00$. The degree of reduction of the biomass relative to NH_3 is:

$$\gamma_B = \frac{1 \times 4 + 1.56 \times 1 - 0.54 \times 2 - 0.16 \times 3}{1} = 4.00$$

Substituting these values into Eq. (4.19):

$$c_{max} = \frac{w\,\gamma_S}{\gamma_B} = \frac{6\,(4.00)}{4.00} = 6.0\ \text{gmol gmol}^{-1}$$

The theoretical maximum biomass yield is therefore the same as the actual biomass yield.

Answer: The biomass yield from substrate of 0.86 g g^{-1} is 100% of the theoretical maximum. *When there is no product formation and no oxygen for electron transfer, all the available electrons from the substrate must go to the biomass.*

(b)
(i)
From Table B.2 (Appendix B), the molecular formula for methanol is CH_4O and the degree of reduction relative to NH_3 is $\gamma_S = 6.00$. The degree of reduction of *Methylophilus methylotrophus* biomass relative to NH_3 is:

$$\gamma_B = \frac{1\times4 + 1.68\times1 - 0.36\times2 - 0.22\times3}{1} = 4.30$$

From Eq. (4.13), $w = 1$ for methanol. Substituting values into Eq. (4.19):

$$c_{max} = \frac{w\,\gamma_S}{\gamma_B} = \frac{1\,(6.00)}{4.30} = 1.40 \text{ gmol gmol}^{-1}$$

From Table B.8 (Appendix B), the molecular weight of methanol is 32.0. The biomass formula weight calculated from the atomic weights in Table B.1 (Appendix B) is 22.55. With 6% ash:

$$\text{Biomass molecular weight} = \frac{22.55}{0.94} = 23.99$$

In mass terms, the maximum possible molar biomass yield of 1.40 gmol $gmol^{-1}$ is equal to (1.40 × 23.99) g biomass per 32.0 g substrate = 1.05 g g^{-1}.

Answer: The maximum possible biomass yield from methanol is 1.05 g g^{-1}. In terms of C atoms, the biomass yield is 1.40 gmol $gmol^{-1}$ as both biomass and substrate have 1 C atom each. In comparison, the C-atom biomass yield from glucose in **(a)** is 1 gmol $gmol^{-1}$. The main reason for the increased yield in **(b)** is the high degree of reduction of methanol compared with glucose.

(ii)
The actual yield of biomass from methanol is $c = 0.42 \times 1.40$ gmol $gmol^{-1} = 0.59$. The oxygen demand can be determined from Eq. (4.16) if biomass remains the only major product so that $f = 0$. Using the parameter values determined in **(b) (i)**:

$$a = {}^1/_4\,(w\,\gamma_S - c\,\gamma_B) = {}^1/_4\,(1\times6.00 - 0.59\times4.30) = 0.87$$

Therefore, 0.87 gmol oxygen is required per gmol methanol. As the molecular weights of methanol and oxygen are the same, the oxygen demand is 0.87 g g^{-1} methanol.

Answer: 0.87 g oxygen is required per g methanol consumed

4.10 Ethanol production by yeast and bacteria

(a)
From Table B.2 (Appendix B), the molecular formula for glucose is $C_6H_{12}O_6$ and the molecular formula for ethanol is C_2H_6O. From Eq. (4.13), the stoichiometric equation for anaerobic growth and product formation is:

$$C_6H_{12}O_6 + b\,NH_3 \rightarrow c\,CH_{1.8}O_{0.5}N_{0.2} + d\,CO_2 + e\,H_2O + f\,C_2H_6O$$

From Table B.8 (Appendix B), the molecular weight of ethanol is 46.1. Using the atomic weights in Table B.1 (Appendix B), the molecular weight of glucose is 180.2 and the biomass molecular weight is 24.6. The values of the stoichiometric coefficients c can be determined from Eq. (4.12) and the yields $Y_{XS} = 0.11$ g g^{-1} for yeast and $Y_{XS} = 0.05$ g g^{-1} for bacteria.

$$\text{For yeast, } c = \frac{Y_{XS}\,(\text{MW substrate})}{\text{MW cells}} = \frac{0.11 \text{ g g}^{-1}\left(180.2 \text{ g gmol}^{-1}\right)}{24.6 \text{ g gmol}^{-1}} = 0.81 \text{ gmol gmol}^{-1}$$

$$\text{For bacteria, } c = \frac{Y_{XS}\,(\text{MW substrate})}{\text{MW cells}} = \frac{0.05 \text{ g g}^{-1}\left(180.2 \text{ g gmol}^{-1}\right)}{24.6 \text{ g gmol}^{-1}} = 0.37 \text{ gmol gmol}^{-1}$$

The other coefficients can be determined using elemental balances.

Yeast
C balance: $6 = c+d+2f = 0.81+d+2f \rightarrow d = 5.19-2f$
H balance: $12+3b = 1.8c+2e+6f = 1.8\times0.81+2e+6f \rightarrow 10.54+3b = 2e+6f$
O balance: $6 = 0.5c+2d+e+f = 0.5\times0.81+2d+e+f \rightarrow 5.595 = 2d+e+f$
N balance: $b = 0.2c = 0.2\times0.81 = 0.16$
Substituting the expression for b from the N balance into the H balances gives:

$$10.54+3\times0.16 = 2e+6f$$

$$e = 5.51-3f$$

Substituting this and the expression for d from the C balance into the O balance gives:

$$5.595 = 2\times(5.19-2f)+(5.51-3f)+f$$

$$6f = 10.295$$

$$f = 1.72$$

Therefore, for yeast, the yield of ethanol from glucose is 1.72 gmol gmol^{-1}.

Bacteria
C balance: $6 = c+d+2f = 0.37+d+2f \rightarrow d = 5.63-2f$
H balance: $12+3b = 1.8c+2e+6f = 1.8\times0.37+2e+6f \rightarrow 11.33+3b = 2e+6f$
O balance: $6 = 0.5c+2d+e+f = 0.5\times0.37+2d+e+f \rightarrow 5.815 = 2d+e+f$
N balance: $b = 0.2c = 0.2\times0.37 = 0.074$
Using the same solution procedure as for yeast, substituting the expression for b from the N balance into the H balances gives:

$$11.33+3\times0.074 = 2e+6f$$

$$e = 5.78-3f$$

Substituting this and the expression for d from the C balance into the O balance gives:

$$5.815 = 2\times(5.63-2f)+(5.78-3f)+f$$

$$6f = 11.225$$

$$f = 1.87$$

Therefore, for bacteria, the yield of ethanol from glucose is 1.87 gmol gmol^{-1}.

Answer: 1.72 gmol gmol^{-1} for yeast; 1.87 gmol gmol^{-1} for bacteria

(b)
The maximum possible ethanol yield can be calculated using Eq. (4.20). From Eq. (4.13), $w = 6$ for glucose and $j = 2$ for ethanol. From Table B.2 (Appendix B), the degree of reduction of glucose relative to NH_3 is $\gamma_S = 4.00$, and the degree of reduction of ethanol relative to NH_3 is $\gamma_P = 6.00$. Using these values in Eq. (4.20) gives:

$$f_{max} = \frac{w\,\gamma_S}{j\,\gamma_P} = \frac{6\,(4.00)}{2\,(6.00)} = 2.0 \text{ gmol gmol}^{-1}$$

Therefore, the actual ethanol yield of 1.72 gmol gmol^{-1} for yeast represents 86% of the theoretical maximum; for bacteria, the actual yield of 1.87 gmol gmol^{-1} represents 94% of the theoretical maximum.

Answer: 86% of the theoretical maximum for yeast; 94% of the theoretical maximum for bacteria

4.11 Detecting unknown products

From Table B.2 (Appendix B), the molecular formula for glucose is $C_6H_{12}O_6$. Assuming that no products other than biomass are formed, from Eq. (4.4), the stoichiometric equation for growth is:

$$C_6H_{12}O_6 + a\,O_2 + b\,NH_3 \rightarrow c\,CH_{1.79}O_{0.56}N_{0.17} + d\,CO_2 + e\,H_2O$$

Using the atomic weights in Table B.1 (Appendix B), the molecular weight of glucose is 180.2, the molecular weight of oxygen is 32.0, and the biomass molecular weight is 25.16. The value of the stoichiometric coefficient c can be determined from the yield $Y_{XS} = 0.37$ g g^{-1} and Eq. (4.12):

$$c = \frac{Y_{XS}(\text{MW substrate})}{\text{MW cells}} = \frac{0.37\ \text{g g}^{-1}\left(180.2\ \text{g gmol}^{-1}\right)}{25.16\ \text{g gmol}^{-1}} = 2.65\ \text{gmol gmol}^{-1}$$

Therefore, 2.65 gmol cells are produced per gmol glucose consumed. Converting the oxygen demand to a molar basis:

$$0.88\ \text{g O}_2\ \text{per g cells} = \frac{0.88\ \text{g O}_2}{1\ \text{g cells}} \cdot \left|\frac{25.16\ \text{g cells}}{1\ \text{gmol cells}}\right| \cdot \left|\frac{1\ \text{gmol O}_2}{32.0\ \text{g O}_2}\right| = 0.69\ \text{gmol O}_2\ \text{gmol}^{-1}\ \text{cells}$$

Combining this with the result for c, the observed oxygen demand a is:

$$a = \frac{0.69\ \text{gmol O}_2}{1\ \text{gmol cells}}\left(\frac{2.65\ \text{gmol cells}}{1\ \text{gmol glucose}}\right) = 1.83$$

From Table B.2 (Appendix B), the degree of reduction of glucose relative to NH_3 is $\gamma_S = 4.00$. The degree of reduction of the biomass relative to NH_3 is:

$$\gamma_B = \frac{1\times 4 + 1.79\times 1 - 0.56\times 2 - 0.17\times 3}{1} = 4.16$$

If no products are formed other than biomass, the theoretical oxygen demand can be determined from Eq. (4.16) with $f = 0$; from Eq. (4.13), $w = 6$ for glucose:

$$a = {}^1/_4\,(w\,\gamma_S - c\,\gamma_B) = {}^1/_4\,(6\times 4.00 - 2.65\times 4.16) = 3.24$$

As the theoretical oxygen demand is significantly higher than that observed, formation of other products acting as electron acceptors is likely to have occurred in the culture.

Answer: Yes

4.12 Medium formulation

Using the atomic weights in Table B.1 (Appendix B), the molecular weight of $(NH_4)_2SO_4$ is 132.1 and the biomass molecular weight is 26.16. Using a basis of 1 litre, production of 25 g cells corresponds to $25/26.16 = 0.956$ gmol cells. As each gmol cells contains 0.25 gmol N, $(0.956 \times 0.25) = 0.239$ gmol N are needed from the medium for biomass synthesis. As $(NH_4)_2SO_4$ is the sole N source and each gmol $(NH_4)_2SO_4$ contains 2 gmol N, $0.239/2 = 0.120$ gmol $(NH_4)_2SO_4$ is required. Multiplying this by the molecular weight, $0.120 \times 132.1 = 15.9$ g $(NH_4)_2SO_4$ are required. The minimum concentration of $(NH_4)_2SO_4$ is therefore 15.9 g l^{-1}.

Answer: 15.9 g l^{-1}

4.13 Oxygen demand for production of recombinant protein

(a)

Recombinant protein can be considered as a product of cell culture even though it is not excreted from the cells; assume that recombinant protein is synthesised in addition to the normal *E. coli* biomass composition. From Table 4.3, the chemical formula for *E. coli* can be taken as $CH_{1.77}O_{0.49}N_{0.24}$; from Table B.2 (Appendix B), the molecular formula for glucose is $C_6H_{12}O_6$. Substituting these formulae into the general stoichiometric equation for growth and product formation, Eq. (4.13), gives:

$$C_6H_{12}O_6 + a\,O_2 + b\,NH_3 \rightarrow c\,CH_{1.77}O_{0.49}N_{0.24} + d\,CO_2 + e\,H_2O + f\,CH_{1.55}O_{0.31}N_{0.25}$$

Using the atomic weights in Table B.1 (Appendix B), the molecular weight of glucose is 180.2, the biomass molecular weight is 25.00, and the recombinant protein molecular weight is 22.03.

Assume that the biomass yield refers to the cell mass without recombinant protein. The value of the stoichiometric coefficient c can be determined from the yield $Y_{XS} = 0.48$ g g^{-1} and Eq. (4.12):

$$c = \frac{Y_{XS}\,(\text{MW substrate})}{\text{MW cells}} = \frac{0.48\ \text{g g}^{-1}\left(180.2\ \text{g gmol}^{-1}\right)}{25.00\ \text{g gmol}^{-1}} = 3.46\ \text{gmol gmol}^{-1}$$

The value of the stoichiometric coefficient f can be determined from the yield $Y_{PS} = 0.20 \times 0.48 = 0.096$ g g^{-1} and Eq. (4.14):

$$f = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW product}} = \frac{0.096\ \text{g g}^{-1}\left(180.2\ \text{g gmol}^{-1}\right)}{22.03\ \text{g gmol}^{-1}} = 0.79\ \text{gmol gmol}^{-1}$$

The ammonia requirement can be determined using an elemental balance for N.
N balance: $b = 0.24\,c + 0.25\,f$
Substituting the above values for c and f into the N balance gives $b = 0.24 \times 3.46 + 0.25 \times 0.79 = 1.03$.

Answer: 1.03 gmol gmol^{-1} glucose

(b)
The oxygen demand can be determined using an electron balance. From Table B.2 (Appendix B), the degree of reduction of glucose relative to NH_3 is $\gamma_S = 4.00$; from Table 4.3, the degree of reduction of *E. coli* relative to NH_3 is $\gamma_B = 4.07$. The degree of reduction of the recombinant protein relative to NH_3 is:

$$\gamma_P = \frac{1 \times 4 + 1.55 \times 1 - 0.31 \times 2 - 0.25 \times 3}{1} = 4.18$$

From Eq. (4.13), $w = 6$ for glucose and $j = 1$ for recombinant protein. Substituting these values into Eq. (4.16) for the theoretical oxygen demand gives:

$$a = {}^1/_4\,(w\,\gamma_S - c\,\gamma_B - f j\,\gamma_P) = {}^1/_4\,(6 \times 4.00 - 3.46 \times 4.07 - 0.79 \times 1 \times 4.18) = 1.65$$

Answer: 1.65 gmol per gmol glucose

(c)
If f in the stoichiometric equation is zero but c remains equal to 3.46, the ammonia requirement can be determined using an elemental balance for N as follows:
N balance: $b = 0.24\,c = 0.24 \times 3.46 = 0.83$
Therefore, in wild-type *E. coli* the ammonia requirement is reduced from 1.03 to 0.83 gmol gmol^{-1} glucose, a decrease of 19%. Eq. (4.16) for the oxygen requirement becomes:

$$a = {}^1/_4\,(w\,\gamma_S - c\,\gamma_B) = {}^1/_4\,(6 \times 4.00 - 3.46 \times 4.07) = 2.48$$

Therefore, the oxygen demand is increased from 1.65 to 2.48 gmol gmol^{-1} glucose, a rise of 50%.

Answer: The ammonia and oxygen requirements for wild-type *E. coli* are 0.83 gmol and 2.48 gmol per gmol glucose, respectively. These values represent a 19% reduction and a 50% increase, respectively, compared with the genetically engineered strain.

4.14 Effect of growth on oxygen demand

The stoichiometric equation for acetic acid production using cell culture must include terms for growth. Based on Eq. (4.13), the stoichiometric equation for growth and product formation is:

$$C_2H_6O + a\,O_2 + b\,NH_3 \rightarrow c\,CH_{1.8}O_{0.5}N_{0.2} + d\,CO_2 + e\,H_2O + f\,C_2H_4O_2$$

From Table B.8 (Appendix B), the molecular weights of ethanol and acetic acid are 46.1 and 60.1, respectively. From the atomic weights in Table B.1 (Appendix B), the biomass molecular weight is 24.63. The value of the stoichiometric coefficient c can be determined from the yield $Y_{XS} = 0.14$ g g^{-1} and Eq. (4.12):

$$c = \frac{Y_{XS}\,(\text{MW substrate})}{\text{MW cells}} = \frac{0.14\ \text{g g}^{-1}\left(46.1\ \text{g gmol}^{-1}\right)}{24.63\ \text{g gmol}^{-1}} = 0.26\ \text{gmol gmol}^{-1}$$

The value of the stoichiometric coefficient f can be determined from the yield $Y_{PS} = 0.92$ g g^{-1} and Eq. (4.14):

$$f = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW product}} = \frac{0.92\ \text{g g}^{-1}\left(46.1\ \text{g gmol}^{-1}\right)}{60.1\ \text{g gmol}^{-1}} = 0.71\ \text{gmol gmol}^{-1}$$

From Table B.2 (Appendix B), the degree of reduction of ethanol relative to NH_3 is $\gamma_S = 6.00$, and the degree of reduction of acetic acid relative to NH_3 is $\gamma_P = 4.00$. The degree of reduction of the biomass relative to NH_3 is :

$$\gamma_B = \frac{1\times 4 + 1.8\times 1 - 0.5\times 2 - 0.2\times 3}{1} = 4.20$$

From Eq. (4.13), $w = 2$ for ethanol and $j = 2$ for acetic acid. Substituting these values into Eq. (4.16) for the theoretical oxygen demand gives:

$$a = {}^1/_4\,(w\,\gamma_S - c\,\gamma_B - f j\,\gamma_P) = {}^1/_4\,(2\times 6.00 - 0.26\times 4.20 - 0.71\times 2\times 4.00) = 1.31$$

Therefore, with growth, 1.31 gmol oxygen are required per gmol glucose consumed, compared with 1 gmol oxygen per gmol glucose without growth. Therefore, with growth, the oxygen demand for acetic acid production is increased by 31%.

Answer: The oxygen demand is increased by 31%.

5

Energy Balances

5.1 Sensible energy change

(a)

From Table B.5 (Appendix B), C_p for *m*-cresol between 25°C and 100°C is 0.551 cal g^{-1} $°C^{-1}$. The specific enthalpy change calculated using Eq. (5.13) is:

$$\Delta h = C_p \Delta T = 0.551 \text{ cal g}^{-1} \text{ °C}^{-1} (100 - 25)\text{°C}$$

$$\Delta h = 41.3 \text{ cal g}^{-1}$$

Answer: 41.3 cal g^{-1}

(b)

From Table B.5 (Appendix B), C_p for ethylene glycol between 10°C and 20°C can be taken as 0.569 cal g^{-1} $°C^{-1}$. The specific enthalpy change calculated using Eq. (5.13) is:

$$\Delta h = C_p \Delta T = 0.569 \text{ cal g}^{-1} \text{ °C}^{-1} (10 - 20)\text{°C}$$

$$\Delta h = -5.69 \text{ cal g}^{-1}$$

Answer: –5.69 cal g^{-1}

(c)

From Table B.6 (Appendix B), C_p for succinic acid between 15°C and 120°C is given by the expression 0.248 + 0.00153 T, where T is in °C and C_p is in cal g^{-1} $°C^{-1}$. The sensible energy change is best determined from the integral of this equation between the limits T = 15°C and T = 120°C:

$$\Delta h = \int_{15°C}^{120°C} C_p \, dT = \int_{15°C}^{120°C} (0.248 + 0.00153\, T)\, dT \text{ cal g}^{-1}$$

$$\Delta h = \left(0.248\, T + \frac{0.00153}{2} T^2\right)\Bigg|_{15°C}^{120°C} \text{ cal g}^{-1}$$

$$\Delta h = 36.9 \text{ cal g}^{-1}$$

Answer: 36.9 cal g^{-1}

(d)

From Table B.3 (Appendix B), the heat capacity of air between 65°C and 150°C is given by the equation:

$$C_p = 28.94 + 0.4147 \times 10^{-2}\, T + 0.3191 \times 10^{-5}\, T^2 - 1.965 \times 10^{-9}\, T^3$$

where C_p is heat capacity in J $gmol^{-1}$ $°C^{-1}$ and T is temperature in °C. The sensible energy change can be determined by evaluating the integral of this expression between the limits T = 150°C and T = 65°C:

$$\Delta h = \int_{150°C}^{65°C} C_p \, dT = \int_{150°C}^{65°C} \left(28.94 + 0.4147 \times 10^{-2}\, T + 0.3191 \times 10^{-5}\, T^2 - 1.965 \times 10^{-9}\, T^3\right) dT \text{ J gmol}^{-1}$$

$$\Delta h = \left(28.94\,T + \frac{0.4147\times10^{-2}}{2}T^2 + \frac{0.3191\times10^{-5}}{3}T^3 - \frac{1.965\times10^{-9}}{4}T^4 \right)\Bigg|_{150°C}^{65°C} \text{ J gmol}^{-1}$$

$$\Delta h = -2500.9 \text{ J gmol}^{-1} = -2.50 \text{ kJ gmol}^{-1}$$

Answer: –2.50 kJ gmol^{-1}

5.2 Heat of vaporisation

The latent heat of vaporisation of water at 33°C is obtained from Table C.1 (Appendix C). Taking the average of the values at 32°C and 34°C, $\Delta h_v = 2423.55$ kJ kg^{-1} at 33°C. From Eq. (5.16):

$$\Delta H = M\,\Delta h_v = 20 \text{ g h}^{-1}\left(2423.55 \text{ kJ kg}^{-1}\right).\left|\frac{1 \text{ kg}}{1000 \text{ g}}\right|$$

$$\Delta H = 48.5 \text{ kJ h}^{-1}$$

Answer: 48.5 kJ h^{-1}

5.3 Steam tables

(a)

The heat of vaporisation of water at 85°C is obtained from Table C.1 (Appendix C). Taking the average of the values at 84°C and 86°C, $\Delta h_v = 2296.05$ kJ kg^{-1} at 85°C.

Answer: 2296.05 kJ kg^{-1}

(b)

From Table C.1 (Appendix C), the enthalpy of liquid water at 10°C relative to the triple point is 42.0 kJ kg^{-1}. The enthalpy of liquid water at 35°C relative to the triple point can be estimated as the average of the values in Table C.1 for 34°C and 36°C = 146.55 kJ kg^{-1}. Using the relationship on p 89, the enthalpy of water at 35°C relative to 10°C is therefore (146.55 – 42.0) kJ kg^{-1} = 104.55 kJ kg^{-1}.

Answer: 104.55 kJ kg^{-1}

(c)

The enthalpy of saturated water vapour at 40°C relative to the triple point can be read directly from Table C.1 (Appendix C) as 2574.4 kJ kg^{-1}.

Answer: 2574.4 kJ kg^{-1}

(d)

The enthalpy of superheated steam at 2.5 atm and 275°C relative to the triple point can be obtained from Table C.3 (Appendix C). To convert the pressure to kPa, from Table A.5 (Appendix A), 1 atm = 1.013×10^5 Pa. Therefore:

$$2.5 \text{ atm} = 2.5 \text{ atm}.\left|\frac{1.013\times10^5 \text{ Pa}}{1 \text{ atm}}\right|.\left|\frac{1 \text{ kPa}}{1000 \text{ Pa}}\right| = 253.3 \text{ kPa}$$

From Table C.3, the enthalpy at 100 kPa and 275°C is 3024 kJ kg^{-1}; the enthalpy at 500 kPa and 275°C is 3013 kJ kg^{-1}. Interpolating between these values gives an enthalpy of 3019.8 kJ kg^{-1} at 253.3 kPa.

Answer: 3019.8 kJ kg^{-1}

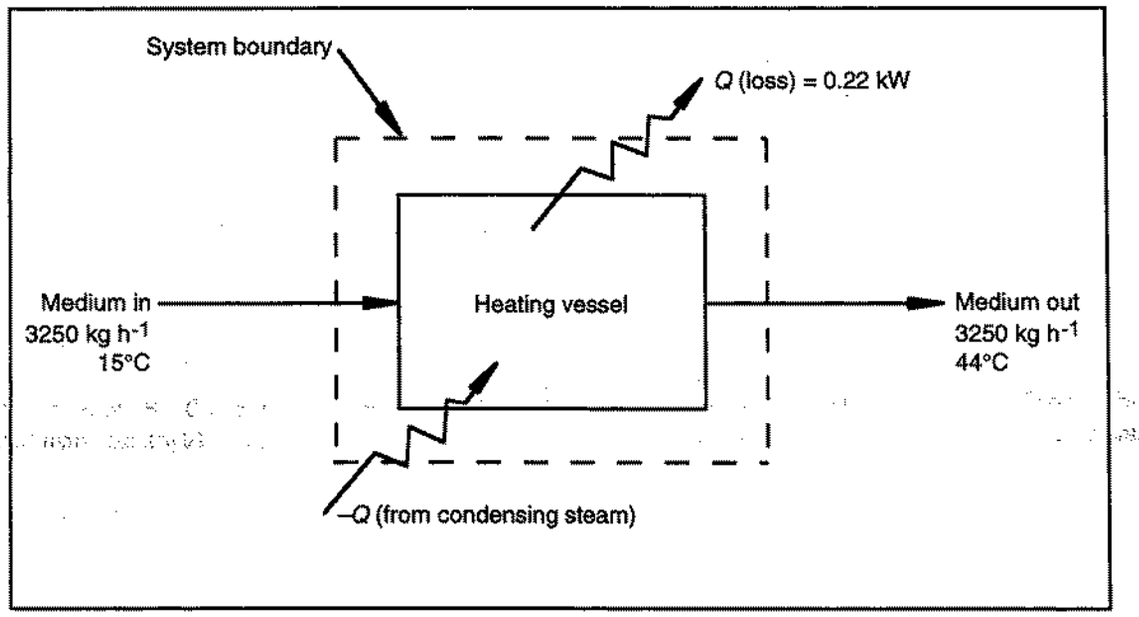
5.4 Pre-heating nutrient medium

1. *Assemble*

(i) Units

kg, h, kJ, °C

(ii) *Flow sheet*



(iii) *System boundary*
The system boundary is shown on the flow sheet.

2. *Analyse*
(i) *Assumptions*
– steady state
– no leaks
– system is homogenous
– condensate temperature is 150°C
– no shaft work
(ii) *Basis*
1 h, or 3250 kg medium in
(iii) *Reference state*
$H = 0$ for water (steam) at its triple point
$H = 0$ for medium at 15°C
(iv) *Extra data*
C_p medium = 0.9 cal g^{-1} $°C^{-1}$ = 0.9 kcal kg^{-1} $°C^{-1}$
Δh_v water at 150°C = 2113.1 kJ kg^{-1} (Table C.2, Appendix C)
1 kcal = 4.187×10^3 J (Table A.7, Appendix A)
1 W = 1 J s^{-1} (Table A.8, Appendix A); therefore, 1 kW = 1 kJ s^{-1}
(v) *Mass balance*
The mass balance is already complete.
(vi) *Energy-balance equation*
At steady state, Eq. (5.9) applies:

$$\underset{\text{input streams}}{\Sigma (M\,h)} - \underset{\text{output streams}}{\Sigma (M\,h)} - Q + W_S = 0$$

3. *Calculate*
(i) *Identify terms in the energy-balance equation*
$W_S = 0$. There are two components for the heat term, Q: Q_{loss} and Q from the condensing steam. With symbol MD = medium, the energy-balance equation becomes:

$$(M\,h)_{\text{MD in}} - (M\,h)_{\text{MD out}} - Q - Q_{loss} = 0$$

$(M\,h)_{\text{MD in}} = 0$ (reference state)
$(M\,h)_{\text{MD out}}$ at 44°C is calculated as a sensible energy change from $H = 0$ at 15°C using Eq. (5.13):

$$(M\,h)_{\text{MD out}} = M\,C_p\,\Delta T = 3250\,\text{kg}\,(0.9\,\text{kcal kg}^{-1}\,°\text{C}^{-1})(44 - 15)°\text{C} = 8.483 \times 10^4\,\text{kcal}$$

Converting to kJ:

$$(M\,h)_{\text{MD out}} = 8.483\times10^{4}\ \text{kcal}\cdot\left|\frac{4.187\times10^{3}\ \text{J}}{1\ \text{kcal}}\right|\cdot\left|\frac{1\ \text{kJ}}{1000\ \text{J}}\right| = 3.55\times10^{5}\ \text{kJ}$$

The rate of heat loss is 0.22 kW. Converting to kJ h^{-1}:

$$0.22\ \text{kW} = 0.22\ \text{kW}\cdot\left|\frac{1\ \text{kJ s}^{-1}}{1\ \text{kW}}\right|\cdot\left|\frac{3600\ \text{s}}{1\ \text{h}}\right| = 792\ \text{kJ h}^{-1}$$

Therefore, on the basis of 1 h, $Q_{\text{loss}} = 792$ kJ. Substituting values into the energy-balance equation gives:

$$0 - 3.55\times10^{5}\ \text{kJ} - Q - 792\ \text{kJ} = 0$$

$$Q = -3.56\times10^{5}\ \text{kJ}$$

Q has a negative value which is consistent with the sign conventions outlined on pp 87–88: heat must be supplied to the system from the surroundings. This heat is provided as the latent heat of vaporisation as saturated steam at 150°C condenses. The enthalpy change from this change of phase is calculated using Eq. (5.16) and must be equal to $-Q$.

$$3.56\times10^{5}\ \text{kJ} = M_{\text{steam}}\,\Delta h_{\text{v}} = M_{\text{steam}}\,(2113.1\ \text{kJ kg}^{-1})$$

$$M_{\text{steam}} = 168\ \text{kg}$$

4. *Finalise*

Answer: 168 kg

5.5 Production of glutamic acid

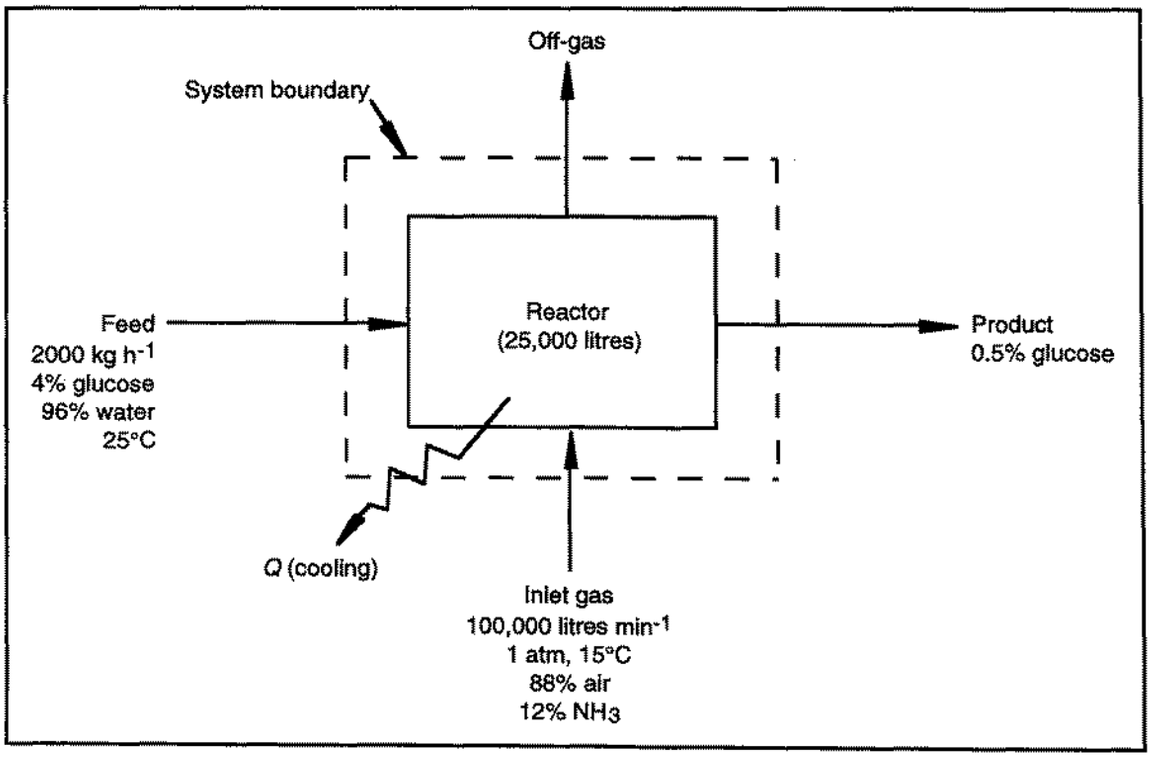
(a)

1. *Assemble*

(i) Units

kg, h, kJ, °C

(ii) *Flow sheet*



(iii) *System boundary*

The system boundary is shown on the flow sheet.

(iv) *Reaction equation*

$$C_6H_{12}O_6 + NH_3 + 1^1/_2\,O_2 \rightarrow C_5H_9NO_4 + CO_2 + 3\,H_2O$$

2. *Analyse*

(i) *Assumptions*

– steady state
– no leaks
– system is homogenous
– solutions are ideal
– inlet air and off-gas are dry
– all excess NH_3 is dissolved in the aqueous phase
– all CO_2 produced leaves in the off-gas
– negligible sensible heat change
– no evaporation
– no shaft work

(ii) *Basis*

1 h, or 2000 kg feed

(iii) *Reference state*

$H = 0$ for water at its triple point
$H = 0$ for feed at 25°C

(iv) *Extra data*

Molecular weights (Table B.1, Appendix B): glucose = 180.2
$NH_3 = 17.0$
$O_2 = 32.0$
$N_2 = 28.0$
glutamic acid = 147.1
$CO_2 = 44.0$
$H_2O = 18.0$
air = 28.8 (see Problem 2.9, Chapter 2)

Ideal gas constant (Table 2.5): $R = 0.082057$ l atm K^{-1} $gmol^{-1}$
Composition of air (p 17): 21% O_2, 79% N_2 by volume
Heats of combustion (Table B.8, Appendix B):

$$\Delta h_c^o \text{ glucose} = -2805.0 \text{ kJ gmol}^{-1}$$

$$\Delta h_c^o \text{ NH}_3 = -382.6 \text{ kJ gmol}^{-1}$$

$$\Delta h_c^o \text{ glutamic acid} = -2244.1 \text{ kJ gmol}^{-1}$$

(v) *Compounds involved in reaction*

Glucose, ammonia, oxygen, glutamic acid, carbon dioxide and water are involved in the reaction.

(vi) *Mass-balance equations*

For glucose, ammonia, oxygen, glutamic acid, carbon dioxide and water, the appropriate mass-balance equation is Eq. (4.2):

$$\text{mass in} + \text{mass generated} = \text{mass out} + \text{mass consumed}$$

For N_2 and total mass, the appropriate mass-balance equation is Eq. (4.3):

$$\text{mass in} = \text{mass out}$$

(vii) *Energy-balance equation*

The modified energy-balance equation, Eq. (5.26), applies:

$$-\Delta H_{rxn} - M_v\,\Delta h_v - Q + W_s = 0$$

3. *Calculate*

(i) *Mass balance*

As gas compositions are normally given in vol%, 88,000 litres air and 12,000 litres NH_3 enter the reactor every min. On a basis of 1 h, the volume of air in is $88{,}000 \times 60 = 5.28 \times 10^6$ litres; the volume of NH_3 in is $12{,}000 \times 60 = 7.20 \times 10^5$ litres. Using the known composition of air, the volume of O_2 in $= 0.21 \times 5.28 \times 10^6 = 1.11 \times 10^6$ litres; the volume of N_2 in $= 0.79 \times 5.28 \times 10^6 = 4.17 \times 10^6$ litres. Converting these gas volumes to moles using the ideal gas law, Eq. (2.32), with the temperature converted from °C to degrees Kelvin using Eq. (2.24):

$$\text{Moles of } O_2 \text{ in} = \frac{pV}{RT} = \frac{1\text{ atm}\left(1.11\times10^6\text{ l}\right)}{0.0820571\text{ atm K}^{-1}\text{ gmol}^{-1}\,(15+273.15)\text{ K}} = 4.69\times10^4\text{ gmol}$$

$$\text{Moles of } N_2 \text{ in} = \frac{pV}{RT} = \frac{1\text{ atm}\left(4.17\times10^6\text{ l}\right)}{0.0820571\text{ atm K}^{-1}\text{ gmol}^{-1}\,(15+273.15)\text{ K}} = 1.76\times10^5\text{ gmol}$$

$$\text{Moles of } NH_3 \text{ in} = \frac{pV}{RT} = \frac{1\text{ atm}\left(7.20\times10^5\text{ l}\right)}{0.0820571\text{ atm K}^{-1}\text{ gmol}^{-1}\,(15+273.15)\text{ K}} = 3.05\times10^4\text{ gmol}$$

These molar quantities can now be converted to masses using the molecular weights:

$$\text{Mass of } O_2 \text{ in} = 4.69\times10^4\text{ gmol}\,.\left|\frac{32.0\text{ g}}{1\text{ gmol}}\right|.\left|\frac{1\text{ kg}}{1000\text{ g}}\right| = 1500.8\text{ kg}$$

$$\text{Mass of } N_2 \text{ in} = 1.76\times10^5\text{ gmol}\,.\left|\frac{28.0\text{ g}}{1\text{ gmol}}\right|.\left|\frac{1\text{ kg}}{1000\text{ g}}\right| = 4928.0\text{ kg}$$

$$\text{Mass of } NH_3 \text{ in} = 3.05\times10^4\text{ gmol}\,.\left|\frac{17.0\text{ g}}{1\text{ gmol}}\right|.\left|\frac{1\text{ kg}}{1000\text{ g}}\right| = 518.5\text{ kg}$$

Therefore, the total mass of inlet gas is 1500.8 + 4928.0 + 518.5 = 6947.3 kg.

The calculation tables below show all known quantities in kg. The total mass of off-gas is denoted *G*; the total mass of product is denoted *P*.

Stream	*In*							
	Glucose	*NH_3*	*O_2*	*N_2*	*Gluconic acid*	*CO_2*	*H_2O*	*Total*
Feed	80	0	0	0	0	0	1920	2000
Inlet gas	0	518.5	1500.8	4928.0	0	0	0	6947.3
Off-gas	–	–	–	–	–	–	–	–
Product	–	–	–	–	–	–	–	–
Total	80	518.5	1500.8	4928.0	0	0	1920	8947.3

Stream	*Out*							
	Glucose	*NH_3*	*O_2*	*N_2*	*Gluconic acid*	*CO_2*	*H_2O*	*Total*
Feed	–	–	–	–	–	–	–	–
Inlet gas	–	–	–	–	–	–	–	–
Off-gas	0	0	?	?	0	?	0	*G*
Product	0.005 *P*	?	0	0	?	0	?	*P*
Total	0.005 *P*	?	?	?	?	?	?	*G* + *P*

N_2 balance

N_2 is a tie component.

$$4928.0\text{ kg } N_2 \text{ in} = N_2 \text{ out}$$

$$N_2 \text{ out} = 4928.0\text{ kg}$$

Glucose balance

80 kg glucose in + 0 kg glucose generated = 0.005 *P* kg glucose out + glucose consumed

$$\text{Glucose consumed} = (80 - 0.005\,P)\text{ kg}$$

Converting the glucose consumed to molar terms:

$$\text{Glucose consumed} = (80-0.005\,P)\text{ kg}\,.\left|\frac{1\text{ kgmol}}{180.2\text{ kg}}\right| = \left(0.444-2.775\times10^{-5}\,P\right)\text{kgmol}$$

From the reaction stoichiometry, conversion of this number of kgmol glucose requires the same number of kgmol NH_3 and 1.5 × the number of kgmol O_2. Converting these molar quantities to masses:

$$NH_3 \text{ consumed} = \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} = \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} \cdot \left|\frac{17.0\,\text{kg}}{1\,\text{kgmol}}\right|$$

$$NH_3 \text{ consumed} = (7.548 - 4.718 \times 10^{-4}\, P)\ \text{kg}$$

$$O_2 \text{ consumed} = 1.5 \times \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} = 1.5 \times \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} \cdot \left|\frac{32.0\,\text{kg}}{1\,\text{kgmol}}\right|$$

$$O_2 \text{ consumed} = (21.312 - 1.332 \times 10^{-3}\, P)\ \text{kg}$$

Similarly, expressions for the masses of glutamic acid, CO_2 and water generated can be determined from the stoichiometry:

$$\text{Glutamic acid generated} = \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} = \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} \cdot \left|\frac{147.1\,\text{kg}}{1\,\text{kgmol}}\right|$$

$$\text{Glutamic acid generated} = (65.312 - 4.082 \times 10^{-3}\, P)\ \text{kg}$$

$$CO_2 \text{ generated} = \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} = \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} \cdot \left|\frac{44.0\,\text{kg}}{1\,\text{kgmol}}\right|$$

$$CO_2 \text{ generated} = (19.536 - 1.221 \times 10^{-3}\, P)\ \text{kg}$$

$$\text{Water generated} = 3 \times \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} = 3 \times \left(0.444 - 2.775 \times 10^{-5}\, P\right) \text{kgmol} \cdot \left|\frac{18.0\,\text{kg}}{1\,\text{kgmol}}\right|$$

$$\text{Water generated} = (23.976 - 1.499 \times 10^{-3}\, P)\ \text{kg}$$

O_2 balance

$$1500.8 \text{ kg } O_2 \text{ in} + 0 \text{ kg } O_2 \text{ generated} = O_2 \text{ out} + (21.312 - 1.332 \times 10^{-3}\, P) \text{ kg } O_2 \text{ consumed}$$

$$O_2 \text{ out} = (1500.8 - (21.312 - 1.332 \times 10^{-3}\, P))\ \text{kg}$$

$$O_2 \text{ out} = (1479.5 + 1.332 \times 10^{-3}\, P)\ \text{kg}$$

CO_2 balance

$$0 \text{ kg } CO_2 \text{ in} + (19.536 - 1.221 \times 10^{-3}\, P) \text{ kg } CO_2 \text{ generated} = CO_2 \text{ out} + 0 \text{ kg } CO_2 \text{ consumed}$$

$$CO_2 \text{ out} = (19.536 - 1.221 \times 10^{-3}\, P)\ \text{kg}$$

Adding together the masses of N_2, O_2 and CO_2 out gives the total mass of off-gas, G:

$$G = 4928.0 \text{ kg } N_2 + (1479.5 + 1.332 \times 10^{-3}\, P) \text{ kg } O_2 + (19.536 - 1.221 \times 10^{-3}\, P) \text{ kg } CO_2$$

$$G = (6427.0 + 1.11 \times 10^{-4}\, P)\ \text{kg}$$

Total mass balance

$$8947.3 \text{ kg total mass in} = (G + P) \text{ kg total mass out}$$

Substituting the above expression for G into the total mass balance:

$$8947.3 \text{ kg} = (6427.0 + 1.11 \times 10^{-4}\, P + P)\ \text{kg}$$

$$2520.3 \text{ kg} = 1.000\, P \text{ kg}$$

$$P = 2520.3 \text{ kg}$$

Therefore, from the total mass balance:

$$G = (8947.3 - 2520.3)\ \text{kg}$$

$$G = 6427.0 \text{ kg}$$

Substituting the result for P into the glucose, O_2 and CO_2 balances gives:

$$\text{Glucose consumed} = (80 - 0.005\,P)\ \text{kg} = 67.40\ \text{kg}$$

$$\text{Glucose out} = 0.005\,P\ \text{kg} = 12.60\ \text{kg}$$

$$O_2\ \text{consumed} = (21.312 - 1.332 \times 10^{-3}\,P)\ \text{kg} = 17.95\ \text{kg}$$

$$O_2\ \text{out} = (1479.5 + 1.332 \times 10^{-3}\,P)\ \text{kg} = 1482.9\ \text{kg}$$

$$CO_2\ \text{out} = CO_2\ \text{generated} = (19.536 - 1.221 \times 10^{-3}\,P)\ \text{kg} = 16.46\ \text{kg}$$

Using the result for P to evaluate the masses of the other reactants and products involved in the reaction:

$$NH_3\ \text{consumed} = (7.548 - 4.718 \times 10^{-4}\,P)\ \text{kg} = 6.36\ \text{kg}$$

$$\text{Glutamic acid generated} = (65.312 - 4.082 \times 10^{-3}\,P)\ \text{kg} = 55.02\ \text{kg}$$

$$\text{Water generated} = (23.976 - 1.499 \times 10^{-3}\,P)\ \text{kg} = 20.20\ \text{kg}$$

These results can be used directly in the energy balance for evaluation of the cooling requirements. However, completion of the mass balance allows the calculations to be checked.

NH_3 balance

$$518.5\ \text{kg}\ NH_3\ \text{in} + 0\ \text{kg}\ NH_3\ \text{generated} = NH_3\ \text{out} + 6.36\ \text{kg}\ NH_3\ \text{consumed}$$

$$NH_3\ \text{out} = 512.14\ \text{kg}$$

Water balance

$$1920\ \text{kg water in} + 20.20\ \text{kg water generated} = \text{water out} + 0\ \text{kg water consumed}$$

$$\text{Water out} = 1940.20\ \text{kg}$$

Glutamic acid balance

$$0\ \text{kg glutamic acid in} + 55.02\ \text{kg glutamic acid generated} = \text{glutamic acid out} + 0\ \text{kg glutamic acid consumed}$$

$$\text{Glutamic acid out} = 55.02\ \text{kg}$$

The Out side of the mass-balance table can now be completed with all quantities in kg.

Stream	*Out*							
	Glucose	*NH_3*	*O_2*	*N_2*	*Gluconic acid*	*CO_2*	*H_2O*	*Total*
Feed	–	–	–	–	–	–	–	–
Inlet gas	–	–	–	–	–	–	–	–
Off-gas	0	0	1482.9	4928.0	0	16.46	0	6427.4
Product	12.60	512.14	0	0	55.02	0	1940.20	2520
Total	12.60	512.14	1482.9	4928.0	55.02	16.46	1940.20	8947.3

All columns and rows of the completed table add up correctly to within round-off error.

(ii) *Energy balance*

$W_s = 0$; $M_v = 0$. Therefore, the energy-balance equation becomes:

$$-\Delta H_{rxn} - Q = 0$$

The heat of reaction is evaluated using Eq. (5.20). As the heat of combustion of CO_2 and H_2O is zero:

$$\Delta H_{rxn} = \left(n\,\Delta h_c^\circ\right)_G + \left(n\,\Delta h_c^\circ\right)_A - \left(n\,\Delta h_c^\circ\right)_{GA}$$

where G = glucose, A = NH_3 and GA = glutamic acid. The n in this equation are the moles of reactant or product involved in the reaction. Converting the masses of reactants and products consumed or generated to moles:

$$\text{Glucose consumed} = 67.40\ \text{kg} = 67.40\ \text{kg} \cdot \left|\frac{1000\ \text{g}}{1\ \text{kg}}\right| \cdot \left|\frac{1\ \text{gmol}}{180.2\ \text{g}}\right| = 374\ \text{gmol}$$

As NH_3 and glutamic acid are involved in the reaction in stoichiometric quantities, 374 gmol NH_3 are consumed and 374 gmol glutamic acid are produced. Substituting these quantities into the heat of reaction equation gives:

$$\Delta H_{rxn} = 374\ \text{gmol}\left(-2805.0\ \text{kJ gmol}^{-1}\right) + 374\ \text{gmol}\left(-382.6\ \text{kJ gmol}^{-1}\right) - 374\ \text{gmol}\left(-2244.1\ \text{kJ gmol}^{-1}\right)$$

$$\Delta H_{rxn} = -3.53 \times 10^5\ \text{kJ}$$

Substituting this result into the energy-balance equation:

$$3.53 \times 10^5\ \text{kJ} - Q = 0$$

$$Q = 3.53 \times 10^5\ \text{kJ}$$

From the sign conventions outlined on pp 87–88, Q positive indicates that heat must be removed from the system.

Answer: 3.53×10^5 kJ h^{-1}

(b)
If cooling were not provided, the heat of reaction would be absorbed as sensible heat by the streams passing through the reactor. For a rough calculation of the effect of this heat on the temperature of the reactor, assume that the 3.53×10^5 kJ h^{-1} is absorbed by 2000 kg h^{-1} aqueous medium and 6947.3 kg h^{-1} gas. Assume that the heat capacity of the aqueous medium is close to that of water = 75.4 J $gmol^{-1}$ $°C^{-1}$ (Table B.3, Appendix B) = 75.4 kJ $kgmol^{-1}$ $°C^{-1}$, and that the heat capacity of the gas stream is equal to that of air = approx. 29 J $gmol^{-1}$ $°C^{-1}$ (Table B.3, Appendix B) = 29 kJ $kgmol^{-1}$ $°C^{-1}$. From Eq. (5.12):

$$\Delta T = \frac{\Delta H}{(M C_p)_{\text{liquid}} + (M C_p)_{\text{gas}}}$$

$$T = \frac{3.53 \times 10^5\ \text{kJ h}^{-1}}{\left(2000\ \text{kg h}^{-1}\right)\left(75.4\ \text{kJ kgmol}^{-1}\ °\text{C}^{-1}\right).\left|\frac{1\ \text{kgmol}}{18.0\ \text{kg}}\right| + \left(6947.3\ \text{kg h}^{-1}\right)\left(29\ \text{kJ kgmol}^{-1}\ °\text{C}^{-1}\right).\left|\frac{1\ \text{kgmol}}{28.8\ \text{kg}}\right|} = 23°\text{C}$$

As a temperature rise of 23°C in the reactor would not be well tolerated by most commercial organisms, provision of adequate cooling for this reaction is an important consideration. Assuming that the usual temperature for the reaction is 25°C, the temperature without cooling would increase to (25 + 23)°C = 48°C.

Answer: 48°C

5.6 Bacterial production of alginate

Alginate production at a rate of 5 kg h^{-1} requires:

$$\frac{5\ \text{kg h}^{-1}}{4\ \text{kg kg}^{-1}} = 1.25\ \text{kg}\ O_2\ \text{h}^{-1}$$

Converting this quantity to gmol using the molecular weight of O_2 = 32.0 (Table B.1, Appendix B):

$$O_2\ \text{required} = 1.25\ \text{kg h}^{-1}.\left|\frac{1000\ \text{g}}{\text{kg}}\right|.\left|\frac{1\ \text{gmol}}{32.0\ \text{g}}\right| = 39.06\ \text{gmol h}^{-1}$$

The heat of reaction for aerobic metabolism is approximately –460 kJ $gmol^{-1}$ O_2 (p 100). Therefore, the heat of reaction for alginate production is:

$$\Delta H_{rxn} = -460\ \text{kJ gmol}^{-1}\left(39.06\ \text{gmol h}^{-1}\right) = -1.80 \times 10^4\ \text{kJ h}^{-1}$$

This result can be used in the modified energy-balance equation, Eq. (5.26):

$$-\Delta H_{rxn} - M_v \Delta h_v - Q + W_s = 0$$

with W_s = 1.5 kW and M_v = 0 (no evaporation). From Table A.8 (Appendix A), 1 W = 1 J s^{-1}; therefore 1 kW = 1 kJ s^{-1} and:

$$1.80 \times 10^4 \text{ kJ h}^{-1} - 0 - Q + 1.5 \text{ kJ s}^{-1} \cdot \left| \frac{3600 \text{ s}}{1 \text{ h}} \right| = 0$$

$$Q = 2.3 \times 10^4 \text{ kJ h}^{-1}$$

From the sign conventions outlined on pp 87–88, Q positive indicates that heat must be removed from the system.

Answer: 2.3×10^4 kJ h^{-1}

5.7 Acid fermentation

From Tables B.8 and B.1 (Appendix B), the molecular formulae and molecular weights are:
sucrose = $C_{12}H_{22}O_{11}$: MW = 342.3
propionic acid = $C_3H_6O_2$: MW = 74.1
acetic acid = $C_2H_4O_2$: MW = 60.1
butyric acid = $C_4H_8O_2$: MW = 88.1
lactic acid = $C_3H_6O_3$: MW = 90.1
From p 75, the biomass molecular formula can be taken as the average, $CH_{1.8}O_{0.5}N_{0.2}$. From the end of Table B.8 (Appendix B), the molecular weight of the biomass is 25.9.

The reaction equation can be obtained by modifying Eq. (4.13) for anaerobic growth and product formation:

$$C_{12}H_{22}O_{11} + b\,NH_3 \rightarrow c\,CH_{1.8}O_{0.5}N_{0.2} + d\,CO_2 + e\,H_2O + f_1\,C_3H_6O_2 + f_2\,C_2H_4O_2 + f_3\,C_4H_8O_2 + f_4\,C_3H_6O_3$$

The biomass yield from substrate $Y_{XS} = 0.12$ g g^{-1}. This value can be used to determine the stoichiometric coefficient c using Eq. (4.12):

$$c = \frac{Y_{XS}\,(\text{MW substrate})}{\text{MW cells}} = \frac{0.12 \text{ g g}^{-1}\,(342.3)}{25.9} = 1.59$$

The coefficients f_1, f_2, f_3 and f_4 can be determined similarly using the product yields and Eq. (4.14):

$$f_1 = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW propionic acid}} = \frac{0.40 \text{ g g}^{-1}\,(342.3)}{74.1} = 1.85$$

$$f_2 = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW acetic acid}} = \frac{0.20 \text{ g g}^{-1}\,(342.3)}{60.1} = 1.14$$

$$f_3 = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW butyric acid}} = \frac{0.05 \text{ g g}^{-1}\,(342.3)}{88.1} = 0.19$$

$$f_4 = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW lactic acid}} = \frac{0.034 \text{ g g}^{-1}\,(342.3)}{90.1} = 0.13$$

Of the remaining coefficients b, d and e, because CO_2 and H_2O do not figure in heat of reaction calculations as their heat of combustion = 0, only b need be determined. This can be done using an elemental balance on N.
N balance: $b = 0.2\,c = 0.2 \times 1.59 = 0.32$.

To calculate the heat of reaction, the heats of combustion of the reactants and products are required from Table B.8 (Appendix B):

Δh_c° sucrose = –5644.9 kJ gmol^{-1}

Δh_c° NH_3 = –382.6 kJ gmol^{-1}

Δh_c° biomass = –552 kJ gmol^{-1}

Δh_c° propionic acid = –1527.3 kJ gmol^{-1}

Δh_c° acetic acid = –874.2 kJ gmol^{-1}

Δh_c° butyric acid = –2183.6 kJ gmol^{-1}

Δh_c° lactic acid = –1368.3 kJ gmol^{-1}

The heat of reaction is determined using Eq. (5.20). As the heat of combustion of CO_2 and H_2O is zero:

$$\Delta H_{rxn} = \left(n\,\Delta h_c^o\right)_S + \left(n\,\Delta h_c^o\right)_A - \left(n\,\Delta h_c^o\right)_B - \left(n\,\Delta h_c^o\right)_{PA} - \left(n\,\Delta h_c^o\right)_{AA} - \left(n\,\Delta h_c^o\right)_{BA} - \left(n\,\Delta h_c^o\right)_{LA}$$

where S = sucrose, A = NH_3, B = biomass, PA = propionic acid, AA = acetic acid, BA = butyric acid, and LA = lactic acid. Using a basis of 1 gmol sucrose, the n in this equation are the stoichiometric coefficients. Substituting values:

$$\begin{aligned}\Delta H_{rxn} = {} & 1\text{ gmol}\left(-5644.9\text{ kJ gmol}^{-1}\right) + 0.32\text{ gmol}\left(-382.6\text{ kJ gmol}^{-1}\right) - 1.59\text{ gmol}\left(-552\text{ kJ gmol}^{-1}\right) \\ & - 1.85\text{ gmol}\left(-1527.3\text{ kJ gmol}^{-1}\right) - 1.14\text{ gmol}\left(-874.2\text{ kJ gmol}^{-1}\right) - 0.19\text{ gmol}\left(-2183.6\text{ kJ gmol}^{-1}\right) \\ & - 0.13\text{ gmol}\left(-1368.3\text{ kJ gmol}^{-1}\right)\end{aligned}$$

$$\Delta H_{rxn} = -474.8\text{ kJ}$$

This ΔH_{rxn} was determined on the basis of 1 gmol sucrose. For 30 kg sucrose consumed over a period of 10 d:

$$\Delta H_{rxn} = 30\text{ kg} \cdot \left|\frac{1000\text{ g}}{1\text{ kg}}\right| \cdot \left|\frac{1\text{ gmol}}{342.3\text{ g}}\right| \left(-474.8\text{ kJ gmol}^{-1}\right) = -4.16 \times 10^4\text{ kJ}$$

The cooling requirements are determined using the modified energy-balance equation, Eq. (5.26). For no evaporation and no shaft work, $M_v = W_s = 0$, so that:

$$Q = -\Delta H_{rxn} = 4.16 \times 10^4\text{ kJ}$$

From the sign conventions outlined on pp 87–88, Q positive means that heat must be removed from the system.

Answer: 4.16×10^4 kJ

5.8 Ethanol fermentation

From Table B.2 (Appendix B), the molecular formula for glucose is $C_6H_{12}O_6$ and the molecular formula for ethanol is C_2H_6O. From Eq. (4.13), the stoichiometric equation under anaerobic conditions is:

$$C_6H_{12}O_6 + b\,NH_3 \rightarrow c\,CH_{1.75}O_{0.58}N_{0.18} + d\,CO_2 + e\,H_2O + f\,C_2H_6O$$

From Table B.8 (Appendix B), the molecular weight of ethanol is 46.1. From the atomic weights in Table B.1 (Appendix B), the molecular weight of glucose is 180.2 and the biomass formula weight is 25.58. Taking into account the 8% ash:

$$\text{Biomass molecular weight} = \frac{25.58}{0.92} = 27.80$$

The value of the stoichiometric coefficient f can be determined from the yield $Y_{PS} = 0.45$ g g^{-1} and Eq. (4.14):

$$f = \frac{Y_{PS}\,(\text{MW substrate})}{\text{MW product}} = \frac{0.45\text{ g g}^{-1}\left(180.2\text{ g gmol}^{-1}\right)}{46.1\text{ g gmol}^{-1}} = 1.76\text{ gmol gmol}^{-1}$$

The other coefficients can be determined using elemental balances.
C balance: $6 = c + d + 2f = c + d + 2 \times 1.76 \rightarrow c = 2.48 - d$
H balance: $12 + 3b = 1.75c + 2e + 6f = 1.75c + 2e + 6 \times 1.76 \rightarrow 1.44 + 3b = 1.75c + 2e$
O balance: $6 = 0.58c + 2d + e + f = 0.58c + 2d + e + 1 \times 1.76 \rightarrow 4.24 = 0.58c + 2d + e$
N balance: $b = 0.18c$
Substituting the expression for c from the C balance into the N balance:

$$b = 0.18\,(2.48 - d) = 0.45 - 0.18\,d$$

Substituting this and the results from the C and N balances into the H balance:

$$1.44 + 3\,(0.45 - 0.18\,d) = 1.75\,(2.48 - d) + 2\,e$$

$$1.21\,d - 1.55 = 2\,e$$

$$e = 0.61\,d - 0.78$$

Substituting the expressions for c, b and e into the O balance:

$$4.24 = 0.58\,(2.48 - d) + 2\,d + (0.61\,d - 0.78)$$

$$3.58 = 2.03\,d$$

$$d = 1.76$$

Substituting this value for d into the expressions for the other coefficients gives $c = 0.72$, $b = 0.13$ and $e = 0.29$. The completed stoichiometric equation is therefore:

$$C_6H_{12}O_6 + 0.13\,NH_3 \rightarrow 0.72\,CH_{1.75}O_{0.58}N_{0.18} + 1.76\,CO_2 + 0.29\,H_2O + 1.76\,C_2H_6O$$

Using a basis of 1 h, 0.4 kg ethanol are produced. Converting this to moles:

$$\text{Moles ethanol produced} = 0.4\text{ kg} \cdot \left|\frac{1000\text{ g}}{1\text{ kg}}\right| \cdot \left|\frac{1\text{ gmol}}{46.1\text{ g}}\right| = 8.68\text{ gmol}$$

From stoichiometry:

$$\text{Moles glucose consumed} = 8.68\text{ gmol} \times \frac{1}{1.76} = 4.93\text{ gmol}$$

$$\text{Moles } NH_3 \text{ consumed} = 8.68\text{ gmol} \times \frac{0.13}{1.76} = 0.64\text{ gmol}$$

$$\text{Moles biomass produced} = 8.68\text{ gmol} \times \frac{0.72}{1.76} = 3.55\text{ gmol}$$

The heats of combustion from Table B.8 (Appendix B) are:

Δh_c° glucose $= -2805.0$ kJ gmol^{-1}

Δh_c° $NH_3 = -382.6$ kJ gmol^{-1}

Δh_c° ethanol $= -1366.8$ kJ gmol^{-1}

From p 101, the heat of combustion of yeast can be taken as -21.2 kJ g^{-1}. The heat of reaction is determined using Eq. (5.20). As the heat of combustion of CO_2 and H_2O is zero:

$$\Delta H_{rxn} = \left(n\,\Delta h_c^\circ\right)_G + \left(n\,\Delta h_c^\circ\right)_A - \left(n\,\Delta h_c^\circ\right)_B - \left(n\,\Delta h_c^\circ\right)_E$$

where G = glucose, A = NH_3, B = biomass and E = ethanol. The n in this equation are the actual moles of reactants and products consumed or produced. Substituting values gives:

$$\Delta H_{rxn} = 4.93\text{ gmol}\left(-2805.0\text{ kJ gmol}^{-1}\right) + 0.64\text{ gmol}\left(-382.6\text{ kJ gmol}^{-1}\right) - 3.55\text{ gmol}\left(-21.2\text{ kJ g}^{-1}\right) \cdot \left|\frac{27.80\text{ g}}{1\text{ gmol}}\right| - 8.68\text{ gmol}\left(-1366.8\text{ kJ gmol}^{-1}\right)$$

$$\Delta H_{rxn} = -117.5\text{ kJ}$$

Using the modified energy-balance equation, Eq. (5.26), with $M_v = 0$ and $W_s = 0$:

$$Q = -\Delta H_{rxn} = 117.5\text{ kJ}$$

From the sign conventions outlined on pp 87–88, Q positive means that heat must be removed from the system; in this case, 117.5 kJ h^{-1} is used to raise the temperature of 2.5 l h^{-1} water from 10°C. The sensible heat change of the water can be calculated from Eq. (5.12). Using a value of 1 kg l^{-1} for the density of water, C_p for water = 75.4 J gmol^{-1} °C^{-1} (Table B.3, Appendix B), and the molecular weight of water = 18.0 (Table B.1, Appendix B):

$$\Delta T = \frac{Q}{M\,C_p} = \frac{117.5\text{ kJ h}^{-1}}{\left(2.5\text{ l h}^{-1} \cdot \left|\frac{1\text{ kg}}{1\text{ l}}\right|\right)\left(75.4\text{ J gmol}^{-1}\text{ °C}^{-1} \cdot \left|\frac{1\text{ gmol}}{18.0\text{ g}}\right| \cdot \left|\frac{1000\text{ g}}{1\text{ kg}}\right| \cdot \left|\frac{1\text{ kJ}}{1000\text{ J}}\right|\right)} = 11.2\text{°C}$$

The final temperature of the water is therefore 10°C + 11.2°C = 21.2°C

Answer: 21.2°C

5.9 Production of bakers' yeast

From Table B.8 (Appendix B), the molecular formula for sucrose is $C_{12}H_{22}O_{11}$. Therefore, from Eq. (4.4), the stoichiometric equation for aerobic cell growth is:

$$C_{12}H_{22}O_{11} + a\,O_2 + b\,NH_3 \rightarrow c\,CH_{1.83}O_{0.55}N_{0.17} + d\,CO_2 + e\,H_2O$$

Using the atomic weights in Table B.1 (Appendix B), the molecular weight of sucrose is 342.3 and the biomass formula weight is 25.04. Taking into account the 5% ash:

$$\text{Biomass molecular weight} = \frac{25.04}{0.95} = 26.36$$

The degree of reduction of the biomass relative to NH_3 is:

$$\gamma_B = \frac{1\times4 + 1.83\times1 - 0.55\times2 - 0.17\times3}{1} = 4.22$$

The degree of reduction of sucrose relative to NH_3 is:

$$\gamma_S = \frac{12\times4 + 22\times1 - 11\times2}{12} = 4.00$$

The value of the stoichiometric coefficient c can be determined from the yield $Y_{XS} = 0.5$ g g^{-1} and Eq. (4.12):

$$c = \frac{Y_{XS}\,(\text{MW substrate})}{\text{MW cells}} = \frac{0.5\text{ g g}^{-1}\left(342.3\text{ g gmol}^{-1}\right)}{26.36\text{ g gmol}^{-1}} = 6.49\text{ gmol gmol}^{-1}$$

The oxygen requirements can be determined from Eq. (4.16) with $f = 0$; from Eq (4.13), $w = 12$ for sucrose:

$$a = {}^1/_4\,(w\,\gamma_S - c\,\gamma_B) = {}^1/_4\,(12\times4.00 - 6.49\times4.22) = 5.15$$

Therefore, 5.15 gmol O_2 are required for each 6.49 gmol biomass produced; this corresponds to 0.79 gmol O_2 per gmol biomass. Converting this oxygen demand to mass terms using the molecular weight of oxygen = 32.0 from Table B.1 (Appendix B):

$$0.79\text{ gmol O}_2\text{ per gmol biomass} = \frac{0.79\text{ gmol O}_2}{1\text{ gmol biomass}} \cdot \left|\frac{32.0\text{ g O}_2}{1\text{ gmol O}_2}\right| \cdot \left|\frac{1\text{ gmol biomass}}{26.36\text{ g biomass}}\right| = 0.96\text{ g g}^{-1}$$

The specific growth rate represents a rate of growth of 0.45 g biomass produced per g biomass per h. As 0.96 g O_2 are required per g biomass produced, the specific rate of O_2 consumption is $0.45 \times 0.96 = 0.43$ g O_2 per g biomass per h. When the biomass concentration is 10 g l^{-1} in a 50,000 litre fermenter, the mass of cells is 10 g l^{-1} × 50,000 l = 5×10^5 g. Therefore, the total rate of O_2 consumption is 0.43 g g^{-1} h^{-1} × 5×10^5 g = 2.15×10^5 g O_2 h^{-1}. Converting this to moles:

$$\text{Rate of O}_2\text{ consumption} = 2.15\times10^5\text{ g h}^{-1} \cdot \left|\frac{1\text{ gmol}}{32.0\text{ g}}\right| = 6.72\times10^3\text{ gmol h}^{-1}$$

From p 100, the heat of reaction for aerobic growth is approximately –460 kJ gmol^{-1} O_2 consumed. Therefore:

$$\Delta H_{rxn} = -460\text{ kJ gmol}^{-1}\,(6.72\times10^3\text{ gmol h}^{-1}) = -3.09\times10^6\text{ kJ h}^{-1}$$

The rate of heat removal from the fermenter is determined using the modified energy-balance equation, Eq. (5.26), with $W_s = M_v = 0$:

$$Q = -\Delta H_{rxn} = 3.09\times10^6\text{ kJ h}^{-1}$$

From the sign conventions outlined on pp 87–88, Q positive confirms that heat is removed from the system.

Answer: 3.09×10^6 kJ h^{-1}

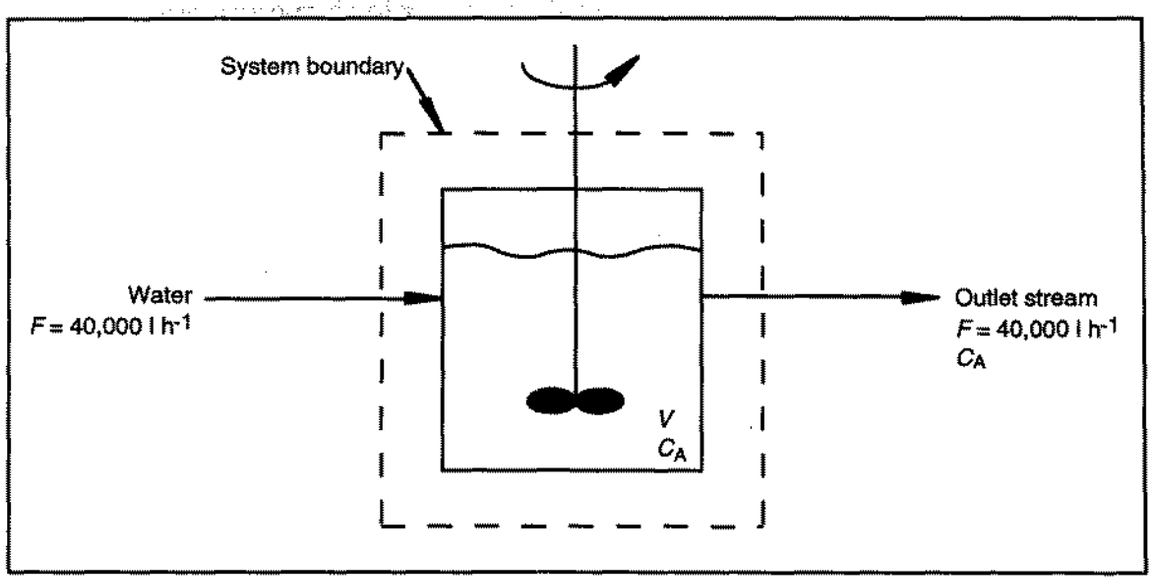
6

Unsteady-State Material and Energy Balances

6.1 Dilution of sewage

(i) *Flow sheet and system boundary*

These are shown in the figure below.



(ii) *Define variables*

V = volume of material in the tank; F = volumetric flow rate into and out of the vessel = 40,000 l h^{-1}; C_A = concentration of suspended solids

(iii) *Assumptions*

– no leaks

– tank is well mixed; therefore C_A in the outlet stream = C_A inside the tank

– density of the solids is the same as that of water = 1 kg l^{-1}

(iv) *Boundary conditions*

At $t = 0$, $V = V_0$ = 440,000 litres + the volume of the solids. The initial mass of solids in the tank is 10,000 kg; therefore, if the density is 1 kg l^{-1}, V_0 = (440,000 + 10,000) l = 450,000 l. At $t = 0$, $C_A = C_{A0}$:

$$C_{A0} = \frac{10{,}000\,\text{kg}}{450{,}000\,\text{l}} = 0.022\,\text{kg l}^{-1}$$

(v) *Total mass balance*

As the volumetric flow rates of material in and out are the same and the density of the inlet and outlet streams are assumed to be equal, the volume of material in the tank is constant (see Example 6.2) so that $V = V_0$ at all times.

(vi) *Solids mass balance*

The general unsteady-state mass-balance equation is Eq. (6.5). As there is no reaction, $R_G = R_C = 0$; in this problem, $\bar{M}_i = 0$, $\bar{M}_o = F\,C_A$ and $M = V\,C_A$. Substituting into Eq. (6.5) gives:

$$\frac{\mathrm{d}(V\,C_A)}{\mathrm{d}t} = -F\,C_A$$

As V is constant, it can be taken outside of the differential:

$$V\frac{dC_A}{dt} = -F\,C_A$$

As F is also constant, the differential equation contains only two variables, C_A and t. Separating variables gives:

$$\frac{dC_A}{C_A} = \frac{-F}{V}\,dt$$

Integrating:

$$\int \frac{dC_A}{C_A} = \int \frac{-F}{V}\,dt$$

Using integration rules (D.27) and (D.24) from Appendix D and combining the constants of integration:

$$\ln C_A = \frac{-F}{V}t + K$$

From the initial condition for C_A, at $t = 0$, $\ln C_{A0} = K$. Substituting this value of K into the equation gives:

$$\ln C_A = \frac{-F}{V}t + \ln C_{A0}$$

$$\ln \frac{C_A}{C_{A0}} = \frac{-F}{V}t$$

$$C_A = C_{A0}\,e^{(-F/V)\,t}$$

Substituting the known values for C_{A0}, F and V:

$$C_A = 0.022\,e^{-0.089\,t}$$

where C_A has units of kg l^{-1} and t has units of h. From this equation, at $t = 5$ h, $C_A = 0.014$ kg l^{-1}.

Answer: 0.014 kg l^{-1}

6.2 Production of fish-protein concentrate

(i) *System*

The system is the whole gutted fish placed in the batch drier at time zero. During drying, the mass of the system decreases as water is removed.

(ii) *Assumptions*

No additional assumptions are required.

(iii) *Boundary conditions*

At $t = 0$, the mass of water in the fish in the drier is equal to M_0.

(iv) *Water mass balance*

The general unsteady-state mass-balance equation is Eq. (6.5). As there is no reaction, $R_G = R_C = 0$. No water is added during drying; therefore $\dot{M}_i = 0$. At any time during drying, as the rate of water removal $\dot{M}_o$ is proportional to the moisture content M, $\dot{M}_o = k\,M$ where k is a constant. Substituting into Eq. (6.5) gives:

$$\frac{dM}{dt} = -k\,M$$

where M and t are the only variables. Separating variables and integrating:

$$\int \frac{dM}{M} = -\int k\,dt$$

Using integration rules (D.27) and (D.24) from Appendix D and combining the constants of integration:

$$\ln M = -k\,t + K$$

At $t = 0$, $M = M_0$; therefore, $K = \ln M_0$. Substituting this value of K into the equation gives:

$$\ln M = -k\,t + \ln M_0$$

$$\ln \frac{M}{M_0} = -k\,t$$

At $t = 20$ min, $M = 0.5\,M_0$. Substituting these values into the equation:

$$\ln \frac{0.5\,M_0}{M_0} = -k\,(20\text{ min})$$

$$\ln 0.5 = -k\,(20\text{ min})$$

$$k = 0.0347\text{ min}^{-1}$$

Therefore:

$$\ln \frac{M}{M_0} = -0.0347\,t$$

where t has units of min. When 95% of the water has been removed, $M = 0.05\,M_0$. Therefore:

$$\ln \frac{0.05\,M_0}{M_0} = -0.0347\,t$$

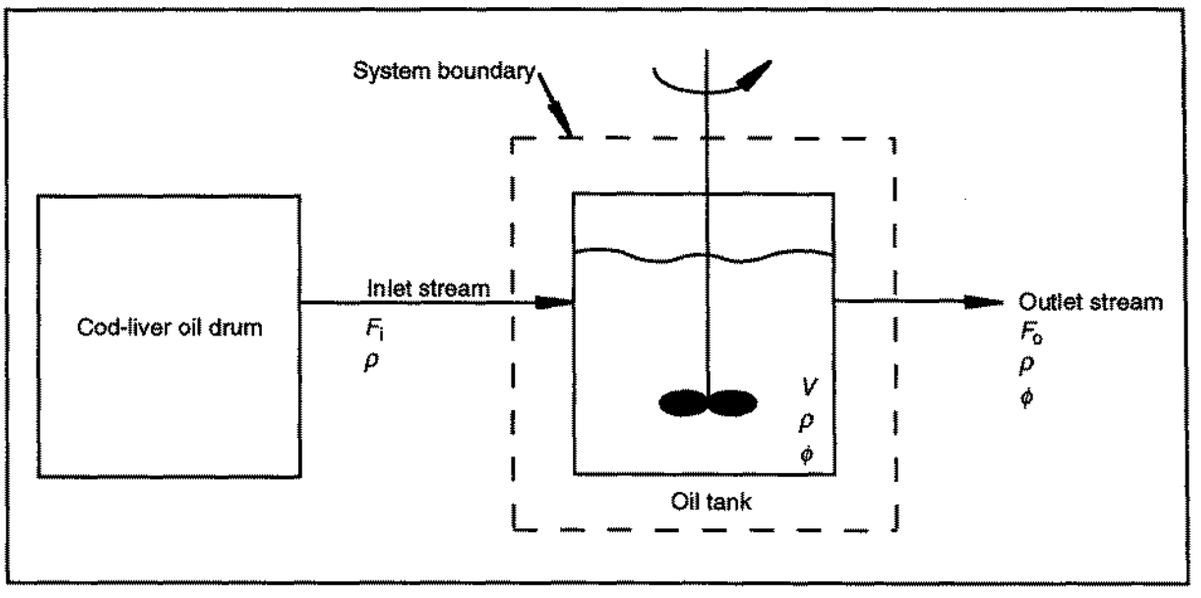
$$\ln 0.05 = -0.0347\,t$$

$$t = 86.3\text{ min} = 86.3\text{ min} \cdot \left| \frac{1\text{ h}}{60\text{ min}} \right| = 1.44\text{ h}$$

Answer: 1.44 h

6.3 Contamination of vegetable oil

(i) *Flow sheet and system boundary*
These are shown in the figure below.



(ii) *Define variables*
V = volume of oil in the tank; F_i = volumetric flow rate of cod-liver oil into the oil tank; F_o = volumetric flow rate of mixed oil out of the oil tank; ϕ = mass fraction of vegetable oil; ρ = density of vegetable and cod-liver oils
(iii) *Assumptions*
– no leaks
– oil tank is well mixed; therefore ϕ in the outlet stream = ϕ inside the tank
– densities of each oil and the oil mixture are the same

(iv) *Boundary conditions*
At $t = 0$ when cod-liver oil first enters the oil tank, $V = V_0 = 60$ litres. At $t = 0$, $\phi = \phi_0 = 1$.

(a)
Total mass balance
The general unsteady-state mass-balance equation is Eq. (6.5). As there is no reaction, $R_G = R_C = 0$. The total mass flow rate into the oil tank is equal to the volumetric flow rate multiplied by the density of the cod-liver oil: $\widetilde{M}_i = F_i \rho$. Similarly, $\widetilde{M}_o = F_o \rho$. The total mass of oil in the tank is equal to the volume of oil multiplied by its density: $M = V \rho$. Substituting these expressions into Eq. (6.5) gives:

$$\frac{d(V\rho)}{dt} = F_i \rho - F_o \rho$$

As ρ is constant it can be taken outside of the differential and cancelled:

$$\rho \frac{dV}{dt} = (F_i - F_o)\rho$$

$$\frac{dV}{dt} = (F_i - F_o)$$

The differential equation contains only two variables, V and t. Separating variables and integrating:

$$dV = (F_i - F_o)\,dt$$

$$\int dV = \int (F_i - F_o)\,dt$$

Using integration rule (D.24) from Appendix D and combining the constants of integration:

$$V = (F_i - F_o)\,t + K$$

From the initial condition for V, at $t = 0$, $K = V_0$. Substituting this value of K into the equation gives:

$$V = (F_i - F_o)\,t + V_0$$

The time between 8 p.m. and 9 a.m. the next morning is 13 h. From the above equation for V, for $t = 13$ h, $F_i = 7.5$ l h^{-1}, $F_o = 4.8$ l h^{-1} and $V_0 = 60$ l, $V = 95.1$ l. As this volume is less than the tank capacity of 100 l, the tank will not overflow.

Answer: No

(b)
Total mass balance
As the volumetric flow rates of oils into and out of the tank are the same and the density of the inlet and outlet streams are assumed to be equal, the volume of oil in the tank is constant (see Example 6.2) so that $V = V_0 = 60$ l at all times.

Vegetable oil mass balance
The general unsteady-state mass-balance equation is Eq. (6.5). As there is no reaction, $R_G = R_C = 0$. No vegetable oil enters the tank; therefore $\widetilde{M}_i = 0$. The mass flow rate of vegetable oil out is $\widetilde{M}_o = F_o \rho \phi$. The mass of vegetable oil in the tank at any time is $M = V \rho \phi$. Substituting these expressions into Eq. (6.5) gives:

$$\frac{d(V\rho\phi)}{dt} = -F_o \rho \phi$$

As both V and ρ are constants, they can be taken outside of the differential and ρ can be cancelled:

$$V\rho \frac{d\phi}{dt} = -F_o \rho \phi$$

$$V \frac{d\phi}{dt} = -F_o \phi$$

As F_o is also constant, the differential equation contains only two variables, ϕ and t. Separating variables and integrating:

$$\frac{d\phi}{\phi} = \frac{-F_o}{V}\,dt$$

$$\int \frac{d\phi}{\phi} = \int \frac{-F_o}{V}\,dt$$

Using integration rules (D.27) and (D.24) from Appendix D and combining the constants of integration:

$$\ln \phi = \frac{-F_o}{V}t + K$$

From the initial condition for ϕ, at $t = 0$, $\ln \phi_0 = K$. Substituting this value of K into the equation gives:

$$\ln \phi = \frac{-F_o}{V}t + \ln \phi_0$$

$$\ln \frac{\phi}{\phi_0} = \frac{-F_o}{V}t$$

$$\phi = \phi_0\, e^{(-F_o/V)\,t}$$

In this problem, $F_o = 4.8\ \mathrm{l\ h^{-1}}$. Substituting the known values for ϕ_0 and V:

$$\phi = 1\, e^{-0.080\,t}$$

where ϕ is dimensionless and t has units of h. The time between 8 p.m. and midnight is 4 h. From the above equation, at $t = 4$ h, $\phi = 0.73$. Therefore, at midnight, the composition of oil in the tank is 73% vegetable oil and 27% cod-liver oil.

Answer: 73% vegetable oil, 27% cod-liver oil

6.4 Batch growth of bacteria

The general unsteady-state mass-balance equation is Eq. (6.5). For a batch culture, $\widetilde{M}_i = \widetilde{M}_o = 0$. For a mass balance on cells, assuming there is no loss of cells from the system, e.g. by lysis, $R_C = 0$. The rate of generation of cells R_G is proportional to the concentration of cells present: $R_G = \mu\ x\ V$ where μ is a constant, x is the cell concentration and V is the culture volume. The total mass of cells M is equal to the culture volume V multiplied by the cell concentration x: $M = V\,x$. Substituting these results into Eq. (6.5) gives:

$$\frac{d(V\,x)}{dt} = \mu\,x\,V$$

Assuming that V is constant throughout the batch culture, it can be taken outside of the differential:

$$V\frac{dx}{dt} = \mu\,x\,V$$

$$\frac{dx}{dt} = \mu\,x$$

The differential equation contains only two variables, x and t. Separating variables and integrating:

$$\frac{dx}{x} = \mu\,dt$$

$$\int \frac{dx}{x} = \int \mu\,dt$$

Using integration rules (D.27) and (D.24) from Appendix D and combining the constants of integration:

$$\ln x = \mu\,t + K$$

Assume an initial condition: at $t = 0$ at the beginning of exponential growth, $x = x_0$. Therefore, at $t = 0$, $\ln x_0 = K$. Substituting this value of K into the equation gives:

$$\ln x = \mu t + \ln x_0$$

$$\ln \frac{x}{x_0} = \mu t$$

When $t = 45$ min, $x = 2\,x_0$. Substituting these values into the equation:

$$\ln \frac{2\,x_0}{x_0} = \mu\,(45\text{ min})$$

$$\ln 2 = \mu\,(45\text{ min})$$

$$\mu = 0.0154\text{ min}^{-1}$$

Therefore:

$$\ln \frac{x}{x_0} = 0.0154\,t$$

or

$$x = x_0\,e^{0.0154\,t}$$

where t has units of min. For $t = 12$ h $= 12 \times 60 = 720$ min:

$$x = 6.54 \times 10^4\,x_0$$

After 12 h, the cell concentration is 6.54×10^4 times the cell concentration at the beginning of exponential growth. If there is a lag phase, the cell concentration at the beginning of exponential growth is usually very close to that at inoculation; therefore, the cell concentration is about 6.54×10^4 times the inoculum level.

Answer: 6.54×10^4 times the inoculum level

6.5 Radioactive decay

The general unsteady-state mass-balance equation is Eq. (6.5). In this problem, $\widehat{M}_i = \widehat{M}_o = 0$. For a mass balance on isotope, $R_G = 0$. The rate of isotope decay R_C is proportional to the concentration of isotope present: $R_C = -k_1\,C\,V$, where k_1 is a constant, C is the isotope concentration and V is the solution volume. The total mass of isotope M is equal to the solution volume V multiplied by the isotope concentration C: $M = V\,C$. Substituting these results into Eq. (6.5) gives:

$$\frac{d(V\,C)}{dt} = -k_1\,C\,V$$

If we assume that the density of the solution is constant during isotope decay, V is constant and can be taken outside of the differential and cancelled:

$$V\frac{dC}{dt} = -k_1\,C\,V$$

$$\frac{dC}{dt} = -k_1\,C$$

The differential equation contains only two variables, C and t. Separating variables and integrating:

$$\frac{dC}{C} = -k_1\,dt$$

$$\int \frac{dC}{C} = \int -k_1\,dt$$

Using integration rules (D.27) and (D.24) from Appendix D and combining the constants of integration:

$$\ln C = -k_1\,t + K$$

Assume an initial condition: at $t = 0$, $C = C_0$. Therefore, at $t = 0$, $\ln C_0 = K$. Substituting this value of K into the equation:

$$\ln C = -k_1 t + \ln C_0$$

$$\ln \frac{C}{C_0} = -k_1 t$$

(a)
When $t = t_h$, the half-life of the isotope, $C = 0.5\ C_0$. Substituting these values into the equation:

$$\ln \frac{0.5\ C_0}{C_0} = -k_1 t_h$$

$$\ln 0.5 = -k_1 t_h$$

$$t_h = \frac{-\ln 0.5}{k_1}$$

Using mathematical rule (D.10) in Appendix D, $-\ln 0.5 = \ln {}^1/_{0.5} = \ln 2$. Therefore:

$$t_h = \frac{\ln 2}{k_1}$$

Answer: Q.E.D.

(b)
From the equation derived in **(a)**:

$$k_1 = \frac{\ln 2}{t_h}$$

For $t_h = 14.3$ days:

$$k_1 = 4.85 \times 10^{-2}\ \text{d}^{-1}$$

Substituting this value of k_1 into the general equation for isotope concentration:

$$\ln \frac{C}{C_0} = -4.85 \times 10^{-2}\ t$$

where t has units of days. For $C = 0.01\ C_0$:

$$\ln \frac{0.01\ C_0}{C_0} = -4.85 \times 10^{-2}\ t$$

$$\ln 0.01 = -4.85 \times 10^{-2}\ t$$

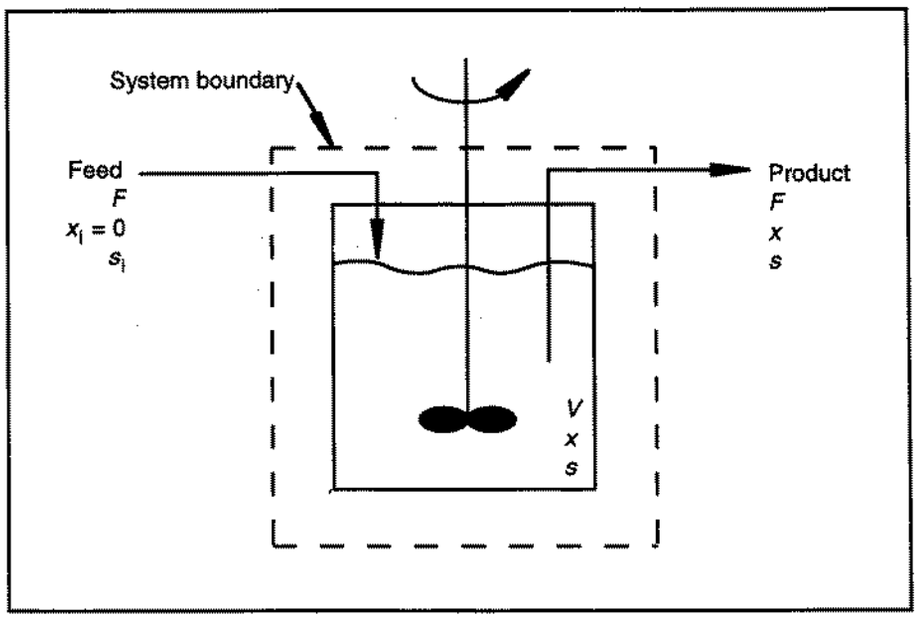
$$t = 95 \text{ days}$$

Answer: 95 days

6.6 Continuous fermentation

(i) *Flow sheet and system boundary*

These are shown for a continuous fermenter in the figure below.



(ii) *Define variables*

V = volume of broth in the fermenter; F = volumetric flow rate into and out of the vessel; x = concentration of cells; s = concentration of substrate; x_i = concentration of cells in the feed; s_i = concentration of substrate in the feed

(iii) *Assumptions*

– no leaks
– fermenter is well mixed; therefore x and s in the outlet stream = x and s, respectively, inside the fermenter
– density of the fermentation broth is the same as that of the feed

(iv) *Boundary condition*

At $t = 0$, $x = x_0$.

(v) *Total mass balance*

As the volumetric flow rates into and out of the fermenter are the same and the density of the inlet and outlet streams are assumed equal, the volume of broth in the fermenter is constant (see Example 6.2) and equal to V at all times.

(a)

Cell mass balance

The general unsteady-state mass-balance equation is Eq. (6.5). As no cells enter in the feed stream, $\bar{M}_i = 0$. The mass flow rate of cells out $\bar{M}_o = F\,x$. The rate of cell generation $R_G = r_X\,V = k_1\,x\,V$. Assuming that cell lysis is negligible, $R_C = 0$. The mass of cells in the fermenter M is equal to $V\,x$. Substituting these terms into Eq. (6.5) gives:

$$\frac{d(V\,x)}{dt} = -F\,x + k_1\,x\,V$$

As V is constant, it can be taken outside of the differential.

$$V\frac{dx}{dt} = -F\,x + k_1\,x\,V$$

Dividing through by V and grouping terms:

$$\frac{dx}{dt} = x\,(k_1 - \frac{F}{V})$$

Answer: $\frac{dx}{dt} = x\,(k_1 - \frac{F}{V})$

(b)

At steady state, $^{dx}/_{dt} = 0$. Therefore, from the equation derived in **(a)**, at steady state k_1 must be equal to $^{F}/_{V}$.

Answer: $k_1 = {}^{F}/_{V}$

(c)
As F, V and k_1 are constants, the differential equation derived in **(a)** contains only two variables, x and t. Separating variables and integrating:

$$\frac{dx}{x} = (k_1 - \frac{F}{V})\,dt$$

$$\int \frac{dx}{x} = \int (k_1 - \frac{F}{V})\,dt$$

Using integration rules (D.27) and (D.24) from Appendix D and combining the constants of integration:

$$\ln x = (k_1 - \frac{F}{V})\,t + K$$

From the initial condition for x, at $t = 0$, $\ln x_0 = K$. Substituting this value of K into the equation gives:

$$\ln x = (k_1 - \frac{F}{V})\,t + \ln x_0$$

$$\ln \frac{x}{x_0} = (k_1 - \frac{F}{V})\,t$$

$$x = x_0\, e^{(k_1 - \frac{F}{V})\,t}$$

Answer: $x = x_0\, e^{(k_1 - \frac{F}{V})\,t}$

(d)
Substituting the parameter values into the equation derived in **(c)**:

$$x = (0.5\ \mathrm{g\,l^{-1}})\, e^{(0.33\ \mathrm{h^{-1}} - \frac{2200\ \mathrm{l\,h^{-1}}}{10{,}000\ \mathrm{l}})\,t}$$

$$x = 0.5\, e^{0.11\,t}$$

where x has units of g l^{-1} and t has units of h. For $x = 4.0$ g l^{-1}:

$$4 = 0.5\, e^{0.11\,t}$$

$$\ln 8 = 0.11\,t$$

$$t = 18.9\ \mathrm{h}$$

Answer: 18.9 h

(e)
Substrate mass balance
The general unsteady-state mass-balance equation is Eq. (6.5). For substrate, the mass flow rates in and out are $\dot{M}_i = F\,s_i$ and $\dot{M}_o = F\,s$. The rate of substrate generation $R_G = 0$; the rate of substrate consumption $R_C = r_S\,V = k_2\,x\,V$. The mass of substrate in the fermenter M is equal to $V\,s$. Substituting these terms into Eq. (6.5) gives:

$$\frac{d(V\,s)}{dt} = F\,s_i - F\,s - k_2\,x\,V$$

As V is constant, it can be taken outside of the differential:

$$V\frac{ds}{dt} = F\,s_i - F\,s - k_2\,x\,V$$

Dividing through by V and grouping terms:

$$\frac{ds}{dt} = \frac{F}{V}(s_i - s) - k_2\,x$$

Substituting the expression for x from **(c)**:

$$\frac{ds}{dt} = \frac{F}{V}(s_i - s) - k_2 x_0 e^{(k_1 - \frac{F}{V})t}$$

In this equation, F, V, k_2 and x_0 are constants and there are only two variables, s and t. However, the variables cannot be easily separated as in the previous problems, making algebraic solution difficult.

Answer: $\frac{ds}{dt} = \frac{F}{V}(s_i - s) - k_2 x_0 e^{(k_1 - \frac{F}{V})t}$

(f)

An equation for $^{ds}/_{dt}$ in terms of x was derived in (e) as:

$$\frac{ds}{dt} = \frac{F}{V}(s_i - s) - k_2 x$$

At steady state, $^{ds}/_{dt} = 0$. Therefore:

$$\frac{F}{V}(s_i - s) = k_2 x$$

$$s_i - s = \frac{V k_2}{F} x$$

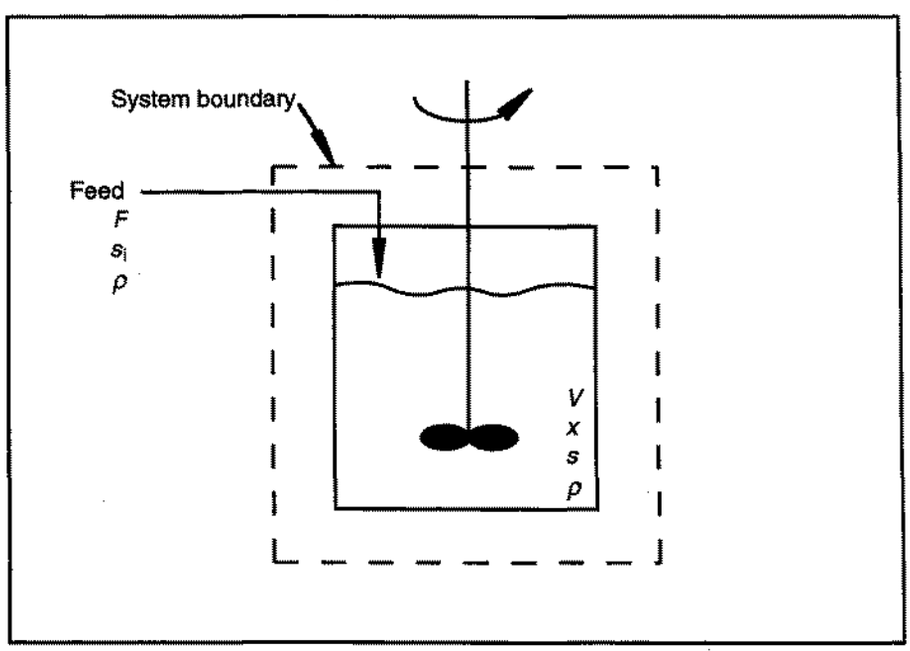
$$s = s_i - \frac{V k_2}{F} x$$

Answer: $s = s_i - \frac{V k_2}{F} x$

6.7 Fed-batch fermentation

(i) *Flow sheet and system boundary*

These are shown for a fed-batch fermenter in the figure below.



(ii) *Define variables*

V = volume of broth in the fermenter; F = volumetric flow rate into the fermenter; x = concentration of cells in the fermenter; s = concentration of substrate in the fermenter; s_i = concentration of substrate in the feed; ρ = density of the feed and fermentation broth

(iii) *Assumptions*

– no leaks
– fermenter is well mixed
– density of the fermentation broth is the same as that of the feed

(iv) *Boundary condition*
At $t = 0$, $V = V_0$.

(a)
Total mass balance
The general unsteady-state mass-balance equation is Eq. (6.5). As total mass cannot be generated or consumed, $R_G = R_C = 0$. No mass leaves the fermenter; therefore $\widetilde{M}_o = 0$. The mass flow rate in $\widetilde{M}_i = \rho F$. The total mass in the fermenter M is equal to $V\rho$. Substituting these terms into Eq. (6.5) gives:

$$\frac{d(V\rho)}{dt} = \rho F$$

As ρ is constant, it can be taken outside of the differential and cancelled:

$$\rho\frac{dV}{dt} = \rho F$$

$$\frac{dV}{dt} = F$$

As F is constant, the differential equation contains only two variables, V and t. Separating variables and integrating:

$$dV = F\,dt$$

$$\int dV = \int F\,dt$$

Using integration rule (D.24) from Appendix D and combining the constants of integration:

$$V = Ft + K$$

From the initial condition for V, at $t = 0$, $V_0 = K$. Substituting this value of K into the equation gives:

$$V = Ft + V_0$$

Answer: $V = Ft + V_0$

(b)
Substrate mass balance
The general unsteady-state mass-balance equation is Eq. (6.5). The mass flow rates of substrate in and out are $\widetilde{M}_i = Fs_i$ and $\widetilde{M}_o = 0$. The rate of substrate generation $R_G = 0$; the rate of substrate consumption $R_C = r_S V = k_1 s V$. The mass of substrate in the fermenter M is equal to Vs. Substituting these terms into Eq. (6.5) gives:

$$\frac{d(Vs)}{dt} = Fs_i - k_1 s V$$

As neither V nor s is constant, both must be kept in the differential as a product. Expanding the differential using the product rule (D.22) from Appendix D:

$$V\frac{ds}{dt} + s\frac{dV}{dt} = Fs_i - k_1 s V$$

Using the equation $dV/dt = F$ derived in **(a)** :

$$V\frac{ds}{dt} + Fs = Fs_i - k_1 s V$$

Grouping terms gives:

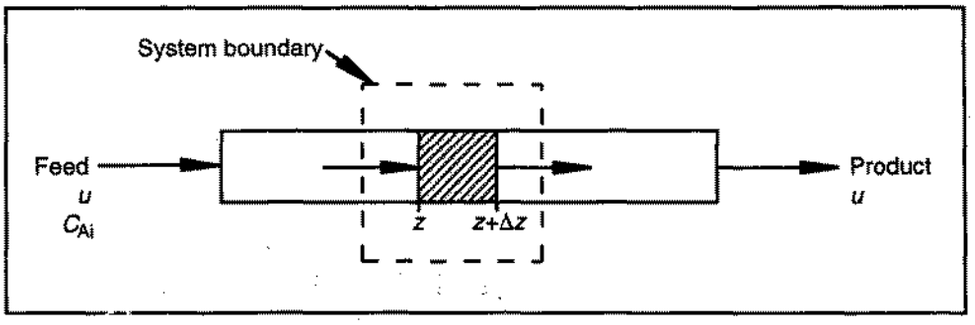
$$\frac{ds}{dt} = \frac{F}{V}(s_i - s) - k_1 s$$

Answer: $\dfrac{ds}{dt} = \dfrac{F}{V}(s_i - s) - k_1 s$

6.8 Plug-flow reactor

(i) *Flow sheet and system boundary*

These are shown for a plug-flow reactor in the figure below.



(ii) *Define variables*

u = fluid linear velocity; z = distance along the reactor; C_A = concentration of reactant; C_{Ai} = concentration of reactant in the feed stream; A = reactor cross-sectional area

(iii) *Assumptions*

– no leaks

– plug flow

(a)

Reactant balance

Consider the system to be a small section of the reactor located between z and $z + \Delta z$. The general unsteady-state mass-balance equation is Eq. (6.5). The rate of entry of reactant into the system is:

$$\dot{M}_i = C_A u A\big|_z$$

where $\big|_z$ means that the parameter values are those at distance z from the front of the reactor. Similarly, the rate at which reactant leaves the system is:

$$\dot{M}_o = C_A u A\big|_{z+\Delta z}$$

Reactant is not generated; therefore $R_G = 0$. The rate of consumption of reactant is given by the equation:

$$R_C = r_C V = r_C A \Delta z$$

where $A \Delta z$ is the volume of the system and r_C is the volumetric rate of reaction. At steady state there is no accumulation in the system and $dM/dt = 0$. Substituting these expressions into Eq. (6.5) gives:

$$0 = C_A u A\big|_z - C_A u A\big|_{z+\Delta z} - r_C A \Delta z$$

As A is constant and does not depend on z, it can be cancelled from each of the terms:

$$0 = C_A u\big|_z - C_A u\big|_{z+\Delta z} - r_C \Delta z$$

Dividing through by Δz:

$$0 = \frac{C_A u\big|_z - C_A u\big|_{z+\Delta z}}{\Delta z} - r_C$$

Taking the limit as Δz approaches zero and applying the definition of the derivative (D.13) in Appendix D:

$$0 = \frac{-d(C_A u)}{dz} - r_C$$

or

$$\frac{d(C_A u)}{dz} = -r_C$$

As the fluid velocity is constant throughout the reactor, u can be taken outside of the differential:

$$u\frac{dC_A}{dz} = -r_C$$

Answer: $u\dfrac{dC_A}{dz} = -r_C$

(b)
Answer: at $z = 0$, $C_A = C_{Ai}$

(c)
If the reaction is first-order, $r_C = k_1 C_A$ where k_1 is the first-order rate constant. The differential equation becomes:

$$u\frac{dC_A}{dz} = -k_1 C_A$$

As u and k_1 are constants, the differential equation contains only two variables, C_A and z. Separating variables and integrating:

$$\frac{dC_A}{C_A} = \frac{-k_1}{u}dz$$

$$\int\frac{dC_A}{C_A} = \int\frac{-k_1}{u}dz$$

Using integration rules (D.27) and (D.24) from Appendix D and combining the constants of integration:

$$\ln C_A = \frac{-k_1}{u}z + K$$

From the initial condition in **(b)**, at $z = 0$, $\ln C_{Ai} = K$. Substituting this value of K into the equation gives:

$$\ln C_A = \frac{-k_1}{u}z + \ln C_{Ai}$$

$$\ln\frac{C_A}{C_{Ai}} = \frac{-k_1}{u}z$$

$$C_A = C_{Ai}\,e^{(-k_1/u)z}$$

Answer: $C_A = C_{Ai}\,e^{(-k_1/u)z}$

(d)
The equation derived in **(c)** is directly analogous to the equation for change in reactant concentration in a batch reactor. As $z = u\,t$ where t is the time taken for the fluid to travel distance z, the above equation can be written as:

$$C_A = C_{Ai}\,e^{-k_1 t}$$

which is the same as the equation for reactant concentration in a batch reactor where C_{Ai} is the concentration at time zero.

Answer: Essentially identical

6.9 Boiling water

The system is the beaker containing water.
(i) *Assumptions*
– no evaporation
– water is well mixed
– no shaft work
– heat capacity is independent of temperature
– heat losses are negligible
– the density of water is constant between 18°C and 100°C

(ii) *Extra data*
Density of water = 1 kg l^{-1}
C_p water = 75.4 J $gmol^{-1}$ $°C^{-1}$ (Table B.3, Appendix B) = 75.4 kJ $kgmol^{-1}$ $°C^{-1}$
Molecular weight of water (Table B.1, Appendix B) = 18.0
1 W = 1 J s^{-1} (Table A.8, Appendix A); therefore, 1 kW = 1 kJ s^{-1}
(iii) *Boundary conditions*
At $t = 0$, $T = T_0 = 18°C$

(a)
The general unsteady-state energy-balance equation is Eq. (6.10). For a batch system, $\bar{M}_i = \bar{M}_o = 0$; also $\bar{W}_s = 0$. Energy is accumulated by the system in the form of sensible heat only; therefore:

$$\frac{dE}{dt} = M C_p \frac{dT}{dt}$$

where M is the mass of water in the beaker and T is its temperature. Substituting these expressions into Eq. (6.10) gives:

$$M C_p \frac{dT}{dt} = -\bar{Q}$$

Answer: $M C_p \frac{dT}{dt} = -\bar{Q}$

(b)
If $\bar{Q}$, C_p and M are constant, T and t are the only variables in the differential equation. Separating variables and integrating:

$$dT = \frac{-\bar{Q}}{M C_p} dt$$

$$\int dT = \int \frac{-\bar{Q}}{M C_p} dt$$

Using integration rule (D.24) from Appendix D and combining the constants of integration:

$$T = \frac{-\bar{Q}}{M C_p} t + K$$

From the initial condition for T, at $t = 0$, $T_0 = K$. Substituting this value of K into the equation gives:

$$T = \frac{-\bar{Q}}{M C_p} t + T_0$$

Using the density of water = 1 kg l^{-1}, the mass of 2 litres of water $M = 2$ kg. Converting the C_p for water to mass terms:

$$C_p = 75.4\,\text{kJ kgmol}^{-1}\,°\text{C}^{-1} \cdot \left|\frac{1\,\text{kgmol}}{18.0\,\text{kg}}\right| = 4.189\,\text{kJ kg}^{-1}\,°\text{C}^{-1}$$

T = the boiling temperature of water = 100°C; $t = 11$ min. Substituting the parameter values into the equation for T:

$$100°\text{C} = \frac{-\bar{Q}}{2\,\text{kg}\left(4.189\,\text{kJ kg}^{-1}\,°\text{C}^{-1}\right)}(11\,\text{min}) + 18°\text{C}$$

$$\bar{Q} = -62.45\,\text{kJ min}^{-1} = -62.45\,\text{kJ min}^{-1} \cdot \left|\frac{1\,\text{min}}{60\,\text{s}}\right| \cdot \left|\frac{1\,\text{kW}}{1\,\text{kJ s}^{-1}}\right| = -1.04\,\text{kW}$$

From the sign conventions outlined on pp 87–88, the negative value for $\bar{Q}$ confirms that heat is added to the system.

Answer: 1.04 kW

6.10 Heating glycerol solution

The system is the stirred tank containing the solution of glycerol in water.

(i) *Assumptions*

– no leaks
– no evaporation
– tank is well mixed
– no shaft work
– heat capacities are independent of temperature between 15°C and 90°C
– ideal solution
– system is adiabatic; therefore heat losses are negligible

(ii) *Extra data*

C_p glycerol = 0.576 cal g^{-1} $°C^{-1}$ (Table B.5, Appendix B) = 0.576 kcal kg^{-1} $°C^{-1}$
C_p water = 75.4 J $gmol^{-1}$ $°C^{-1}$ (Table B.3, Appendix B) = 75.4 kJ $kgmol^{-1}$ $°C^{-1}$
Molecular weight of water (Table B.1, Appendix B) = 18.0
1 kcal = 4.187×10^3 J (Table A.7, Appendix A) = 4.187 kJ
1 W = 1 J s^{-1} (Table A.8, Appendix A); therefore, 1 kW = 1 kJ s^{-1}

(iii) *Boundary conditions*

At $t = 0$, $T = T_0 = 15°C$

(a)

The general unsteady-state energy-balance equation is Eq. (6.10). For a batch system, $\bar{M}_i = \bar{M}_o = 0$; also $\bar{W}_s = 0$. Energy is accumulated by the system in the form of sensible heat only; therefore:

$$\frac{dE}{dt} = M C_p \frac{dT}{dt}$$

where M is the mass of glycerol solution in the tank and T is its temperature. Substituting these expressions into Eq. (6.10) gives:

$$M C_p \frac{dT}{dt} = -\bar{Q}$$

M has two components, glycerol and water, which have different heat capacities. Therefore, this equation can be written:

$$\left(M_W C_{pW} + M_G C_{pG}\right) \frac{dT}{dt} = -\bar{Q}$$

where M_W is the mass of water in the tank, M_G is the mass of glycerol, C_{pW} is the heat capacity of water, and C_{pG} is the heat capacity of glycerol.

Answer: $\left(M_W C_{pW} + M_G C_{pG}\right) \frac{dT}{dt} = -\bar{Q}$

(b)

If $\bar{Q}$, C_{pW}, C_{pG}, M_W and M_G are constant, T and t are the only variables in the differential equation. Separating variables and integrating:

$$dT = \frac{-\bar{Q}}{\left(M_W C_{pW} + M_G C_{pG}\right)} dt$$

$$\int dT = \int \frac{-\bar{Q}}{\left(M_W C_{pW} + M_G C_{pG}\right)} dt$$

Using integration rule (D.24) from Appendix D and combining the constants of integration:

$$T = \frac{-\bar{Q}}{\left(M_W C_{pW} + M_G C_{pG}\right)} t + K$$

From the initial condition for T, at $t = 0$, $T_0 = K$. Substituting this value of K into the equation gives:

$$T = \frac{-\bar{Q}}{\left(M_W C_{pW} + M_G C_{pG}\right)} t + T_0$$

Answer: $T = \dfrac{-\bar{Q}}{\left(M_W C_{pW} + M_G C_{pG}\right)} t + T_0$

(c)
The mass of glycerol in the tank $M_G = 45$ kg; the mass of water $M_W = 55$ kg. Converting the C_p for water to mass terms:

$$C_{pW} = 75.4\,\text{kJ kgmol}^{-1}\,^\circ\text{C}^{-1} \cdot \left|\frac{1\,\text{kgmol}}{18.0\,\text{kg}}\right| = 4.189\,\text{kJ kg}^{-1}\,^\circ\text{C}^{-1}$$

Converting the C_p for glycerol to kJ:

$$C_{pG} = 0.576\,\text{kcal kg}^{-1}\,^\circ\text{C}^{-1} \cdot \left|\frac{4.187\,\text{kJ}}{1\,\text{kcal}}\right| = 2.412\,\text{kJ kg}^{-1}\,^\circ\text{C}^{-1}$$

The rate of heat input to the system is 0.88×2.5 kW = 2.2 kW = 2.2 kJ s^{-1}. From the sign conventions outlined on pp 87–88, $\bar{Q}$ must be negative as heat is added to the system; therefore, $\bar{Q} = -2.2$ kJ s^{-1}. Substituting the parameter values into the equation for T with $T_0 = 15^\circ$C and $T = 90^\circ$C:

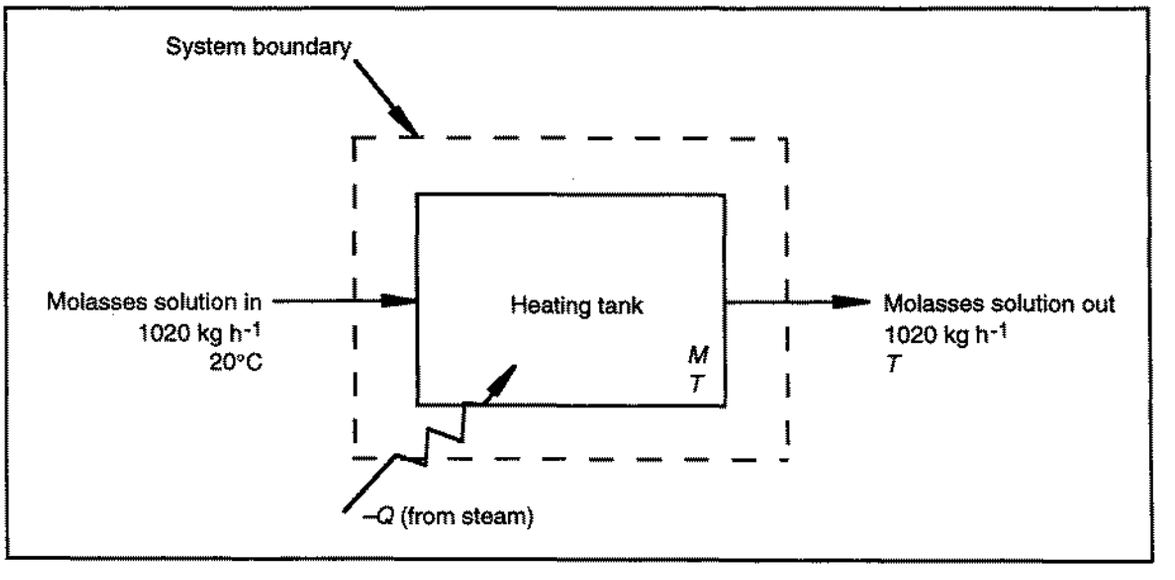
$$90^\circ\text{C} = \frac{2.2\,\text{kJ s}^{-1}}{55\,\text{kg}\left(4.189\,\text{kJ kg}^{-1}\,^\circ\text{C}^{-1}\right) + 45\,\text{kg}\left(2.412\,\text{kJ kg}^{-1}\,^\circ\text{C}^{-1}\right)} t + 15^\circ\text{C}$$

$$t = 1.16 \times 10^4\,\text{s} = 1.16 \times 10^4\,\text{s} \cdot \left|\frac{1\,\text{h}}{3600\,\text{s}}\right| = 3.2\,\text{h}$$

Answer: 3.2 h

6.11 Heating molasses

(i) *Flow sheet and system boundary*
These are shown in the figure below.



(ii) *Assumptions*
– no leaks
– no evaporation
– tank is well mixed; therefore the temperature of the molasses solution out is the same as in the tank
– no shaft work
– heat capacity is independent of temperature
– negligible heat losses

– condensate from the steam leaves at saturation conditions
(iii) *Reference state*
$T_{ref} = 20°C$; $H = 0$ for molasses solution at 20°C
(iv) *Extra data*
$C_p = 0.85$ kcal kg^{-1} °C^{-1}
1 psi = 6.895 × 10^3 Pa (Table A.5, Appendix A) = 6.895 kPa
Converting 40 psi to kPa:

$$40 \text{ psi} = 40 \text{ psi} \cdot \left| \frac{6.895 \text{ kPa}}{1 \text{ psi}} \right| = 275.8 \text{ kPa}$$

The temperature of saturated steam at 275.8 kPa interpolated from Table C.2 (Appendix C) = 130.7°C.
(v) *Boundary conditions*
At $t = 0$, $T = T_0 = 20°C$
(vi) *Total mass balance*
As the mass flow rates into and out of the tank are the same, the mass of molasses solution in the tank M is constant and equal to 5000 kg at all times.

(a)
The general unsteady-state energy-balance equation is Eq. (6.10). From the reference state, $h_i = 0$. The value of h_o relative to the reference state is equal to the sensible heat absorbed by the molasses solution between T_{ref} and the exit temperature T. From Eq. (5.13):

$$h_o = \Delta h = C_p (T - T_{ref})$$

$\bar{W}_s = 0$. The rate at which the molasses solution is heated is given by the equation:

$$\bar{Q} = -UA (T_{steam} - T)$$

From the sign conventions outlined on pp 87–88, $\bar{Q}$ must be negative as heat is added to the system. Energy is accumulated in the form of sensible heat only; therefore:

$$\frac{dE}{dt} = M C_p \frac{dT}{dt}$$

Substituting these expressions into Eq. (6.10) gives:

$$M C_p \frac{dT}{dt} = -\bar{M}_o C_p (T - T_{ref}) + UA (T_{steam} - T)$$

After rearranging, the differential equation is:

$$\frac{dT}{dt} = \frac{UA T_{steam} + \bar{M}_o C_p T_{ref}}{M C_p} - \left(\frac{\bar{M}_o C_p + UA}{M C_p} \right) T$$

Answer: $\dfrac{dT}{dt} = \dfrac{UA T_{steam} + \bar{M}_o C_p T_{ref}}{M C_p} - \left(\dfrac{\bar{M}_o C_p + UA}{M C_p} \right) T$

(b)
As $U, A, T_{steam}, \bar{M}_o, C_p, T_{ref}$ and M are all constant, T and t are the only variables in the differential equation derived in **(a)**. Substituting the known numerical values for the parameters:

$$\frac{dT}{dt} = \frac{190 \text{ kcal m}^{-2} \text{ h}^{-1} \text{ °C}^{-1} (1.5 \text{ m}^2) (130.7°C) + 1020 \text{ kg h}^{-1} \left(0.85 \text{ kcal kg}^{-1} \text{ °C}^{-1}\right) (20°C)}{5000 \text{ kg} \left(0.85 \text{ kcal kg}^{-1} \text{ °C}^{-1}\right)}$$
$$- \left(\frac{1020 \text{ kg h}^{-1} \left(0.85 \text{ kcal kg}^{-1} \text{ °C}^{-1}\right) + 190 \text{ kcal m}^{-2} \text{ h}^{-1} \text{ °C}^{-1} (1.5 \text{ m}^2)}{5000 \text{ kg} \left(0.85 \text{ kcal kg}^{-1} \text{ °C}^{-1}\right)} \right) T$$

$$\frac{dT}{dt} = 12.84 - 0.271 T$$

where T has units °C and t has units h. Separating variables and integrating:

$$\frac{dT}{12.84 - 0.271\,T} = dt$$

$$\int \frac{dT}{12.84 - 0.271\,T} = \int dt$$

Using integration rules (D.28) and (D.24) from Appendix D and combining the constants of integration:

$$\frac{-1}{0.271} \ln (12.84 - 0.271\,T) = t + K$$

From the initial condition for T, at $t = 0$:

$$K = \frac{-1}{0.271} \ln (12.84 - 0.271\,T_0)$$

Applying the numerical value of $T_0 = 20°C$, $K = -7.395$. Substituting this value for K into the equation gives:

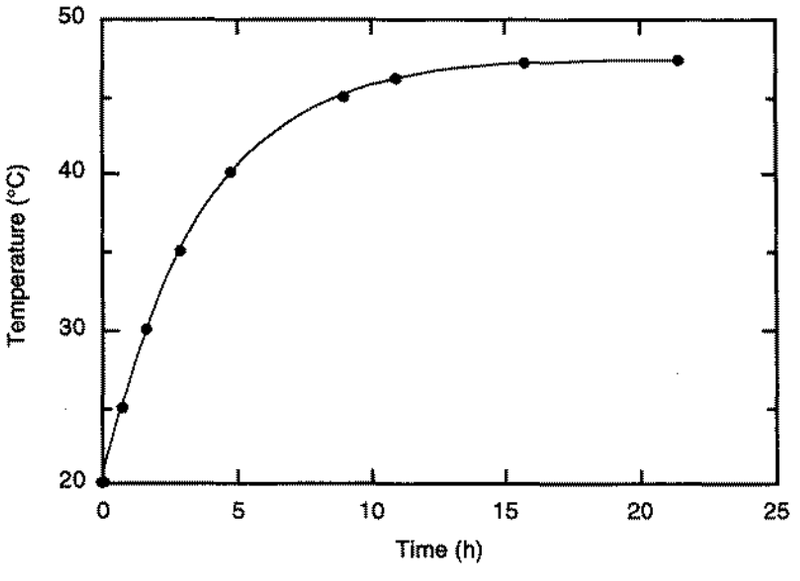
$$\frac{-1}{0.271} \ln (12.84 - 0.271\,T) + 7.395 = t$$

Answer: $\dfrac{-1}{0.271} \ln (12.84 - 0.271\,T) + 7.395 = t$

(c)
Values of t corresponding to various temperatures in the tank can be calculated from the equation derived in **(b)**.

Temperature, T (°C)	Time, t (h)
20	0.00
25	0.74
30	1.68
35	2.93
40	4.84
45	9.01
46	11.0
47	15.8
47.3	21.5

The results are shown in the figure below.



(d)
From the equation derived in **(b)**, as the logarithm of zero and negative numbers is not defined (p 413), the theoretical maximum temperature that can be achieved in the tank occurs when $12.84 - 0.271\,T = 0$; i.e. when $T = 47.4°C$.

Answer: 47.4°C

(e)
The temperature changes constantly with time; therefore, strictly speaking, there is no steady state. For practical purposes, however, the temperature approaches a constant value after about 16 h.

Answer: About 16 h

(f)
From the calculation table in **(c)**, the temperature reaches 40°C after 4.84 h.

Answer: 4.84 h

6.12 Pre-heating culture medium

The system is the glass fermenter containing nutrient medium.
(i) *Assumptions*
– no evaporation
– fermenter is well mixed
– no shaft work
– heat capacities are independent of temperature between 15°C and 36°C
– heat losses are negligible
(ii) *Extra data*
C_p glass vessel = 0.20 cal g^{-1} $°C^{-1}$ = 0.20 kcal kg^{-1} $°C^{-1}$
C_p medium = 0.92 cal g^{-1} $°C^{-1}$ = 0.92 kcal kg^{-1} $°C^{-1}$
1 W = 1.433×10^{-2} kcal min^{-1} (Table A.8, Appendix A)
(iii) *Boundary conditions*
At $t = 0$, $T = T_0 = 15°C$
(iv) *Energy balance*
The general unsteady-state energy-balance equation is Eq. (6.10). For a batch system, $\bar{M}_i = \bar{M}_o = 0$; also $\bar{W}_s = 0$. Energy is accumulated by the system in the form of sensible heat only; therefore:

$$\frac{dE}{dt} = M C_p \frac{dT}{dt}$$

M has two components, the glass vessel and the medium, which have different heat capacities. Therefore, this equation can be written:

$$\frac{dE}{dt} = \left(M_V C_{pV} + M_M C_{pM}\right) \frac{dT}{dt}$$

where M_V is the mass of the glass vessel, M_M is the mass of the medium, C_{pV} is the heat capacity of the vessel, and C_{pM} is the heat capacity of the medium. Substituting into Eq. (6.10) gives:

$$\left(M_V C_{pV} + M_M C_{pM}\right) \frac{dT}{dt} = -\bar{Q}$$

As $\bar{Q}$, C_{pV}, C_{pM}, M_V and M_M are constant, T and t are the only variables in the differential equation. Separating variables and integrating:

$$dT = \frac{-\bar{Q}}{\left(M_V C_{pV} + M_M C_{pM}\right)} dt$$

$$\int dT = \int \frac{-\bar{Q}}{\left(M_V C_{pV} + M_M C_{pM}\right)} dt$$

Using integration rule (D.24) from Appendix D and combining the constants of integration:

$$T = \frac{-\bar{Q}}{\left(M_V C_{pV} + M_M C_{pM}\right)} t + K$$

From the initial condition for T, at $t = 0$, $T_0 = K$. Substituting this value of K into the equation gives:

$$T = \frac{-\bar{Q}}{\left(M_V C_{pV} + M_M C_{pM}\right)} t + T_0$$

The rate of heat input to the system is 450 W. Converting this to kcal min^{-1}:

$$\bar{Q} = -450\,\text{W} = -450\,\text{W}. \left| \frac{1.433 \times 10^{-2}\,\text{kcal min}^{-1}}{1\,\text{W}} \right| = -6.45\,\text{kcal min}^{-1}$$

From the sign conventions outlined on pp 87–88, $\bar{Q}$ must be negative as heat is added to the system. Substituting this and the other parameter values into the equation for T with $T_0 = 15°C$ and $T = 36°C$:

$$36°C = \frac{6.45\,\text{kcal min}^{-1}}{12.75\,\text{kg}\left(0.20\,\text{kcal kg}^{-1}\,°C^{-1}\right) + 7.5\,\text{kg}\left(0.92\,\text{kcal kg}^{-1}\,°C^{-1}\right)} t + 15°C$$

$$t = 30.8\,\text{min}$$

Answer: 30.8 min

6.13 Water heater

The system is the tank containing the water.

(i) *Assumptions*

– no leaks
– no evaporation
– tank is well mixed
– no shaft work
– heat capacity is independent of temperature
– condensate from the steam leaves at saturation conditions

(ii) *Extra data*

C_p water = 75.4 J $gmol^{-1}$ $°C^{-1}$ (Table B.3, Appendix B) = 75.4 kJ $kgmol^{-1}$ $°C^{-1}$
Molecular weight of water (Table B.1, Appendix B) = 18.0
1 kcal = 4.187×10^3 J (Table A.7, Appendix A) = 4.187 kJ

(iii) *Boundary conditions*

At $t = 0$, $T = T_0 = 24°C$

(iv) *Energy balance*

The general unsteady-state energy-balance equation is Eq. (6.10). For a batch system, $\bar{M}_i = \bar{M}_o = 0$; also $\bar{W}_s = 0$. Energy is accumulated by the system in the form of sensible heat only; therefore:

$$\frac{dE}{dt} = M C_p \frac{dT}{dt}$$

where M is the mass of water in the tank and T is its temperature. Substituting these expressions into Eq. (6.10) gives:

$$M C_p \frac{dT}{dt} = -\bar{Q}$$

(a)

There are two components to $\bar{Q}$: the rate of heating from the steam, and the rate of heat loss to the surrounding air:

$$\bar{Q} = U_1 A_1 (T - T_{air}) - U_2 A_2 (T_{steam} - T)$$

This equation reflects the sign conventions outlined on pp 87–88: the term for the heat loss to the atmosphere is positive to indicate heat removal from the system, while the term for heat input from the steam is negative. Substituting into Eq. (6.10) gives:

$$M C_p \frac{dT}{dt} = U_2 A_2 (T_{steam} - T) - U_1 A_1 (T - T_{air})$$

$$\frac{dT}{dt} = \frac{U_2 A_2 (T_{steam} - T) - U_1 A_1 (T - T_{air})}{M C_p}$$

As U_1, A_1, U_2, A_2, T_{steam}, T_{air}, C_p and M are all constant, T and t are the only variables in the differential equation. Substituting the known parameter values:

$$\frac{dT}{dt} = \frac{220\ \text{kcal m}^{-2}\ \text{h}^{-1}\ ^\circ\text{C}^{-1}\ (0.3\ \text{m}^2)\ (130 - T)^\circ\text{C} - 25\ \text{kcal m}^{-2}\ \text{h}^{-1}\ ^\circ\text{C}^{-1}\ (0.9\ \text{m}^2)\ (T - 20)^\circ\text{C}}{1000\ \text{kg}\left(75.4\ \text{kJ kgmol}^{-1}\ ^\circ\text{C}^{-1} \cdot \left|\frac{1\ \text{kgmol}}{18.0\ \text{kg}}\right| \cdot \left|\frac{1\ \text{kcal}}{4.187\ \text{kJ}}\right|\right)}$$

$$\frac{dT}{dt} = 9.026 - 0.088\,T$$

where T has units of °C and t has units of h. Separating variables and integrating:

$$\frac{dT}{9.026 - 0.088\,T} = dt$$

$$\int \frac{dT}{9.026 - 0.088\,T} = \int dt$$

Using integration rules (D.28) and (D.24) from Appendix D and combining the constants of integration:

$$\frac{-1}{0.088} \ln (9.026 - 0.088\,T) = t + K$$

From the initial condition for T, at $t = 0$:

$$K = \frac{-1}{0.088} \ln (9.026 - 0.088\,T_0)$$

Applying the numerical value of $T_0 = 24°\text{C}$, $K = -21.97$. Substituting this value for K into the equation gives:

$$\frac{-1}{0.088} \ln (9.026 - 0.088\,T) + 21.97 = t$$

From this equation, for $T = 80°\text{C}$, $t = 14.2$ h.

Answer: 14.2 h

(b)
If heat losses can be neglected, $\dot{Q}$ has only one component and Eq. (6.10) becomes:

$$M C_p \frac{dT}{dt} = U_2 A_2 (T_{steam} - T)$$

$$\frac{dT}{dt} = \frac{U_2 A_2 (T_{steam} - T)}{M C_p}$$

Substituting the known numerical values:

$$\frac{dT}{dt} = \frac{220\ \text{kcal m}^{-2}\ \text{h}^{-1}\ ^\circ\text{C}^{-1}\ (0.3\ \text{m}^2)\ (130 - T)^\circ\text{C}}{1000\ \text{kg}\left(75.4\ \text{kJ kgmol}^{-1}\ ^\circ\text{C}^{-1} \cdot \left|\frac{1\ \text{kgmol}}{18.0\ \text{kg}}\right| \cdot \left|\frac{1\ \text{kcal}}{4.187\ \text{kJ}}\right|\right)}$$

$$\frac{dT}{dt} = 8.576 - 0.066\,T$$

where T has units of °C and t has units of h. Separating variables and integrating:

$$\frac{dT}{8.576 - 0.066\,T} = dt$$

$$\int \frac{dT}{8.576 - 0.066\,T} = \int dt$$

Using integration rules (D.28) and (D.24) from Appendix D and combining the constants of integration:

$$\frac{-1}{0.066} \ln (8.576 - 0.066\,T) = t + K$$

From the initial condition for T, at $t = 0$:

$$K = \frac{-1}{0.066} \ln (8.576 - 0.066\,T_0)$$

Applying the numerical value of $T_0 = 24°C$, $K = -29.47$. Substituting this value for K into the equation gives:

$$\frac{-1}{0.066} \ln (8.576 - 0.066\,T) + 29.47 = t$$

From this equation, for $T = 80°C$, $t = 11.4$ h. Therefore, from (**a**), the time saved is $(14.2 - 11.4)$ h $= 2.8$ h, which corresponds to 20% of the time required when heat losses occur.

Answer: 2.8 h, or 20% of the time required when heat losses occur

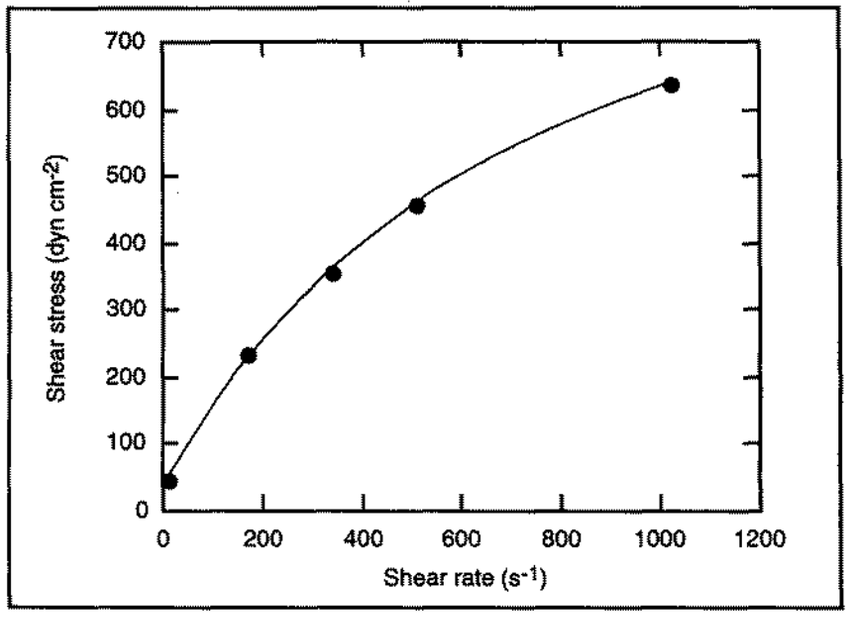
7

Fluid Flow and Mixing

7.1 Rheology of fermentation broth

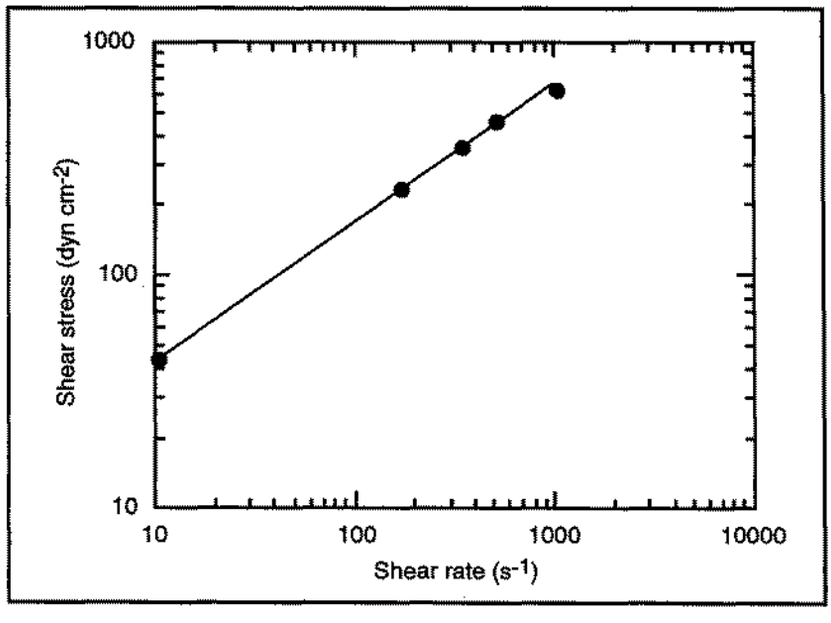
(a)

The rheogram is obtained by plotting shear stress against shear rate.



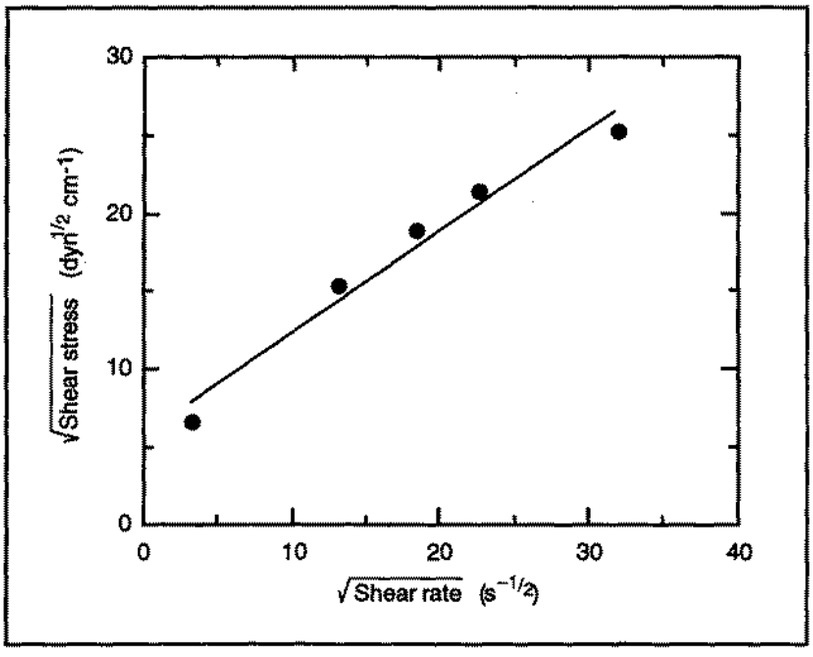
(b)

The rheogram in **(a)** is similar to those for pseudoplastic and Casson plastic fluids in Figure 7.7. It is not clear from the rheogram whether the fluid exhibits a yield stress at $\dot{\gamma}=0$ or not; therefore, both non-Newtonian models are worth checking. From the equation in Figure 7.7 for pseudoplastic fluids, a plot of shear stress versus shear rate on log–log coordinates would be expected to give a straight line. This plot is shown below.



The equation for the straight line in the log–log plot is $\tau = 11.43\,\dot{\gamma}^{0.587}$. Therefore, the flow behaviour index $n = 0.587$ and the consistency index $K = 11.43$ dyn s^n cm^{-2}. The sum of squares of the residuals for this data fit is 1137.

From the equation in Figure 7.7 for Casson plastic fluids, a plot of the square root of shear stress versus the square root of shear rate on linear coordinates would be expected to give a straight line. This plot is shown below.



The equation for the straight line in the plot is $\tau^{1/2} = 5.93 + 0.648\,\dot{\gamma}^{1/2}$. From the equation in Figure 7.7, this means that $K_p = 0.648$ dyn$^{1/2}$ $s^{1/2}$ cm^{-1} and the yield stress $\tau_0 = 35.2$ dyn cm^{-2}. The sum of squares of the residuals for this data fit is 8986.

Comparison of the residuals from the two models suggests that the equation for a pseudoplastic fluid is the better fit.

Answer: Pseudoplastic fluid: $n = 0.587$; $K = 11.43$ dyn s^n cm^{-2}

(c)
The apparent viscosity for a pseudoplastic fluid is given by Eq. (7.8) and can be calculated using the parameter values determined in **(b)**.
(i)

$$\mu_a = K\dot{\gamma}^{n-1} = 11.43\,\text{dyn s}^{0.587}\,\text{cm}^{-2}\,(15\,\text{s}^{-1})^{0.587-1} = 3.7\,\text{dyn s cm}^{-2}$$

Answer: 3.7 dyn s cm^{-2}

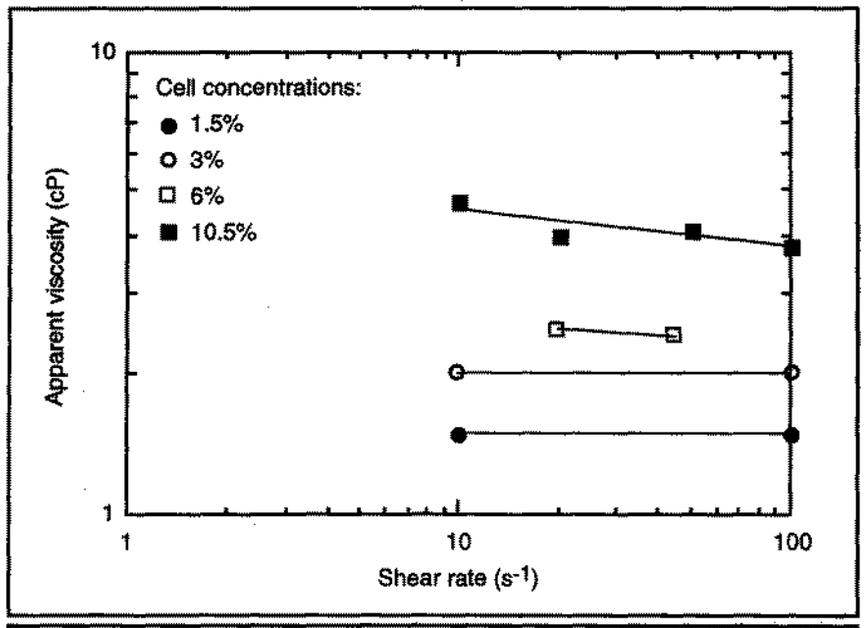
(ii)

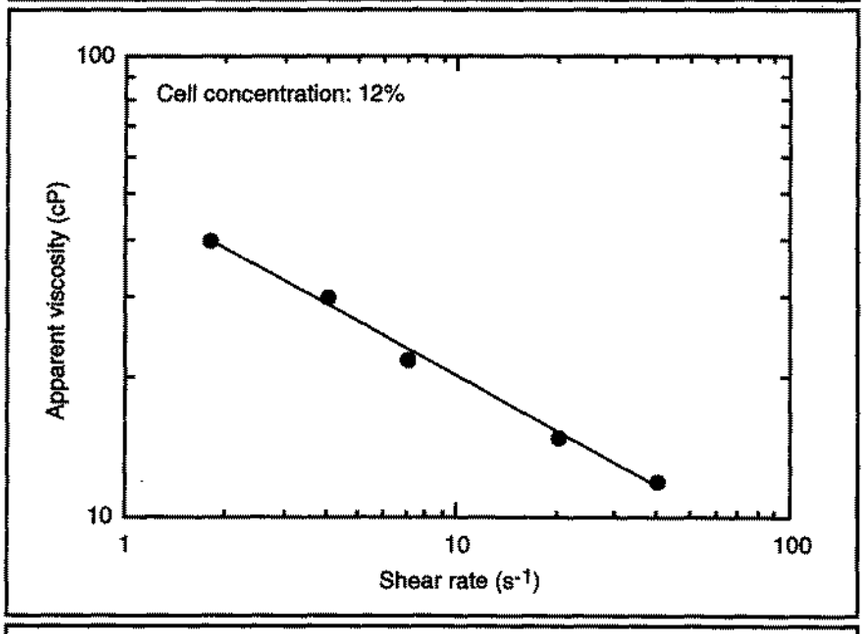
$$\mu_a = K\dot{\gamma}^{n-1} = 11.43\,\text{dyn s}^{0.587}\,\text{cm}^{-2}\,(200\,\text{s}^{-1})^{0.587-1} = 1.3\,\text{dyn s cm}^{-2}$$

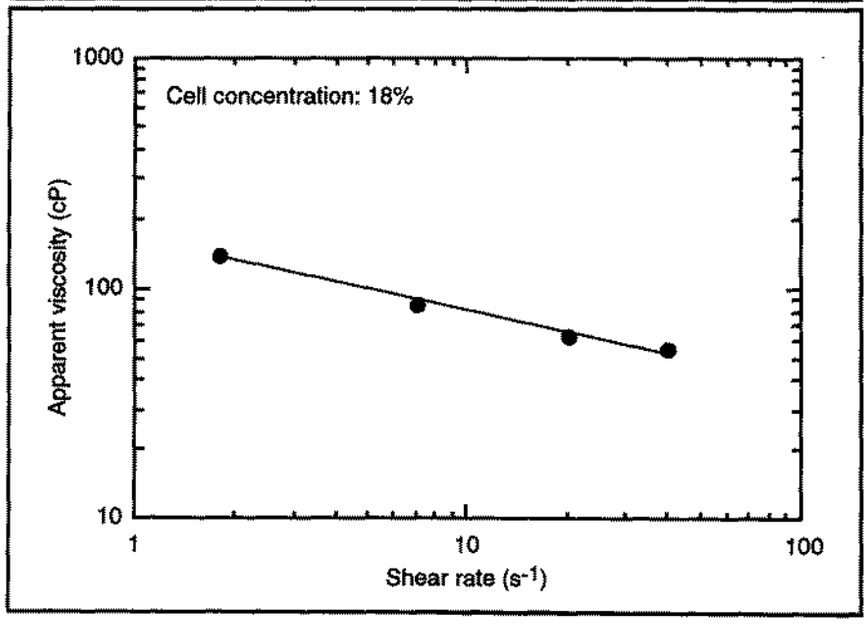
Answer: 1.3 dyn s cm^{-2}

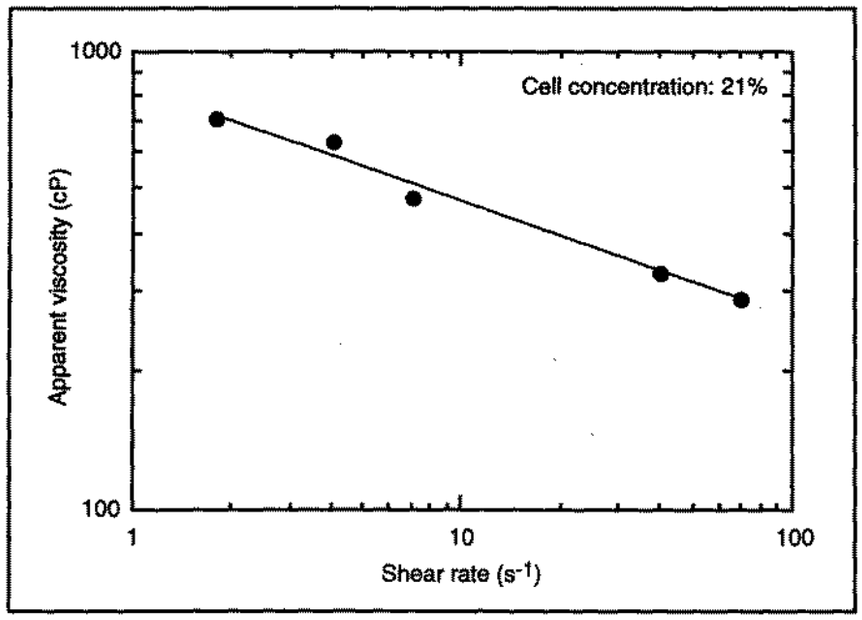
7.2 Rheology of yeast suspensions

From Eq. (7.8), for pseudoplastic fluids, a plot of apparent viscosity versus shear rate on log–log coordinates can be expected to give a straight line. Log–log plots for the different cell concentrations are shown below.


10
Cell concentrations:
1.5%
3%
6%
10.5%
Apparent viscosity (cP)
1
1
10
100
Shear rate (s-1)


100
Cell concentration: 12%
Apparent viscosity (cP)
10
1
10
100
Shear rate (s-1)

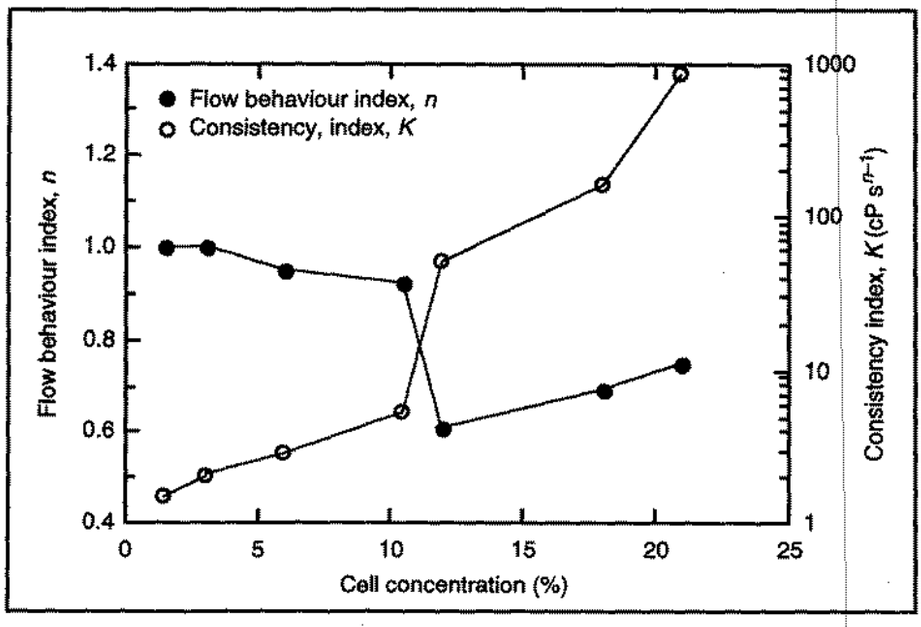

1000
Cell concentration: 18%
Apparent viscosity (cP)
100
10
1
10
100
Shear rate (s-1)



The equations and parameter values for the straight lines in each plot are listed below.

Cell concentration (%)	Equation	Flow behaviour index, n	Consistency index, K (cP s^{n-1})
1.5	$\mu_a = 1.5\,\dot{\gamma}^0 = 1.5$	1	1.5
3	$\mu_a = 2.0\,\dot{\gamma}^0 = 2.0$	1	2.0
6	$\mu_a = 2.91\,\dot{\gamma}^{-0.050}$	0.95	2.9
10.5	$\mu_a = 5.38\,\dot{\gamma}^{-0.076}$	0.92	5.4
12	$\mu_a = 50.1\,\dot{\gamma}^{-0.395}$	0.61	50
18	$\mu_a = 162\,\dot{\gamma}^{-0.307}$	0.69	160
21	$\mu_a = 833\,\dot{\gamma}^{-0.251}$	0.75	830

K and n are plotted as a function of cell concentration below.



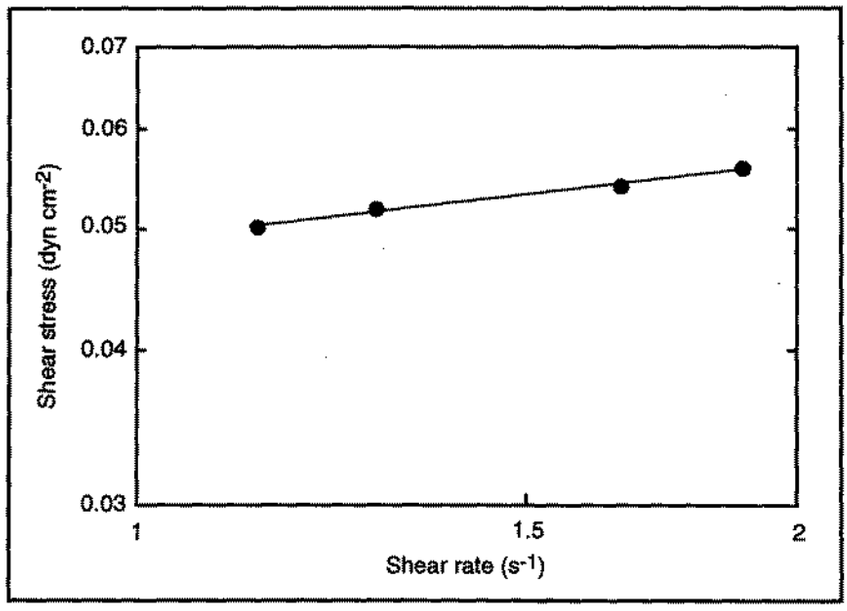
The cell broth is Newtonian up to a cell concentration of about 2%, then becomes pseudoplastic. The flow behaviour index continues to decrease until a cell concentration of about 12% is reached. The consistency index rises throughout the culture with increasing cell concentration.

7.3 Impeller viscometer

(a)

If the rheology can be described using a power-law model, a plot of shear stress versus shear rate on log–log coordinates can be expected to give a straight line. Values of shear stress and shear rate can be determined from torque and stirrer speed data using Eqs (7.11) and (7.12) with k =10.2 and D_i = 4 cm = 4 × 10^{-2} m. The results are listed below and plotted below.

Stirrer speed (s^{-1})	Torque (N m)	Shear stress (N m^{-2})	Shear rate (s^{-1})
0.185	3.57×10^{-6}	0.0559	1.89
0.163	3.45×10^{-6}	0.0540	1.66
0.126	3.31×10^{-6}	0.0518	1.29
0.111	3.20×10^{-6}	0.0501	1.13



The equation for the straight line on the log–log plot is $\tau = 0.049\,\dot{\gamma}^{0.20}$. From Eq. (7.7), this means that the flow behaviour index n is 0.20 and the consistency index K is 0.049 N s^n m^{-2}.

Answer: Yes: $n = 0.20$; $K = 0.049$ N s^n m^{-2}

(b)

The impeller Reynolds number for pseudoplastic fluids is defined in Eq. (7.23). If Re_i at the highest stirrer speed is in the laminar range, conditions at the other stirrer speeds are also laminar. Therefore, for N_i = 0.185 s^{-1} and using the unit conversion factor 1 N = 1 kg m s^{-2} (Table A.4, Appendix A):

$$Re_i = \frac{N_i^{2-n} D_i^2 \rho}{k^{n-1} K} = \frac{(0.185\,\mathrm{s^{-1}})^{2-0.20} (0.04\,\mathrm{m})^2\, 1000\,\mathrm{kg\,m^{-3}}}{10.2^{0.20-1}\left(0.049\,\mathrm{N\,s^{0.20}\,m^{-2}} \cdot \left|\frac{1\,\mathrm{kg\,m\,s^{-2}}}{1\,\mathrm{N}}\right|\right)} = 10.0$$

This value for Re_i is at the upper limit of the laminar range (p 137); therefore, we can conclude that flow at this and lower stirrer speeds is laminar.

Answer: Flow is laminar.

(c)
From Table 7.4, $k = 30$ for a helical-ribbon impeller. From Figure 7.25, flow with this type of impeller is laminar up to about $Re_i = 10^3$. These conditions allow use of higher stirrer speeds within the laminar region than is possible using a turbine impeller. Solving for N_i from Eq. (7.23) and substituting values of n and K for the fluid from **(a)**, the maximum stirrer speed at $Re_i = 30$ can be calculated as:

$$N_i^{2-n} = \frac{Re_i\, k^{n-1}\, K}{D_i^2\, \rho}$$

$$N_i^{2-0.20} = \frac{10^3\left(30^{0.20-1}\right)\left(0.049\ \text{N s}^{0.20}\ \text{m}^{-2} \cdot \left|\frac{1\ \text{kg m s}^{-2}}{1\ \text{N}}\right|\right)}{(0.04\ \text{m})^2\ 1000\ \text{kg m}^{-3}}$$

$$N_i^{1.8} = 2.02\ \text{s}^{-1.8}$$

$$N_i = 1.48\ \text{s}^{-1}$$

Therefore, from Eq. (7.11), as $\dot{\gamma} = 30\, N_i$ for a helical-ribbon impeller, shear rates up to $30 \times 1.48 = 44\ \text{s}^{-1}$ can be used. This is a considerable improvement on the restricted range of up to about $1.9\ \text{s}^{-1}$ as determined in **(a)** for the upper limit of laminar flow with a Rushton turbine impeller.

Answer: The shear rate range can be extended from $1.9\ \text{s}^{-1}$ with a Rushton turbine impeller to $44\ \text{s}^{-1}$ with a helical-ribbon impeller.

7.4 Particle suspension and gas dispersion

The Zwietering equation is an equation in numerics (p 12); therefore, the parameter values used in the equation must have the units specified. $D_p = 10\ \mu\text{m} = 10 \times 10^{-6}$ m; $g = 9.8\ \text{m s}^{-2}$ (p 16); $D_i = 30$ cm $= 0.3$ m; $\rho_L =$ density of water $= 1000\ \text{kg m}^{-3}$. The dynamic viscosity of water μ_L at 20°C is about 1 cP (p 133); from Table A.9 (Appendix A), this is the same as $\mu_L = 10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1}$. Using the definition of kinematic viscosity from p 133:

$$\nu_L = \frac{\mu_L}{\rho_L} = \frac{10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1}}{1000\ \text{kg m}^{-3}} = 10^{-6}\ \text{m}^2\ \text{s}^{-1}$$

The density of the cells ρ_p is $1.04\ \text{g cm}^{-3}$. Converting to kg m^{-3}:

$$\rho_p = 1.04\ \text{g cm}^{-3} \cdot \left|\frac{100\ \text{cm}}{1\ \text{m}}\right|^3 \cdot \left|\frac{1\ \text{kg}}{1000\ \text{g}}\right| = 1040\ \text{kg m}^{-3}$$

Substituting values in the correct units into the Zwietering equation:

$$N_i^* = C\, \nu_L^{0.1}\, D_p^{0.2} \left(\frac{g\,(\rho_p - \rho_L)}{\rho_L}\right)^{0.45} D_i^{-0.85}\, x^{0.13}$$

$$N_i^* = 7.7\left(10^{-6}\right)^{0.1}\left(10 \times 10^{-6}\right)^{0.2}\left(\frac{9.8\,(1040-1000)}{1000}\right)^{0.45}(0.3)^{-0.85}\,(40)^{0.13} = 0.57\ \text{s}^{-1}$$

This minimum stirrer speed for suspension of the cells can be compared with the minimum stirrer speed for dispersion of air bubbles. Taking the average minimum tip speed of $2.0\ \text{m s}^{-1}$ for bubble dispersion:

$$N_i = \frac{\text{tip speed}}{\pi D_i} = \frac{2.0\ \text{m s}^{-1}}{\pi\,(0.3\ \text{m})} = 2.1\ \text{s}^{-1}$$

The stirrer speed for air dispersion is 3.7 times higher than for cell suspension. As power in the turbulent regime is proportional to the stirrer speed cubed, about 50 times more power is required to disperse the air bubbles than to keep the cells in suspension.

Answer: Bubble dispersion requires significantly more power than cell suspension.

7.5 Scale-up of mixing system

(a)

The Reynolds number for a Newtonian fluid in a stirred vessel is defined in Eq. (7.2). Using this equation to evaluate the maximum viscosity μ under turbulent conditions with the density of water $\rho = 1000\ \text{kg m}^{-3}$:

$$\mu = \frac{N_i D_i^2 \rho}{Re_i} = \frac{\left(800\ \text{min}^{-1} \cdot \left|\frac{1\ \text{min}}{60\ \text{s}}\right|\right)\left((5\ \text{cm})^2 \cdot \left|\frac{1\ \text{m}}{100\ \text{cm}}\right|^2\right) 1000\ \text{kg m}^{-3}}{10^4} = 3.3 \times 10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1}$$

From Table A.9 (Appendix A), $1\ \text{kg m}^{-1}\ \text{s}^{-1} = 10^3$ cP; therefore:

$$\mu = 3.3 \times 10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1} \cdot \left|\frac{10^3\ \text{cP}}{1\ \text{kg m}^{-1}\ \text{s}^{-1}}\right| = 3.3\ \text{cP}$$

Answer: 3.3 cP

(b)

The tip speed in the laboratory equipment is:

$$\text{Tip speed} = \pi N_i D_i = \pi\left(800\ \text{min}^{-1} \cdot \left|\frac{1\ \text{min}}{60\ \text{s}}\right|\right)\left(5\ \text{cm} \cdot \left|\frac{1\ \text{m}}{100\ \text{cm}}\right|\right) = 2.1\ \text{m s}^{-1}$$

In the large-scale vessel, $D_i = 15 \times 5\ \text{cm} = 75\ \text{cm}$. If the tip speed is kept the same after scale-up:

$$N_i = \frac{\text{tip speed}}{\pi D_i} = \frac{2.1\ \text{m s}^{-1}}{\pi\left(75\ \text{cm} \cdot \left|\frac{1\ \text{m}}{100\ \text{cm}}\right|\right)} = 0.89\ \text{s}^{-1}$$

Using this value of N_i to calculate the maximum viscosity as in **(a)**:

$$\mu = \frac{N_i D_i^2 \rho}{Re_i} = \frac{0.89\ \text{s}^{-1}\left((75\ \text{cm})^2 \cdot \left|\frac{1\ \text{m}}{100\ \text{cm}}\right|^2\right) 1000\ \text{kg m}^{-3}}{10^4} = 0.05\ \text{kg m}^{-1}\ \text{s}^{-1}$$

Converting to cP:

$$\mu = 0.05\ \text{kg m}^{-1}\ \text{s}^{-1} \cdot \left|\frac{10^3\ \text{cP}}{1\ \text{kg m}^{-1}\ \text{s}^{-1}}\right| = 50\ \text{cP}$$

Answer: Scale-up increases the maximum viscosity for turbulent conditions to 50 cP.

7.6 Effect of viscosity on power requirements

(a)

$D_i = 1$ m; $\rho = 1\ \text{g cm}^{-3} = 1000\ \text{kg m}^{-3}$.

(i)

The viscosity of water at 20°C is about 1 cP (p 133); from Table A.9 (Appendix A), this is the same as $10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1}$. Substituting parameter values into Eq. (7.2):

$$Re_i = \frac{N_i D_i^2 \rho}{\mu} = \frac{\left(90\ \text{min}^{-1} \cdot \left|\frac{1\ \text{min}}{60\ \text{s}}\right|\right)(1\ \text{m})^2\ 1000\ \text{kg m}^{-3}}{10^{-3}\ \text{kg m}^{-1}\ \text{s}^{-1}} = 1.5 \times 10^6$$

From Figure 7.24 for a Rushton turbine, this value of Re_i corresponds to turbulent flow and N_P' can be taken as 5.8. The power required is evaluated using Eq. (7.20):

$$P = N_P' \rho N_i^3 D_i^5 = 5.8\left(1000\ \text{kg m}^{-3}\right)\left(90\ \text{min}^{-1} \cdot \left|\frac{1\ \text{min}}{60\ \text{s}}\right|\right)^3 (1\ \text{m})^5 = 1.96 \times 10^4\ \text{kg m}^2\ \text{s}^{-3}$$

From Table A.8 (Appendix A), 1 W = 1 kg m^2 s^{-3}; therefore:

$$P = 1.96 \times 10^4 \text{ W} = 19.6 \text{ kW}$$

Answer: 19.6 kW

(ii)
For a viscosity of 100×10^{-3} kg m^{-1} s^{-1} = 0.1 kg m^{-1} s^{-1}, Re_i is:

$$Re_i = \frac{N_i D_i^2 \rho}{\mu} = \frac{\left(90 \text{ min}^{-1} \cdot \left|\frac{1 \text{ min}}{60 \text{ s}}\right|\right)(1 \text{ m})^2 \, 1000 \text{ kg m}^{-3}}{0.1 \text{ kg m}^{-1} \text{ s}^{-1}} = 1.5 \times 10^4$$

From Figure 7.24 for a Rushton turbine, this value of Re_i is still within the turbulent regime so that N_P' is again 5.8. Therefore, the power required is the same as that calculated in **(i)**: P = 19.6 kW.

Answer: 19.6 kW

(iii)
For a viscosity of $10^4 \times 10^{-3}$ kg m^{-1} s^{-1} = 10 kg m^{-1} s^{-1}, Re_i is:

$$Re_i = \frac{N_i D_i^2 \rho}{\mu} = \frac{\left(90 \text{ min}^{-1} \cdot \left|\frac{1 \text{ min}}{60 \text{ s}}\right|\right)(1 \text{ m})^2 \, 1000 \text{ kg m}^{-3}}{10 \text{ kg m}^{-1} \text{ s}^{-1}} = 150$$

From Figure 7.24 for a Rushton turbine, this value of Re_i is within the transition regime. N_P read from Figure 7.24 is about 3.5. The power required is evaluated using Eq. (7.18):

$$P = N_P \rho N_i^3 D_i^5 = 3.5\left(1000 \text{ kg m}^{-3}\right)\left(90 \text{ min}^{-1} \cdot \left|\frac{1 \text{ min}}{60 \text{ s}}\right|\right)^3 (1 \text{ m})^5 = 1.18 \times 10^4 \text{ kg m}^2 \text{ s}^{-3}$$

From Table A.8 (Appendix A), 1 W = 1 kg m^2 s^{-3}; therefore:

$$P = 1.18 \times 10^4 \text{ W} = 11.8 \text{ kW}$$

Answer: 11.8 kW

(b)
From Figure 7.24 for a Rushton turbine, turbulence with N_P' = 5.8 is achieved at a minimum Reynolds number of about 10^4. For a viscosity of 1000×10^{-3} kg m^{-1} s^{-1} = 1 kg m^{-1} s^{-1}, the stirrer speed required can be determined from Eq. (7.2):

$$N_i = \frac{Re_i \mu}{D_i^2 \rho} = \frac{10^4\left(1 \text{ kg m}^{-1} \text{ s}^{-1}\right)}{(1 \text{ m})^2 \, 1000 \text{ kg m}^{-3}} = 10 \text{ s}^{-1}$$

Using this result in Eq. (7.20):

$$P = N_P' \rho N_i^3 D_i^5 = 5.8\left(1000 \text{ kg m}^{-3}\right)\left(10 \text{ s}^{-1}\right)^3 (1 \text{ m})^5 = 5.80 \times 10^6 \text{ kg m}^2 \text{ s}^{-3}$$

From Table A.8 (Appendix A), 1 W = 1 kg m^2 s^{-3}; therefore:

$$P = 5.80 \times 10^6 \text{ W} = 5.80 \times 10^3 \text{ kW}$$

Answer: 5.80×10^3 kW

7.7 Electrical power required for mixing

D_i = 7 cm = 0.07 m; ρ = 1000 kg m^{-3}. The viscosity of water at 20°C is about 1 cP (p 133); from Table A.9 (Appendix A), this is the same as 10^{-3} kg m^{-1} s^{-1}. Substituting parameter values into Eq. (7.2):

$$Re_i = \frac{N_i D_i^2 \rho}{\mu} = \frac{\left(900\,\text{min}^{-1}\cdot\left|\frac{1\,\text{min}}{60\,\text{s}}\right|\right)(0.07\,\text{m})^2\,1000\,\text{kg m}^{-3}}{10^{-3}\,\text{kg m}^{-1}\,\text{s}^{-1}} = 7.4\times10^4$$

From Figure 7.24 for a Rushton turbine, this value of Re_i corresponds to turbulent flow and N_P' can be taken as 5.8. The power required is evaluated using Eq. (7.20):

$$P = N_P' \rho N_i^3 D_i^5 = 5.8\left(1000\,\text{kg m}^{-3}\right)\left(900\,\text{min}^{-1}\cdot\left|\frac{1\,\text{min}}{60\,\text{s}}\right|\right)^3(0.07\,\text{m})^5 = 32.9\,\text{kg m}^2\,\text{s}^{-3}$$

From Table A.8 (Appendix A), 1 W = 1 kg m^2 s^{-3}; therefore:

$$P = 32.9\,\text{W}$$

This value is considerably lower that the electrical power consumed by the stirrer motor. Much of the remainder of the electrical power is converted into heat within the motor housing.

Answer: 32.9 W; a significant fraction of the electrical power is dissipated as heat within the motor housing

7.8 Mixing time with aeration

D_i = 0.67 m; ρ = density of water = 1000 kg m^{-3}. From Table A.9 (Appendix A), 1 cP = 10^{-3} kg m^{-1} s^{-1}; therefore, μ = 4 cP = 4 × 10^{-3} kg m^{-1} s^{-1}. For a cylindrical tank of diameter D_T = 2 m and height H= 2 m, the volume V is:

$$V = \pi\left(\frac{D_T}{2}\right)^2 H = \pi\left(\frac{2\,\text{m}}{2}\right)^2 2\,\text{m} = 6.28\,\text{m}^3$$

(a)

If the maximum specific power consumption is 1.5 kW m^{-3}, the maximum power P is:

$$P = \left(1.5\,\text{kW m}^{-3}\right)6.28\,\text{m}^3 = 9.42\,\text{kW}$$

From Table A.8 (Appendix A), 1 W = 1 kg m^2 s^{-3}; therefore 1 kW = 1000 kg m^2 s^{-3}. Converting P to kg m^2 s^{-3}:

$$P = 9.42\,\text{kW} = 9.42\,\text{kW}\cdot\left|\frac{1000\,\text{kg m}^2\,\text{s}^{-3}}{1\,\text{kW}}\right| = 9.42\times10^3\,\text{kg m}^2\,\text{s}^{-3}$$

Assume for now that the fluid flow is turbulent and N_P' = 5.8. The stirrer speed can be evaluated using Eq. (7.20):

$$N_i^3 = \frac{P}{N_P'\rho D_i^5} = \frac{9.42\times10^3\,\text{kg m}^2\,\text{s}^{-3}}{5.8\left(1000\,\text{kg m}^{-3}\right)(0.67\,\text{m})^5} = 12.0\,\text{s}^{-3}$$

$$N_i = 2.29\,\text{s}^{-1} = 137\,\text{rpm}$$

Check that this stirrer speed provides turbulent mixing conditions by evaluating the Reynolds number. Re_i for Newtonian fluids in a stirred vessel is defined in Eq. (7.2):

$$Re_i = \frac{N_i D_i^2 \rho}{\mu} = \frac{\left(2.29\,\text{s}^{-1}\right)(0.67\,\text{m})^2\,1000\,\text{kg m}^{-3}}{4\times10^{-3}\,\text{kg m}^{-1}\,\text{s}^{-1}} = 2.6\times10^5$$

From Figure 7.24 for a Rushton turbine, this value of Re_i is well into the turbulent regime; therefore the value for N_P' assumed above is valid.

For high Re_i, the mixing time can be calculated using Eq. (7.16):